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# Surgery & Male Health

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## Out of the League
## Family Medicine

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## TOPIC: Anal Fissure

**Anal fissure: Summary**

- An anal fissure is a tear or ulcer in the lining of the anal canal which causes pain on defecation. It is classified as:
  - Acute — if present for less than 6 weeks.
  - Chronic — if present for 6 weeks or longer.
  - Primary — if there is no clear underlying cause.
  - Secondary — if there is a clear underlying cause, such as constipation, inflammatory bowel disease, sexually transmitted infection, or colorectal cancer.
- Anal fissures are common. Peak incidence occurs in people aged 15–40 years but anal fissures can occur at any age. Primary anal fissures are uncommon in elderly people and warrant further investigation for an underlying cause.
- Clinical features of anal fissure include anal pain with defecation (with or without bright red rectal bleeding) and anal spasm.
- External examination of the anus may reveal a linear split in the anal mucosa.
  - Acute anal fissures are usually superficial with well-demarcated edges.
  - Chronic anal fissures are wider and deeper with muscle fibers visible in the base. The edges are often swollen, and a skin tag may be visible at the end of the fissure.
  - Primary anal fissures usually occur in the posterior midline, although about 10% of cases occur anteriorly (especially in women).
  - Secondary anal fissures should be suspected if fissures have an irregular outline, are multiple, or occur laterally.
  - Referral for examination under anaesthesia may be needed if the diagnosis is unclear or if anal spasm and pain make diagnosis difficult.
- Management of anal fissure includes:
  - Referral to secondary care if a serious underlying cause (such as rectal cancer or inflammatory bowel disease) is suspected.
  - Giving dietary and lifestyle advice to ensure stools are soft and easy to pass (such as a high fiber diet and increased fluid intake) and to aid healing of the fissure (such as good anal hygiene).
  - Offering simple analgesia (paracetamol or ibuprofen) and advising on measures to reduce pain (such as soaking in a shallow, warm bath) — a short course of topical anaesthetic can also be considered for adults with extreme pain on defecation.
  - For adults whose symptoms have persisted for 1 week or more without improvement considering a 6–8 weeks course of rectal glyceryl trinitrate (GTN) ointment.
  - For people with secondary anal fissure, managing the underlying cause, where possible, or referring the person to secondary care (using clinical judgement to determine the urgency) if a serious underlying cause (such as rectal cancer) is suspected.

- For adults with primary anal fissure, review should be arranged at 6–8 weeks (or sooner if necessary).
    - People whose anal fissure has healed should be advised to continue with the dietary and lifestyle measures to reduce the risk of recurrence.
    - People with unhealed anal fissure despite adherence to dietary and lifestyle measures should be referred to a general or colorectal surgeon.
- For children, specialist advice should be sought if an anal fissure has not healed after 2 weeks (or sooner if there is significant pain).

**Causes:**

- Primary anal fissures do not have a clear underlying cause.
    - Acute anal fissure can occur due to trauma from passing hard or loose stools, but many cases are not associated with constipation or diarrhoea.
    - Other aetiological theories suggest that development of anal fissure may be associated with:
        - A persistent increase in internal anal sphincter tone which leads to reduction in arterial blood flow, local ischaemia, increased risk of tearing, and poor healing of the resulting fissure.
        - Local deficiency in nitric oxide synthase which is needed for production of nitric oxide (the main mediator of internal anal sphincter relaxation).
- Secondary anal fissures can present with atypical features and are associated with an underlying condition. For example:
    - Constipation — passage of hard stool tears the anal mucosa.
    - Conditions which can cause ulceration of the anal mucosa such as:
        - Inflammatory bowel disease.
        - Sexually transmitted infections such as HIV, syphilis, and herpes simplex.
        - Colorectal cancer.
        - Dermatological conditions such as psoriasis and pruritus ani.
        - Bacterial, fungal, or viral skin infections.
    - Anal trauma, for example from previous anal surgery or anal sex.
    - Adverse drug effects, for example opioid analgesics, nicorandil, and chemotherapy.
    - Pregnancy and childbirth (due to pressure on the perineum). Fissures related to childbirth are most commonly located in the anterior midline.

**Prognosis:**

- Primary anal fissures
    - Around half of all acute primary anal fissures resolve within 6–8 weeks of conservative treatment (dietary/lifestyle measures and pain relief).
    - Spontaneous resolution may occur within 2 weeks (often before medical attention is sought).
    - Approximately 40% of people who present with an acute anal fissure progress to chronic anal fissure.
    - Chronic primary anal fissures are more resistant to conservative measures and often more intensive secondary care treatments are needed.

    - Recurrence of anal fissure is common, especially if dietary and lifestyle measures are not adhered to.
- Secondary anal fissures
    - Prognosis of a secondary anal fissure is dependent on the prognosis of the underlying cause.

**Complications:**

Complications of anal fissure include:

- Failure to heal/progression to chronic fissure.
- Recurrent fissure.
- Anorectal fistula.
- Infection and/or abscess.
- Faecal impaction — avoidance of defecation to prevent anal pain may worsen constipation and lead to faecal impaction.
- Reduced quality of life due to pain (especially in people with chronic or recurrent anal fissure).

**Diagnosis:**

- Take a history.
    - Ask about symptoms.
        - Anal pain always occurs with passing a stool — pain is severe and sharp and often followed by deep burning pain that persists for several hours.
        - Bleeding may occur with defecation. When present, it is usually seen as a small quantity of bright red blood on the stool or toilet paper.
        - A tearing sensation on passing stool may also be described.
        - In children, anal fissure typically presents with pain and/or crying during defecation and rectal bleeding.
    - Ask about features of an underlying cause including:
        - Dietary and bowel habits (for example constipation, diarrhoea, or any recent changes). Children often have constipation — pain on defecation may lead to stool retention and harder bulkier stools which cause further damage to the anal mucosa when passed.
        - Previous anorectal trauma (including anal surgery and obstetric history).
        - Associated symptoms (such as abdominal pain or weight loss).
        - Family history of possible underlying causes (such as colorectal cancer or inflammatory bowel disease).
- Examine the person.
    - Assess the anal fissure. Ask the person to lie comfortably in a lateral position and gently part the buttocks. Anal spasm and pain may prevent full visualization of the fissure.
        - Acute anal fissures are typically superficial with well-demarcated edges.
        - Chronic anal fissures are wider and deeper with muscle fibers visible in the base. The edges are often swollen, and a skin tag may be visible at the end of the fissure.

        - Primary anal fissures are usually singular and occur in the posterior midline of the anus, although a few cases may be seen in the anterior midline (especially in women).
        - Secondary anal fissures may have an irregular outline, be multiple, or occur laterally — these require further investigation to identify the underlying cause.
    - Exclude a thrombosed haemorrhoid, which may present similarly.
    - If a secondary anal fissure is suspected, assess for features of an underlying cause, such as:
        - Constipation
        - Colorectal cancer
        - A sexually transmitted infection (such as HIV or syphilis)
        - Inflammatory bowel disease
    - Consider the possibility of sexual abuse if a child has an anal fissure and possible causes (such as constipation or inflammatory bowel disease) have been excluded.
    - Digital rectal examination is not recommended in primary care to diagnose anal fissure.
        - Consider referral for examination under anaesthesia if the diagnosis is unclear or if spasm and pain make diagnosis difficult.
- Be aware that anal fissure is less common in older adults — consider alternative diagnoses.

**Differential diagnosis:**

- Anal pain with or without bright red rectal bleeding may also occur with thrombosed haemorrhoids.
- Other causes of anal pain with or without bleeding include:
    - Inflammatory bowel disease.
    - Sarcoidosis.
    - Infection for example tuberculosis, HIV, or syphilis.
    - Malignancy for example anal carcinoma or lymphoma.
    - Peri-anal abscess.

**Management:**
**Person with anal fissure in PHC:**

- Consider the need for referral.
    - Refer using a suspected cancer pathway referral (for an appointment within 2 weeks) if rectal cancer is suspected.
    - Refer to an appropriate specialist (with urgency dependent on clinical judgement) if another serious underlying cause, such as inflammatory bowel disease or a sexually transmitted infection (such as HIV infection), is suspected (for example if fissure looks atypical on examination).
    - Consider referring to a general or colorectal surgeon if anal fissure occurs in an elderly person.
- For all people with anal fissure:
    - Ensure stools are soft and easy to pass.
        - If the anal fissure is caused by constipation

        - If the person is not constipated, advise adequate dietary fibre intake by eating a balanced diet containing whole grains, fruits, and vegetables; this should be done gradually to minimize flatulence and bloating. Also advise that adequate fluid intake is particularly important with an increased fibre diet to maintain soft, well-lubricated stools and to prevent intestinal obstruction.
    - Give lifestyle advice to aid healing of the anal fissure.
        - Discuss the importance of correct anal hygiene, especially in children. The anal region should be kept clean and dry to aid healing and avoid complications of anal fissure (such as infection or abscess).
        - Advise against 'stool withholding' and undue straining during bowel movements, both of which can worsen anal fissure.
    - Manage pain.
        - Offer simple analgesia (paracetamol or ibuprofen) if there is prolonged burning pain following defecation. Avoid opioid analgesics as they can cause constipation and exacerbate symptoms.
        - Advise that sitting in a shallow, warm bath several times a day (if possible, particularly after a bowel movement) may help relieve pain.
        - For an adult with extreme pain on defecation, consider prescribing a short course (a few days) of a topical anaesthetic (lidocaine 5% ointment) for use before passing a stool. Advise the person to apply 1–2 mL when required.
- For an adult with a primary anal fissure who has had symptoms for 1 week or more without improvement:
    - In addition to the lifestyle measures described above, consider prescribing rectal glyceryl trinitrate (GTN) 0.4% ointment (provided there are no contraindications) to aid healing.
    - Advise the person to apply twice a day for 6–8 weeks.
        - Around 25% of people experience headache as a side effect which may affect compliance.
        - GTN ointment should be avoided in children, and during pregnancy and breastfeeding
- For people with secondary anal fissure for whom referral is not indicated:
    - Manage the underlying cause, where possible.
- Give follow-up advice:
    - For a primary anal fissure:
        - In a child, advise the parents/carers to return if the fissure is unhealed after 2 weeks, or earlier if the child is in a lot of pain.
        - In an adult, review if the fissure is unhealed after 6–8 weeks, or earlier if needed (for example if the person develops intolerable adverse effects from rectal GTN ointment).
        - Advise all people that they should continue with the dietary and lifestyle measures when the fissure has healed to reduce the risk of recurrence.
    - For a secondary anal fissure, follow-up arrangements will depend on the underlying cause.

**Person with an unhealed or recurrent anal fissure:**

- Following initial management:
    - If anal fissure is unhealed in a child after 2 weeks (or earlier if the child is in a lot of pain):
        - Seek specialist advice/refer to a paediatrician.
    - If anal fissure is unhealed in an adult after 6–8 weeks but the person is asymptomatic or has had notable symptomatic improvement, consider (based on clinical judgement):
        - Prescribing a second course of rectal glyceryl trinitrate (GTN) 0.4% ointment with review after a further 6–8 weeks, or
        - Discussion with/referral to colorectal surgery.
    - If anal fissure is unhealed in an adult after 6–8 weeks and there is no notable symptomatic improvement, confirm that rectal GTN ointment was used for 6–8 weeks.
        - If there was inadequate adherence to treatment with rectal GTN ointment (for example it was stopped early due to intolerable headache), seek specialist advice on alternatives (for example topical diltiazem 2% [unlicensed product]).
        - If there was adequate adherence to initial treatment, refer the person to a general or colorectal surgeon for investigation and treatment.
    - If anal fissure recurs after the initial episode has healed ask about continuation with dietary and lifestyle measures:
        - If adherence has been inadequate, reinforce the importance of doing so, and manage the anal fissure.
        - If adherence was adequate, refer the person to a general or colorectal surgeon for investigation and consideration of more intensive treatment.
    - If at any time an anal fissure appears atypical or scarring or skin tags develop — refer to secondary care for further investigation and management.

**Contraindications and cautions with rectal glyceryl trinitrate ointment**

- Do not prescribe rectal glyceryl trinitrate (GTN) 0.4% ointment to:
    - People with:
        - Postural hypotension, hypotension, or uncorrected hypovolaemia — the use of GTN in such states could cause severe hypotension or shock.
        - Increased intracranial pressure (for example head trauma or cerebral haemorrhage) or inadequate cerebral circulation.
        - Migraine or recurrent headache — there is an increased risk of developing headache with treatment.
        - Aortic or mitral stenosis.
        - Hypertrophic obstructive cardiomyopathy.
        - Constrictive pericarditis or pericardial tamponade.
        - Marked anaemia.
        - Closed-angle glaucoma.
    - People who are:

        - Younger than 18 years of age — the manufacturer states that rectal GTN 0.4% ointment is not recommended for use in this age group due to a lack of data on safety and efficacy.
        - Taking drugs that may interact such as phosphodiesterase type 5 (PDE5) inhibitors for example sildenafil and other long-acting GTN products, isosorbide dinitrate.
- Prescribe rectal GTN 0.4% ointment with caution to:
    - People with:
        - Severe hepatic or renal disease.
        - Cardiac disorders (such as acute myocardial infarction or heart failure) — careful clinical and haemodynamic monitoring must be used to avoid hypotension and tachycardia.
    - People who are:
        - Elderly — they may be more susceptible to the development of postural hypotension.
        - Pregnant — the manufacturer advises against the use of rectal GTN in pregnancy because animal studies are inconclusive about its effects on fetal and postnatal development. However, the British National Formulary (BNF) states that it is not known to be harmful during pregnancy.
        - Breastfeeding — the manufacturer advises against the use of rectal GTN in breastfeeding women due to a lack of evidence and potential harmful effects on a breastfed infant. However, the UK Specialist Pharmacy Services states that it is compatible with breastfeeding, but the woman should be advised to wash her hands immediately after use and before handling the infant.

**Adverse effects are associated with rectal glyceryl trinitrate ointment**

- Dose-related headache is the most common adverse effect, experienced by more than half of all people using rectal glyceryl trinitrate (GTN) 0.4% ointment.
- Other adverse effects include dizziness, light-headedness, blurred vision, nausea, and uncommonly, diarrhoea, anal discomfort, vomiting, rectal bleeding, pruritus, anal burning and itching, and tachycardia.
    - If bleeding associated with haemorrhoids increases, treatment should be stopped.
- Rarely, hypotension may occur, causing dizziness, syncope, and light-headedness. Advise the person:
    - To get up slowly from lying or sitting down to minimize postural hypotension. This is particularly important for people taking diuretics.
    - That the hypotensive effects of GTN may be enhanced by alcohol.
- As rectal GTN may cause dizziness, light-headedness, blurred vision, headache or tiredness in some people, especially on first use, the manufacturer recommends giving advice on driving or operating machinery.

## TOPIC: Haemorrhoids

**Summary**

- Haemorrhoids (also known as piles) are abnormally swollen vascular mucosal cushions in the anal canal.
- Haemorrhoids are classed as external or internal, depending on their origin in relation to the dentate line. The dentate line is situated 2 cm from the anal verge and marks the transition between the upper and lower anal canal.
    - External haemorrhoids originate below the dentate line and are covered by modified squamous epithelium (anoderm), which is richly innervated with pain fibres. External haemorrhoids can therefore be itchy and painful.
    - Internal haemorrhoids arise above the dentate line and are covered by columnar epithelium, which have no pain fibres. Internal haemorrhoids are therefore not sensitive to touch, temperature, or pain (unless they become strangulated).
- Complications of haemorrhoids include ulceration; skin tags; maceration of the perianal skin; ischaemia, thrombosis, or gangrene; and rarely, perianal sepsis and anaemia from bleeding.
- Management includes:
    - Ensuring stools are soft and easy to pass.
    - Prescribing laxative treatment if the person is constipated.
    - Giving lifestyle advice to aid healing of the haemorrhoid, such as minimizing straining and maintaining good anal hygiene.
    - Offering symptomatic relief with simple analgesia and/or topical haemorrhoidal preparations.
- Admission or referral should be arranged (using clinical judgement to determine the urgency) if:
    - The diagnosis is unclear or a serious pathology is suspected.
    - There are severe symptoms which cannot be managed in primary care.
    - The person does not respond to conservative treatment.
    - There are recurrent symptoms which do not respond to primary care treatment.
- Secondary care treatments for haemorrhoids may be non-surgical or surgical, depending on the severity of symptoms and the degree of prolapse.
    - Non-surgical treatments include rubber band ligation, injection sclerotherapy, infrared coagulation/photocoagulation, and bipolar diathermy and direct-current electrotherapy.
    - Surgical treatments include haemorrhoidectomy, stapled haemorrhoidectomy, and haemorrhoidal artery ligation.

**Background information**
**Definition**

- Haemorrhoids (also known as piles) are abnormally swollen vascular mucosal cushions in the anal canal.
    - In the anus, there are three vascular mucosal cushions which help maintain anal continence. These are typically described as being present at the left lateral, right posterior, and right anterior positions (that is, at 3, 7, and 11 o'clock), but there is considerable individual variation.

    - When these mucosal cushions become enlarged and symptomatic, they are called haemorrhoids.
- Haemorrhoids are classed as external or internal, depending on their origin in relation to the dentate line. The dentate line is situated 2 cm from the anal verge and marks the transition between the upper and lower anal canal.
    - External haemorrhoids originate below the dentate line and are covered by modified squamous epithelium (anoderm), which is richly innervated with pain fibres. External haemorrhoids can therefore be itchy and painful.
    - Internal haemorrhoids arise above the dentate line and are covered by columnar epithelium, which have no pain fibres. Internal haemorrhoids are therefore not sensitive to touch, temperature, or pain (unless they become strangulated).
    - Internal haemorrhoids are further graded by degree of prolapse (although these classifications do not always reflect the severity of the symptoms):
        - First degree (Grade 1) — haemorrhoids project into the lumen of the anal canal but do not prolapse.
        - Second degree (Grade 2) — haemorrhoids protrude beyond the anal canal on straining but spontaneously reduce when straining is stopped.
        - Third degree (Grade 3) — haemorrhoids protrude outside the anal canal and reduce fully on manual pressure.
        - Fourth degree (Grade 4) — haemorrhoids protrude outside the anal canal and cannot be reduced.
    - People can have internal and external haemorrhoids at the same time.
- Complications of haemorrhoids include:
    - Perianal thrombosis.
    - Incarceration of prolapsing haemorrhoidal tissue — the tissue is unable to be reduced into the anal canal, causing severe pain.
    - Ulceration — from thrombosis of external haemorrhoids.
    - Skin tags — from repeated episodes of haemorrhoid dilatation and thrombosis. This causes enlargement of the overlying skin, problems with hygiene, and secondary irritation.
    - Maceration of the perianal skin — due to mucus discharge.
    - Ischaemia, thrombosis, or gangrene in fourth degree internal haemorrhoids — progressive venous engorgement and incarceration of the acutely inflamed haemorrhoid leads to thrombosis, infarction, and severe pain.
    - Anal stenosis.
    - Perianal or pelvic sepsis (rare).
    - Anaemia from continuous or excessive bleeding (rare).
- The prognosis of haemorrhoids is usually good.
    - Many symptomatic episodes settle with conservative measures, with low rates of recurrence.
    - Residual or recurrent symptoms are more likely in people with continuing predisposing factors, such as constipation and straining while trying to pass stool.
    - About 10% of people will require surgery to alleviate their symptoms.

    - Recurrence rate after surgery is around 13%
- Factors which contribute to the development of haemorrhoids include:
    - Constipation.
    - Straining while trying to pass stools.
    - Ageing (weakening of the support structures makes haemorrhoids more likely to prolapse).
    - Conditions that cause raised intra-abdominal pressure (such as pregnancy, childbirth, ascites, or a pelvic mass).
    - Chronic cough.
    - Heavy lifting.
    - Exercising.
    - Hereditary factors (possibly due to a congenital weakness of the venous walls).
    - A low fiber-diet.
- These factors lead to increased pressure within the submucosal arteriovenous plexus and ultimately contribute to swelling of the cushions, laxity of the supporting connective tissue, and protrusion into and through the anal canal.
- Women who experienced spontaneous pushing and slow delivery of the baby's head had fewer symptoms of haemorrhoids at 3 weeks postnatally.

**How should I diagnose haemorrhoids?**

- Take a history.
    - Ask about symptoms, including the duration and severity, the relationship between symptoms and defecation, and any exacerbating or relieving factors.
        - Bright red, painless rectal bleeding is the most common symptom of haemorrhoids. It typically occurs with defecation and is seen as streaks on the toilet paper, in the toilet bowl, and/or outside of the stool (but not mixed with it).
        - Anal itching or irritation may occur with prolapsing internal haemorrhoids (from prolapse-associated moisture, mucus secretion, and/or faecal incontinence) and with external haemorrhoids (from difficulty with perianal hygiene as a result of skin tags or extensive external haemorrhoids).
        - A feeling of rectal fullness, discomfort, or of incomplete evacuation on bowel movements may be present if prolapse occurs with straining and in people with large haemorrhoids.
        - Soiling (due to mucus discharge or impaired continence) may also occur.
        - Pain is not usually reported with internal haemorrhoids unless they prolapse and becomes strangulated, resulting in intense pain. External haemorrhoids are usually painless unless thrombosis occurs (the haemorrhoidal tissue becomes engorged and forms clots), resulting in an acutely painful mass in the rectum.
        - A person with haemorrhoids may be asymptomatic, or there may be a history of intermittent symptoms lasting from a few days to a few weeks.
        - In a symptomatic person, it may be difficult to determine whether symptoms are due to internal or external haemorrhoids, especially since people can have internal and external haemorrhoids at the same time.

    - Ask about a family history of, and the presence of 'red flag' symptoms for, anal or colorectal cancer.
    - Ask about the presence of predisposing factors for haemorrhoids, such as constipation.
- Examine the person. Ask them to lie comfortably in the left lateral position, the prone jack-knife position, or the lithotomy position. Offer the person the option of a chaperone.
    - Inspect the perianal area by gently spreading the buttocks.
        - The perineum may appear normal if there is a non-prolapsed internal haemorrhoid.
        - The perineum may be macerated from chronic mucus discharge causing local irritation.
        - If internal haemorrhoids have prolapsed, straining may cause them to appear at the anal verge, and bluish, soft bulging vessels covered by mucosa may be seen.
        - If external haemorrhoids are asymptomatic, a bluish bulging of the blood vessel beneath the skin may be seen.
        - An acutely thrombosed external haemorrhoid may present as a purplish, oedematous, tense, tender, subcutaneous perianal mass.
    - Check for the presence of skin tags or other anal pathology, such as fissures, fistulas, or perianal masses.
    - Perform a digital rectal examination (if this is possible without causing pain).
        - Although internal haemorrhoids are not palpable, it is important to exclude differential diagnoses, such as a rectal prolapse.
    - A rigid anoscope, proctoscope, or rectoscope can be used to make a diagnosis of haemorrhoids, classify their severity, and exclude sinister pathology. Where the facilities or expertise are not available to perform this investigation in primary care, the person may need to be referred for assessment.
        - Haemorrhoids appear as pink swellings of the mucosa.
        - Even if haemorrhoids are seen, this does not mean that other pathology is not present.
        - Serious pathology cannot be discounted at any age. Refer if there is doubt about the diagnosis.
- Consider the need for further investigations.
    - Arrange a full blood count if there are signs of anaemia in a person that has experienced significant rectal bleeding.

**Differential diagnosis of haemorrhoids include:**

- Adenomatous polyps — bleeding, diarrhoea, and constipation may be present. Rarely a rectal mass may be palpable.
- Anal fissure
- Anal or colorectal cancer
- Anorectal fistula — may present with anal discharge, itch, intermittent swelling, bleeding, and pain. An external opening may be seen and a tract or cord may be palpable on digital rectal examination. Induration laterally or posteriorly suggests deep perianal or ischiorectal extension.

- Anorectal melanoma — there may be rectal bleeding, perianal pain, and/or a perianal mass.
- Condylomata acuminata (warts) — may resemble skin tags and bleed or cause irritation.
- Diverticular disease
- Inflammatory bowel disease.
- Perianal abscess — which can cause perianal pain, swelling, fever, marked tenderness, induration, redness, and fluctuation.
- Portal hypertension — can cause varices of the anal canal which are distinct from haemorrhoids.
- Pruritus ani and associated causes
- Rectal prolapse — a protruding mass and possible faecal incontinence.
- Sexually transmitted infections (gonorrhoea, syphilis, or chancroid)

**How should I manage a person with haemorrhoids?**

- Consider the need for admission or referral.
  - Consider admitting people with:
    - Extremely painful, acutely thrombosed external haemorrhoids who present within 72 hours of onset (reduction or excision may be needed).
    - Internal haemorrhoids which have prolapsed and become swollen, incarcerated, and thrombosed (haemorrhoidectomy may be needed).
    - Perianal sepsis (a rare but life-threatening complication).
  - Refer using the suspected cancer pathway referral (for an appointment within 2 weeks) if anal or colorectal cancer is suspected.
  - Refer to an appropriate specialist (using clinical judgement to determine the urgency) if another serious pathology, such as inflammatory bowel disease or a sexually transmitted infection, is suspected.
  - Refer for non-urgent assessment and management, people with:
    - Fourth degree or third-degree haemorrhoids which are too large for non-operative measures (haemorrhoidectomy may be needed).
    - Perianal haematoma (a blue or dark coloured swelling at the anal verge) if symptoms are for less than 24 hours duration for clot evaluation.
    - Combined internal and external haemorrhoids with severe symptoms (surgery may be required).
    - Thrombosed haemorrhoids when bleeding is problematic, or there is chronic irritation or leakage.
    - Large skin tags (surgical excision may be required).
- Ensure stools are soft and easy to pass.
  - If the person is constipated, treat the constipation
  - If the person is not constipated:
    - Advise adequate dietary fibre intake by eating a balanced diet containing whole grains, fruits, and vegetables; this should be done gradually to minimize flatulence and bloating.

        - Advise that adequate fluid intake is particularly important with an increased fibre diet to maintain soft, well-lubricated stools and to prevent intestinal obstruction.
- Give lifestyle advice to aid healing of the haemorrhoid.
    - Advise on the importance of correct anal hygiene. The anal region should be kept clean and dry to aid healing and reduce irritation and itching. Recommend careful perianal cleansing and to pat (rather than rub) the area dry.
    - Advise against 'stool withholding' and undue straining during bowel movements, both of which can worsen the condition.
- Manage any symptoms.
    - Offer simple analgesia (such as paracetamol) for pain relief. Avoid opioid analgesics (such as codeine) as they can cause constipation, and avoid nonsteroidal anti-inflammatory drugs (NSAIDs) if rectal bleeding is present.
    - Note: haemorrhoidal conditions are a type of condition for which over the counter preparations should no longer be routinely prescribed.
- Advise the person that when the haemorrhoid has healed, they should continue with dietary and lifestyle measures to reduce the risk of recurrence.
- Refer to secondary care for further investigation and management:
    - People who do not respond to conservative treatment.
    - People with recurrent symptoms who do not respond to primary care management.

**What treatments are available in secondary care?**

- Secondary care treatments for haemorrhoids may be non-surgical or surgical, depending on the severity of symptoms and the degree of prolapse.
- Non-surgical treatments include:
    - Rubber band ligation:
        - A band is applied to the base of the haemorrhoid. The strangulated haemorrhoid becomes necrotic and sloughs off. The underlying tissue undergoes fixation by fibrotic wound healing. Up to three haemorrhoids can be banded at one visit.
        - This is currently the best available outpatient treatment of haemorrhoids. About 2 in 10 people will require a second banding within 6 months for symptom control.
        - Minor complications include haemorrhoid thrombosis, band displacement, mild bleeding, and formation of mucosal ulcers.
    - Injection sclerotherapy:
        - Phenol in oil is injected into the submucosa of the rectum, around the pedicles of the haemorrhoids. It induces a fibrotic reaction which obliterates the haemorrhoidal vessels, causing atrophy of the haemorrhoids.
        - Provides short-term benefit in most people.
        - Complications include pelvic infection and erectile dysfunction due to incorrectly sited injections (rare).
    - Infrared coagulation/photocoagulation:

        - This involves using infrared energy to produce an area of submucosal fibrosis leading to mucosal fixation and a reduction in the tendency to prolapse.
        - It may be as effective as rubber band ligation and injection sclerotherapy in the treatment of first degree and second-degree haemorrhoids.
    - Bipolar diathermy and direct-current electrotherapy:
        - Causes coagulation and fibrosis after local application of heat.
        - Success rates are similar to those of infrared coagulation, and complication rates are low.
- Surgical treatments include:
    - Haemorrhoidectomy:
        - Only symptomatic haemorrhoids are excised as conserves the sensitive anoderm for continence.
        - Complications include post-operative urinary retention, secondary haemorrhage 7–10 days after the operation (from the vascular pedicle or from the edges of the wound), anal stricture, abscess, fistula, formation of skin tags, infection, pseudopolyps, and faecal incontinence.
        - Surgical haemorrhoidectomy confers the best long-term effect with less than 20% symptom recurrence.
    - Stapled haemorrhoidectomy:
        - A circular stapling gun is used to excise a doughnut of mucosa from the upper anal canal and lift the haemorrhoidal cushions back within the canal.
        - Although termed haemorrhoidectomy, it is more accurately termed a haemorrhoidopexy as the haemorrhoids are not excised but relocated within the anal canal.
        - Retroperitoneal sepsis, rectal perforation, anovaginal fistula, and substantial haemorrhage are rare, but serious, complications.
    - Haemorrhoidal artery ligation:
        - Using a proctoscope, the haemorrhoidal arteries are ligated with sutures (above the dentate line) to remove the flow of blood to the haemorrhoids. For larger prolapsing haemorrhoids, an adjunctive mucosal plication procedure is done; the prolapsing mucosa is plicated up to the level of the dentate line where it is fixed by ligation of the plicating sutures (haemorrhoidopexy).
        - Recent evidence questions the efficacy of this when compared to conventional treatments. Frequent repeat procedures are required to achieve symptom improvement or cure and also reduce the cost-effectiveness of this treatment.

**Prescribing information**
Important aspects of prescribing information relevant to primary healthcare are covered in this section specifically for the drugs recommended in this CKS topic.

**Topical haemorrhoidal preparations**
**Which topical haemorrhoidal preparations are available and which should I prescribe?**

- Haemorrhoidal preparations usually contain multiple ingredients, including astringent, lubricants, antiseptic, local anaesthetic, and/or corticosteroids.

    - There is no evidence that any topical haemorrhoidal preparations are more effective than another. The choice of preparation should therefore be based on the risk of adverse effects and the person's symptoms and preference.
        - Preparations containing mild astringents or lubricants relieve local irritation and are less likely to cause skin sensitization.
        - Preparations containing local anaesthetics may alleviate pain, burning, and itching, but can cause sensitization of the anal skin. Lidocaine is the preferred topical anaesthetic because others, including tetracaine, cinchocaine (dibucaine), and pramocaine (pramoxine), are more irritant.
        - Preparations containing corticosteroids may reduce inflammation and pain, but prolonged use may lead to skin atrophy, contact dermatitis, and skin sensitization. Local infection (for example herpes simplex infection or perianal thrush) must be excluded before use.
- They include creams and ointments (generally used for external haemorrhoids) and suppositories (generally used for internal haemorrhoids).
    - No topical haemorrhoidal preparations are licensed for use during pregnancy or breastfeeding. However, the potential risk of harms to the pregnant woman and/or fetus is likely to be less with simple, soothing products than with those containing corticosteroid and/or local anaesthetic (which are generally not recommended by the manufacturers for use during pregnancy due to inadequate safety evidence).

If a topical haemorrhoidal preparation is prescribed or recommended, advise the person on the safe use of the preparation.

**What should I advise on the safe use of topical haemorrhoidal preparations?**

- If a topical haemorrhoidal preparation is prescribed or recommended, advise the person:
    - On how to use it:
        - Generally, haemorrhoidal preparation should be used in the morning and night, and after a bowel movement.
        - Anaesthetic-containing preparations should only be used for a few days because they may cause sensitization of the anal skin.
        - Corticosteroid-containing preparations should be used for no longer than 7 days because prolonged use may lead to skin atrophy, contact dermatitis, and skin sensitization. Excessive application should be avoided, as systematic absorption can occur through the rectal mucosa.
    - That local adverse effects may occur on application, such as burning, erythema, exfoliation, or irritation (frequency unknown).
    - That they should seek advice from a pharmacist before buying over the counter topical haemorrhoidal preparations as they may contain similar ingredients to the one prescribed (including corticosteroid and/or local anaesthetic).

TOPIC: Abdominal swelling

**The table below gives characteristic exam question features for conditions causing abdominal swelling**

| Condition | Characteristic exam feature |
|---|---|
| Pregnancy | Young female<br>Amenorrhoea |
| Intestinal obstruction | History of malignancy/previous operations<br>Vomiting<br>Not opened bowels recently<br>'Tinkling' bowel sounds |
| Ascites | History of alcohol excess, cardiac failure |
| Urinary retention | History of prostate problems<br>Dullness to percussion around suprapubic area |
| Ovarian cancer | Older female<br>Pelvic pain<br>Urinary symptoms e.g. urgency<br>Raised CA125<br>Early satiety, bloating |

TOPIC: Abdominal wall hernias

The classical surgical definition of a hernia is the protrusion of an organ or the fascia of an organ through the wall of the cavity that normally contains it.
**Risk factors for abdominal wall hernias include:** Obesity, ascites, increasing age and surgical wounds
**Features:** palpable lump, cough impulse, pain, obstruction more common in femoral hernia, strangulation: may compromise the bowel blood supply leading to infarction
**Types of abdominal wall hernias:**

| Type of hernia | Details |
|---|---|
| **Inguinal hernia** | Inguinal hernias account for 75% of abdominal wall hernias. Around 95% of patients are male; men have around a 25% lifetime risk of developing an inguinal hernia.<br>Above and medial to pubic tubercle<br>Strangulation is rare |
| **Femoral hernia** | Below and lateral to the pubic tubercle<br>More common in women, particularly multiparous ones<br>High risk of obstruction and strangulation<br>Surgical repair is required |
| **Umbilical hernia** | Symmetrical bulge under the umbilicus |
| **Paraumbilical hernia** | Asymmetrical bulge - half the sac is covered by skin of the abdomen directly above or below the umbilicus |
| **Epigastric hernia** | Lump in the midline between umbilicus and the xiphisternum<br>Risk factors include extensive physical training or coughing (from lung diseases), obesity |
| **Incisional hernia** | May occur in up to 10% of abdominal operations |
| **Spigelian hernia** | Also known as lateral ventral hernia<br>Rare and seen in older patients<br>A hernia through the spigelian fascia (the aponeurotic layer between the rectus abdominis muscle medially and the semilunar line laterally) |
| **Obturator hernia** | A hernia which passes through the obturator foramen. More common in females and typical presents with bowel obstruction |
| **Richter hernia** | A rare type of hernia where only the antimesenteric border of the bowel herniates through the fascial defect<br><br>Richter's hernia can present with strangulation without symptoms of obstruction |

**Abdominal wall hernias in children:**

| Type of hernia | Details |
|---|---|
| **Congenital inguinal hernia** | Indirect hernias resulting from a patent processus vaginalis<br>Occur in around 1% of term babies. More common in premature babies and boys<br>60% are right sided, 10% are bilaterally<br>Should be surgically repaired soon after diagnosis as at risk of incarceration |
| **Infantile umbilical hernia** | Symmetrical bulge under the umbilicus<br>More common in premature and Afro-Caribbean babies<br>The vast majority resolve without intervention before the age of 4-5 years<br>Complications are rare |

## TOPIC: Hydatid cysts

Hydatid cysts are endemic in Mediterranean and Middle Eastern countries. They are caused by the tapeworm parasite *Echinococcus granulosus*. An outer fibrous capsule is formed containing multiple small daughter cysts. These cysts are allergens which precipitate a **type 1 hypersensitivity reaction**.

**Clinical features are as follows:**

- Up to 90% of cysts occur in the liver and lungs
- Can be asymptomatic, or symptomatic if cysts > 5cm in diameter
- Morbidity caused by cyst bursting, infection and organ dysfunction (biliary, bronchial, renal and cerebrospinal fluid outflow obstruction)
- In biliary rupture, there may be the classical triad of; biliary colic, jaundice, and urticaria

**Investigation:**

- **imaging**
  - ultrasound if often used first-line
  - CT is the best investigation to differentiate hydatid cysts from amoebic and pyogenic cysts
- **serology**
  - useful for primary diagnosis and for follow-up after treatment
  - wide variety of different antibody/antigen tests available

Surgery is the mainstay of treatment (the cyst walls must not be ruptured during removal and the contents sterilised first).

CT scan showing a hydatid cyst of the liver.

## TOPIC: Renal Stones

Urolithiasis is a condition arising from the formation of renal calculi when the urine is supersaturated with salt and minerals such as calcium oxalate, struvite (ammonium magnesium phosphate), uric acid and cystine. 80% of stones contain calcium. These urinary tract stones vary considerably in size from small 'gravel-like' stones to large staghorn calculi. The calculi may stay in the position in which they are formed, or migrate down the urinary tract, producing symptoms along the way. Studies suggest that the initial factor involved in the formation of a urinary tract stone may be the presence of nanobacteria that form a calcium phosphate shell.

The other factor that leads to urolithiasis/urinary tract stone production is the formation of Randall's plaques. Calcium oxalate precipitates form in the basement membrane of the thin loops of Henle; these eventually accumulate in the subepithelial space of the renal papillae, leading to a Randall's plaque and eventually a calculus

**Risk Factors:**

- Anatomical anomalies in the kidneys and/or urinary tract - eg, horseshoe kidney, ureteral stricture.
- Family history of urinary tract stones. Stone formation is twice as likely to occur in people who have a first-degree relative with stones.
- Hypertension.
- Gout.
- Hyperparathyroidism.
- Immobilisation.
- Relative dehydration.
- Metabolic disorders which increase excretion of solutes - eg, chronic metabolic acidosis, hypercalciuria, hyperuricosuria.
- Chronic lead and cadmium exposure are associated with stone formation.
- Deficiency of citrate in the urine.
- Cystinuria (an autosomal-recessive aminoaciduria).
- Drugs - eg, diuretics such as triamterene and calcium/vitamin D supplements.
- More common occurrence in hot climates, due to the effect on fluid status and urine volume.
- A diet with excessive intake of oxalate, urate, sodium, and animal protein.
- Increased risk of stones in higher socio-economic groups.
- Obesity - the association seems to be greater in women than in men. It is thought that low urine pH and uric acid stones and an association with hypercalciuria could account for an increased risk of uric acid and/or calcium stones in obese people.
- Contamination - as demonstrated by a spate of melamine-contaminated infant milk formula.

**Risk for recurrent stones:**

- Type of stone - people with calcium-containing stones, uric acid and ammonium urate stones, and infection stones (such as struvite stones).
- Positive family history of stones.
- Previous stone formation.
- Early onset of stones.

**Presentation:**

- Many urinary tract stones are asymptomatic and discovered during investigations for other conditions.

- The classical features of renal colic are sudden severe pain. It is usually caused by stones in the kidney, renal pelvis or ureter, causing dilatation, stretching and spasm of the ureter. In most cases no cause is found:
    - Pain starts in the loin about the level of the costovertebral angle (but sometimes lower) and moves to the groin, with tenderness of the loin or renal angle, sometimes with haematuria.
    - If the stone is high and distends the renal capsule then pain will be in the flank but as it moves down pain will move anteriorly and down towards the groin.
    - A stone that is moving is often more painful than a stone that is static.
    - The pain radiates down to the testis, scrotum, labia or anterior thigh.
    - Whereas the pain of biliary or intestinal colic is intermittent, the pain of renal colic is more constant but there are often periods of relief or just a dull ache before it returns. The pain may change as the stone moves. The patient is often able to point to the place of maximal pain and this has a good correlation with the current site of the stone.
- **Other symptoms which may be present include:**
    - Rigors and fever.
    - Dysuria.
    - Haematuria.
    - Urinary retention.
    - Nausea and vomiting.

**Examination:**

- The patient with colic of any sort writhes around in agony. This is in contrast to the patient with peritoneal irritation who lies still.
- The patient is apyrexial in uncomplicated renal colic (pyrexia suggests infection and the body temperature is usually very high with pyelonephritis).
- Examination of the abdomen can sometimes reveal tenderness over the affected loin. Bowel sounds may be reduced. This is common with any severe pain.
- There may be severe pain in the testis but the testis should not be tender.
- Blood pressure may be low.
- Full and thorough abdominal examination is essential to check for other possible diagnoses - eg, acute appendicitis, ectopic pregnancy, aortic aneurysm.

**Differential diagnosis:**

This depends upon the position of the pain and the presence or absence of pyrexia and includes:

- Biliary colic.
- Dissection of an aortic aneurysm: beware the patient who presents with features of renal colic for the first time over the age of 60 years. This may be dissection of aortic aneurysm leading to ruptured aortic aneurysm.
- Pyelonephritis: very high temperature. Pain is unlikely to radiate to the groin.
- Acute pancreatitis.
- Acute appendicitis.
- Perforated peptic ulcer.
- Epididymo-orchitis or torsion of the testis: very tender testis.
- Sinister causes of back pain: usually tender over vertebrae.
- Drug misuse: there are reports of people with fictitious stories of renal colic, designed to obtain an injection of pethidine. These patients tend to be abusive when offered anything other than pethidine.
- Münchhausen's syndrome.

## Management

The British Association of Urological Surgeons (BAUS) published guidelines in 2018 on the management of acute ureteric/renal colic.

**Initial management of renal colic**

**Medication**

- the BAUS recommend an NSAID as the analgesia of choice for renal colic
- whilst diclofenac has been traditionally used the increased risk of cardiovascular events with certain NSAIDs (e.g. diclofenac, ibuprofen) should be considered when prescribing
- the CKS guidelines suggest for patients who require admission: '*Administer a parenteral analgesic (such as intramuscular diclofenac) for rapid relief of severe pain*'
- BAUS no longer endorse the use of alpha-adrenergic blockers to aid ureteric stone passage routinely. They do however acknowledge a recently published meta-analysis advocates the use of α-blockers for one month for patients amenable to conservative management, with greatest benefit amongst those with larger stones between 5mm to 10mm.

**Initial investigations:**

- urine dipstick and culture
- serum creatinine and electrolytes: check renal function
- FBC / CRP: look for associated infection
- calcium/urate: look for underlying causes, clotting if percutaneous intervention planned and blood cultures if pyrexial or other signs of sepsis

**Imaging:**

- BAUS now recommend that non-contrast CT KUB should be performed on all patients, within 14 hours of admission
- if a patient has a fever, a solitary kidney or when the diagnosis is uncertain an immediate CT KUB should be performed. In the case of an uncertain diagnosis, this is to exclude other diagnoses such as ruptured abdominal aortic aneurysm
- CT KUB has a sensitivity of 97% for ureteric stones and a specificity of 95%
- ultrasound still has a role but given the wider availability of CT now and greater accurary it is no longer recommend first-line. The sensitivity of ultrasound for stones is around 45% and specificity is around 90%.

## Management of renal stones

Stones < 5 mm will usually pass spontaneously. Lithotripsy and nephrolithotomy may be for severe cases.

Most renal stones measuring less than 5mm in maximum diameter will typically pass within 4 weeks of symptom onset. More intensive and urgent treatment is indicated in the presence of ureteric obstruction, renal developmental abnormality such as horseshoe kidney and previous renal transplant. Ureteric obstruction due to stones together with infection is a surgical emergency and the system must be decompressed. Options include nephrostomy tube placement, insertion of ureteric catheters and ureteric stent placement.

In the non-emergency setting, the preferred options for treatment of stone disease include extra corporeal shock wave lithotripsy, percutaneous nephrolithotomy, ureteroscopy, open surgery remains an option for selected cases. However, minimally invasive options are the most popular first-line

treatment.

**Shockwave lithotripsy**

- A shock wave is generated external to the patient, internally cavitation bubbles and mechanical stress lead to stone fragmentation. The passage of shock waves can result in the development of solid organ injury. Fragmentation of larger stones may result in the development of ureteric obstruction. The procedure is uncomfortable for patients and analgesia is required during the procedure and afterwards.

**Ureteroscopy**

- A ureteroscope is passed retrograde through the ureter and into the renal pelvis. It is indicated in individuals (e.g. pregnant females) where lithotripsy is contraindicated and in complex stone disease. In most cases a stent is left in situ for 4 weeks after the procedure.

**Percutaneous nephrolithotomy**

- In this procedure, access is gained to the renal collecting system. Once access is achieved, intra corporeal lithotripsy or stone fragmentation is performed and stone fragments removed.

**Therapeutic selection**

| Disease | Option |
|---|---|
| Stone burden of less than 2cm in aggregate | Lithotripsy |
| Stone burden of less than 2cm in pregnant females | Ureteroscopy |
| Complex renal calculi and staghorn calculi | Percutaneous nephrolithotomy |
| Ureteric calculi less than 5mm | Manage expectantly |

**Prevention of renal stones:**
Calcium stones may be due to hypercalciuria, which is found in up to 5-10% of the general population.

- high fluid intake
- low animal protein, low salt diet (a low calcium diet has not been shown to be superior to a normo-calcemic diet)

**Oxalate stones:**

- cholestyramine reduces urinary oxalate secretion
- pyridoxine reduces urinary oxalate secretion

**Uric acid stones:**

- allopurinol, urinary alkalinization e.g. oral bicarbonate

TOPIC: Appendicitis

**Summary**

- Appendicitis describes an acute inflammation of the appendix (a small, narrow tube that is connected to the caecum).
- The aetiology and pathogenesis of acute appendicitis remain largely unknown, but it is thought to be caused by luminal obstruction in about 50% of cases.
    - The main causes of luminal obstruction are thought to be faecolith (hard mass of faecal matter), lymphoid hyperplasia during an infection, impacted stool, foreign body, or rarely appendiceal or caecal tumour.
    - Luminal obstruction leads to distension of the appendix owing to increased mucus production, bacterial overgrowth, and suppurative inflammation. This results in impaired lymphatic and venous drainage from the appendix, with eventual ischaemia and necrosis, and potential perforation.
- Acute appendicitis is the most common abdominal surgical emergency in the world.
    - It can occur at any age, with peak incidence usually in the second or third decade of life, with reduced prevalence at the extremes of age.
- Appendicitis may be uncomplicated (non-perforating) or complicated (perforating).
    - A delay or misdiagnosis of appendicitis can result in additional severe complications from perforation, such as abscess formation, peritonitis, sepsis, intra-abdominal adhesions, and bowel obstruction.
- Although the natural progression of acute appendicitis results in perforation, not all cases follow this course and some will resolve spontaneously.
- A diagnosis of appendicitis should be suspected if there is:
    - Periumbilical or epigastric pain that worsens, and migrates to the right lower quadrant over 24–48 hours. It is typically aggravated by movement.
    - Low-grade fever, general malaise, and anorexia.
    - Nausea, vomiting, and sometimes constipation or diarrhoea.
    - Tenderness in the right lower quadrant on abdominal examination, which may be worse on coughing or hopping in children.
    - Abdominal distension, guarding, rebound tenderness or percussion tenderness, or absent bowel sounds (which may all suggest peritonitis).
    - A palpable abdominal mass (which may suggest an appendix mass or abscess).
    - Note: appendicitis may present atypically, particularly in the very young, elderly, and in pregnancy, and the anatomical position of the appendix may vary resulting in different clinical presentations.
- Examine the person:
    - Facial flushing, dry tongue, halitosis, low-grade fever (less than 38°C), and/or tachycardia.
    - Tenderness in the right lower quadrant on abdominal examination.
        - The site of maximal tenderness is typically over 'McBurney's point', which lies two-thirds of the way along a line drawn from the umbilicus to the anterior superior iliac spine.
        - Children may present with right lower quadrant pain on coughing or hopping.

    - Abdominal distension, guarding (muscular rigidity), and rebound tenderness or percussion tenderness may suggest peritonitis. Absent bowel sounds may indicate ileus or peritonitis associated with perforation.
    - A palpable abdominal mass may suggest an appendix mass or abscess.
    - Consider testing for peritoneal signs specific for acute appendicitis:
        - Rovsing's sign — palpation of the left lower quadrant increases the pain felt in the right lower quadrant.
        - Psoas sign — passive extension of the right thigh with the person in the left lateral position elicits pain in the right lower quadrant.
        - Obturator sign — passive internal rotation of the flexed right thigh elicits pain in the right lower quadrant.
    - Consider performing a groin and scrotal examination, where appropriate, to exclude an alternative cause, such as incarcerated hernia or testicular torsion.
    - Consider performing a pelvic examination, where appropriate, to exclude an alternative gynaecological cause for symptoms.
    - Note: do not routinely perform a rectal examination to confirm the diagnosis in primary care.
- Be aware of possible atypical presentations of acute appendicitis.
    - Children — infants and young children may present with non-specific abdominal pain and anorexia, and may appear withdrawn.
    - Older age — there may be minimal pain or fever; may present with acute confusion or shock.
    - Pregnancy — there may be displacement of the appendix by the gravid uterus. In the later stages of pregnancy, may present with right upper quadrant or right flank pain. Nausea and vomiting may be mistaken for pregnancy-related symptoms.
    - The anatomical position of the appendix can vary considerably in non-pregnant people:
        - A retrocaecal/retrocolic appendix may present with right loin pain and tenderness, and a positive psoas test. Muscular rigidity and tenderness to deep palpation are often absent because of protection from the overlying caecum.
        - A pre-ileal and post-ileal appendix may present with vomiting and diarrhoea (due to irritation of the distal ileum).
        - A subcaecal and pelvic appendix may present with suprapubic pain and urinary frequency; diarrhoea and tenesmus may be present owing to rectal irritation; abdominal tenderness may be lacking, but rectal or vaginal tenderness may be present on the right side; microscopic haematuria and leucocytes may be present on urine dipstick testing.
        - A long appendix with tip inflammation in the left lower quadrant may cause pain in that region.
- Consider the need for additional investigations to exclude an alternative cause, depending on clinical judgement.
    - Full blood count — neutrophil-predominant leucocytosis is present in 80–90% of people with appendicitis.

- C-reactive protein (CRP) — raised levels may be present, but normal levels do not exclude a diagnosis of appendicitis.
- Urine dipstick test — to help exclude a urinary tract infection (UTI).
  - Be aware that as the appendix often lies in close proximity to the urinary tract, 40% of people with acute appendicitis may have leucocytes in their urine.
- Pregnancy test — to exclude ectopic pregnancy and other pregnancy-related conditions.

**Management:**
**person with suspected appendicitis**

- If a person presents with suspected acute appendicitis:
  - Arrange emergency hospital admission for specialist assessment and management. In particular, have a very low threshold for admitting:
    - People with suspected complications.
    - Pregnant women.
    - Elderly people.
    - Infants and young children.
- If there is a low index of suspicion of acute appendicitis at initial presentation and hospital admission is not considered immediately necessary:
  - Consider arranging additional investigations in primary care to exclude alternative conditions, depending on clinical judgement.
  - Advise the person and/or parents of symptoms to be aware of, and when to seek urgent medical review.

**Specialist assessment and management**

Specialist management of suspected acute appendicitis may include:

- Imaging investigations
  - Imaging studies in people with a clinical suspicion of acute appendicitis can reduce the negative appendectomy rate, which has been reported to be as high as 15%. Ultrasonography, abdominal computed tomography (CT) and magnetic resonance imaging (MRI) are most commonly used.
    - Selective imaging may be used when the diagnosis is uncertain given the risks, benefits, costs, and time delay to surgery associated with imaging.
    - Ultrasound should be the preferred imaging modality in children, pregnant, and breastfeeding women.
    - MRI scans are mainly reserved for pregnant women when ultrasound is non-diagnostic.
- Non-operative management
  - After negative imaging, a non-operative management strategy with intravenous fluids and antibiotics can be a safe and effective approach in selected patients with uncomplicated acute appendicitis.
  - The reported rate of recurrence after non-surgical treatment for perforated acute appendicitis and phlegmon (pus-producing swelling) ranges from 12–24%.

    - Appendix mass may be treated surgically or conservatively with intravenous antibiotics and possible percutaneous drainage. Perforated appendicitis may be managed non-operatively with antibiotics, with or without interval appendicectomy.
- Operative management
    - Removal of the appendix is the gold standard treatment for uncomplicated appendicitis.
    - In people with progressive or persistent pain, explorative laparoscopy is recommended to establish/exclude the diagnosis of acute appendicitis or alternative diagnoses.
        - Laparoscopic appendicectomy offers significant advantages over open appendicectomy in terms of less pain, lower incidence of surgical site infection, decreased length of hospital-stay, earlier return to work, overall costs, and better quality of life scores.
        - Possible postoperative complications include small bowel obstruction, superficial wound infection, intra-abdominal abscess, stump leakage, and stump appendicitis.

## TOPIC: Cholecystitis – acute

**Definition**

- Acute cholecystitis is an inflammation of the gallbladder. It usually occurs when a gallstone completely obstructs the gallbladder neck or cystic duct.
- Pathological classifications of acute cholecystitis include:
  - Oedematous cholecystitis — occurs after 2–4 days of obstruction. The gallbladder tissue is intact histologically, with oedema in the subserosal layer.
  - Necrotizing cholecystitis — occurs after 3–5 days of obstruction and is characterized by oedematous changes in the gallbladder, with areas of haemorrhage and necrosis. Necrosis does not involve the full thickness of the gall bladder wall.
  - Suppurative cholecystitis — occurs after 7–10 days of obstruction and is characterized by thickened gallbladder wall with white cell infiltration, intra-wall abscesses, and necrosis. May result in perforation of the gallbladder and a pericholecystic abscess formation.
  - Chronic cholecystitis — occurs after repeated episodes of mild attacks and is characterized by mucosal atrophy and fibrosis of the gallbladder wall. Acute-on-chronic cholecystitis refers to an acute infection that has occurred in chronic cholecystitis.
- Acalculous cholecystitis is inflammation of the gallbladder which occurs in the absence of gallstones. It typically occurs in critically ill people due to a combination of risk factors.

**Causes**

- Gallstones account for 90–95% of cases of acute cholecystitis:
  - Obstruction of the gallbladder neck or cystic duct by a gallstone causes bile to become trapped in the gallbladder, resulting in irritation and increased pressure in the gallbladder.
  - If the obstruction is partial and of short duration, the person experiences biliary colic, characterized by severe pain and tenderness of the right side of the abdomen and/or back). Acute cholecystitis occurs when the obstruction is complete and prolonged (usually several hours).
  - Trauma caused by the gallstone stimulates the release of prostaglandins, which mediates the inflammatory response.
  - As the gallbladder becomes more distended and inflamed, blood flow and lymphatic drainage are compromised, leading to mucosal ischemia. Secondary bacterial infection of the bile may occur.
  - Continued inflammation, ischemia, and/or infection can result in necrosis and gallbladder perforation.
- Acalculous cholecystitis (gallbladder inflammation without gallstones) is seen in about 5–14% of people who present with acute cholecystitis.
  - The exact cause is poorly understood. However, functional cystic duct obstruction is often present and is associated with bile stasis or thickening.
  - It typically occurs in critically ill people due to a combination of risk factor that result in bile stasis (such as trauma, severe systemic illness, and prolonged total parenteral nutrition use) or bile thickening (such as fever and dehydration).
  - Other factors may be involved, such as a predisposition to bacterial colonization of static gallbladder bile.

**Risk factors**

- Gallstones account for 90–95% of cases of acute cholecystitis.
    - Risk factors for gallstones include increasing age, female gender, obesity, and a low fibre diet.
- Acalculous cholecystitis typically occurs in critically ill people due to a combination of factors that result in bile stasis (due to gallbladder hypomotility/dysmotility) or bile thickening (due to dehydration).
    - Conditions associated with biliary stasis or thickening include:
        - Sepsis.
        - Extensive trauma.
        - Burns.
        - Major surgery.
        - Prolonged fasting or starvation.
        - Prolonged total parenteral nutrition use.
    - Other risk factors for acalculous cholecystitis include:
        - Diabetes, end-stage renal disease, congestive heart failure/coronary artery disease, and peripheral vascular disease.
        - Drug treatments, for example cyclosporin, can decrease bile acid secretion, leading to sludge formation. Ceftriaxone is secreted into bile and can precipitate with calcium, forming biliary sludge.
        - Infection, such as Epstein-Barr virus (EBV) infection. In people who are HIV positive, cholangiopathy due to infection with microsporidia species can occur.

**Complications**

- Without treatment, acute cholecystitis may resolve spontaneously within 1–7 days. However, 25–30% of people will require surgery or develop complications, such as:
    - Necrosis of the gallbladder wall (gangrenous cholecystitis).
    - Perforation of the gallbladder — occurs in about 10% of people with acute cholecystitis and has a mortality rate of up to 30%.
    - Biliary peritonitis — occurs with the entry into the peritoneal cavity of bile leakage due to various causes, including cholecystitis-induced gallbladder perforation.
    - Pericholecystic abscess — perforation of the gallbladder wall is covered by the surrounding tissues along with the formation of abscesses around the gallbladder.
    - Fistula (between the gallbladder and duodenum) — usually caused by a large gallbladder stone eroding through the wall of the gallbladder into the duodenum. If the stone is large in size, the person can develop gallstone ileus with the stone causing mechanical small bowel obstruction at the ileocecal valve.
    - Jaundice — occurs in up to 10% of people with cholecystitis and may be due to biliary tract inflammation and oedema, pressure from a distended gallbladder, or a stone in the common bile duct or impacted in the gallbladder neck (Mirizzi's syndrome).
    - Sepsis — could develop if there is superimposed bacterial infection.

- The mortality rate of acalculous cholecystitis ranges from 10–50% (which is much higher than the expected 4% mortality observed in people with acute cholecystitis). In people who are critically ill with acalculous cholecystitis and perforation or gangrene, mortality can be as high as 50–60%.

**Diagnosis**

- Suspect acute cholecystitis if a person presents with:
  - Sudden-onset, constant, severe pain in the upper right quadrant, lasting several hours.
  - Tenderness, with or without guarding, in the right upper quadrant.
- If acute cholecystitis is suspected, assess the person to:
  - Identify other symptoms and signs of acute cholecystitis.
  - Identify risk factors for acute cholecystitis and acute acalculous cholecystitis.
  - Exclude differential diagnoses, such as gallstones and peptic ulcer disease.
  - Identify complications, such as an abscess, perforated gallbladder, sepsis, and jaundice.
- Take a detailed history.
  - Ask about the nature of the pain.
    - A constant pain present for several hours is consistent with acute cholecystitis. The duration of pain can be shorter if the gallstone returns into the gallbladder lumen or passes into the duodenum.
    - The pain is severe, steady, and may radiate to the back.
    - Referred pain from the gallbladder may be felt in the right shoulder or interscapular region.
  - Ask about the presence of other symptoms, such as:
    - Fever/chills — may be present as a symptom of infection.
    - Nausea — can occur in conjunction with severe pain and can be a prominent symptom of a stone in the common bile duct.
    - Vomiting — sometimes associated with biliary disease but not specific for cholecystitis. It can be a prominent symptom of a stone in the common bile duct.
    - Anorexia — associated with biliary disease but not specific for cholecystitis.
  - Ask about current or previous episodes of gallstones or biliary colic.
    - Over 90% of people with acute cholecystitis have gallstones. See the CKS topic on Gallstones for more information.
    - About 50% of the people who have had one episode of biliary pain will have another within 12 months.
  - Ask about the presence of risk factors for acalculous cholecystitis, such as recent severe illness or trauma.
- Perform a physical examination.
  - Look for signs of sepsis, such as extreme pain, mottled skin, fever or reduced core temperature, rigors or chills, and/or shortness of breath.

        - Suspect sepsis based on acute deterioration in a person in whom there is clinical evidence or strong suspicion of infection.
        - The person may present with non-specific or non-localized symptoms (such as being acutely unwell with a normal temperature) or there may be severe signs with evidence of multi-organ dysfunction and shock.
    - Examine the person's abdomen. Palpate for:
        - Right upper quadrant tenderness.
        - Right upper quadrant mass — a distended, tender gallbladder may be palpable as a distinct mass in 30–40% of people.
        - Presence of Murphy's sign (inspiration is inhibited by pain on palpitation when the examiner's hand is positioned along the costal margin). Be aware that Murphy's sign is unreliable in older adults and is difficult to elicit in people who are critically ill.
    - Look for other signs of complication, such as:
        - Jaundice — occurs in up to 10% of people with cholecystitis and may be due to biliary tract inflammation and oedema, pressure from a distended gallbladder, or a stone in the common bile duct or impacted in the gallbladder neck (Mirizzi's syndrome).
        - Abscess formation and/or gallbladder perforation — the person may present with persistent fever, a palpable mass, severe localized and/or generalized pain/tenderness.
    - Acalculous cholecystitis can present with fever, jaundice, and abdominal tenderness and is usually a diagnosis of exclusion. It often occurs in critically ill people.
- Differential diagnoses of acute cholecystitis include:
    - Acute cholangitis — presents with jaundice, fever, chills, and abdominal pain.
    - Acute coronary syndrome
    - Acute pancreatitis
    - Appendicitis
    - Gastrointestinal cancer
    - Gastro-oesophageal reflux disease
    - Hepatitis
    - Peptic ulcer disease
    - Right lower lobe pneumonia
    - Symptomatic gallstones

**How should I manage a person with suspected acute cholecystitis?**

- If acute cholecystitis is suspected, admit the person to hospital for:
    - Confirmation of the diagnosis, including abdominal ultrasound and blood tests (such as a white blood cell count, C-reactive protein, and serum amylase).
    - Monitoring (for example blood pressure, pulse, and urinary output).
    - Treatment (may include intravenous fluids, antibiotics, and analgesia).
    - Surgical assessment for cholecystectomy.

## TOPIC: Breast cancer

**Definition**

- Familial breast cancer occurs in women with more family members with breast or related cancers than would have been expected by chance alone. It may be that members of such a family carry gene that cause or, more likely, contribute to the development of breast cancer.
- A family history of breast cancer is a strong risk factor for developing the disease:
    - The risk increases with the number of relatives affected and the age at diagnosis of the relative (the younger the age of diagnosis the greater the risk), and is modified by other breast cancer risk factors, including age at menopause, parity, oral contraception, hormone replacement therapy, and breastfeeding.
    - A person's risk can be calculated based on their family history by the use of specialist computer programmes in secondary care, or in specialist genetic clinics.
- It is important to note that:
    - Most women will never develop breast cancer and most of those who do will not have familial breast cancer. Given the high prevalence of breast cancer, many people have a relative with the disease, but this may be due to chance rather than to genetic or shared lifestyle factors.
    - Most women with a family history of breast cancer are not at a substantially increased risk of breast cancer.

**Genetic risk factors**

- A hereditary cancer syndrome is a genetic predisposition to certain types of cancer.
    - It is caused by a germline mutation, inherited from either parent, resulting in a significantly elevated risk of cancer compared with that of the general population that does not have a mutation in a cancer susceptibility gene.
    - Onset at an early age, a high incidence of bilateral disease, and an association with other malignancies (such as ovarian cancer) usually characterize hereditary breast cancer.
- Most inherited cases (more than 90%) of breast (and ovarian) cancer are associated with mutations in two genes: BRCA1 and BRCA2.
- Other gene mutations identified include:
    - TP53 mutation on chromosome 17 — most women with this mutation develop breast cancer by 50 years of age. TP53 is also associated with sarcomas of childhood, leukaemias, adrenocortical carcinomas, and brain tumours (Li–Fraumeni syndrome).
    - PTEN gene (Cowden's syndrome) — this predisposes to breast, thyroid, and uterine cancers, and hamartomatous lesions of the skin.
    - ATM, CHEK2, BRIP1, and PALB2 genes — these confer moderate risk of breast cancer.
- It is important to emphazise that.
    - Some people may carry the abnormal gene without developing cancer themselves.
    - Most women who have a family history of breast cancer do not have a specific identifiable gene mutation. Some of these family histories will be due to lower-penetrance genes not yet discovered.

**Management**
**Take a family history of breast cancer when:**

    - A person has concerns about their family history of breast cancer.
    - A person has breast symptoms.
    - It is clinically relevant, for example:
        - In women over 35 years of age using an oral contraceptive pill.
        - In women being considered for long-term hormone replacement therapy (HRT).
- Document a family tree including the person and their first-degree relatives (mother, father, daughter, son, sister, and brother) and second-degree relatives (grandparents, grandchildren, aunt, uncle, niece, nephew, half-sister, and half-brother).
    - Ask about:
        - All cancer diagnoses (including breast cancer).
        - People in whom a faulty gene (such as BRCA1, BRCA2, or TP53) has been identified.
        - Age at diagnosis.
        - Presence of bilateral disease.
        - Male breast cancer.
        - Multiple cases in the family (particularly on one side).
        - Jewish ancestry.
        - Other related early onset tumours, such as ovarian, pancreatic, and prostate cancer, sarcoma, glioma, and adrenal carcinoma.
- Seek specialist advice from secondary care or a specialist genetics service if:
    - Any of the following are present in the family history in addition to breast cancers in relatives not fulfilling the above criteria:
        - Bilateral breast cancer.
        - Male breast cancer.
        - Ovarian cancer.
        - Jewish ancestry.
        - Sarcoma in a relative younger than 45 years of age.
        - Glioma or childhood adrenal cortical carcinomas.
        - Complicated patterns of multiple cancers at a young age.
        - Two or more relatives with breast cancer on the father's side of the family.
    - There is uncertainty about whether or not to refer.
    - The person is not sufficiently reassured by the information provided.
- Refer directly to a specialist genetics service if a high-risk predisposing gene mutation (such as BRCA1, BRCA2, or TP53) has been identified.

**Hormonal contraception can be considered?**

- For women with a family history of breast cancer, the following may, therefore, be used without restriction (UKMEC category 1):
    - The copper-bearing intrauterine device (Cu-IUD).
    - The levonorgestrel-releasing intrauterine system (LNG-IUS, for example, Mirena®).

    - Etonogestrel-only implant (Nexplanon®).
    - Depot medroxyprogesterone acetate (Depo-Provera®, SAYANA PRESS®).
    - Progestogen-only pill (POP).
    - Combined hormonal contraceptives (CHC).
        - Inform women aged over 35 years with a family history of breast cancer that there is an increased risk of breast cancer associated with taking the oral contraceptive pill, given that their absolute risk increases with age.
        - Advice to women up to age 35 years with a family history of breast cancer should be in keeping with general health advice on the use of the oral contraceptive pill.
- For women who are known carriers of a gene mutation associated with breast cancer (such as BRCA1 or BRCA2):
    - The Cu-IUD may be used without restriction (UKMEC category 1).
    - The following methods may be generally used (UKMEC category 2):
        - POP.
        - Depot medroxyprogesterone acetate (Depot-Provera®, SAYANA PRESS®).
        - Etonogestrel-only implant (Nexplanon®).
        - LNG-IUS.
    - If the CHC is being considered (UKMEC category 3), discuss with (or refer the woman to) a specialist genetics service, as views are conflicting on whether or not the protective effects of CHC against ovarian cancer outweigh the increased risk of breast cancer.

**Manage a woman with a family history of breast cancer who wishes to take hormone replacement therapy?**

- If the woman is at low risk of breast cancer (that is, she does not fulfil the referral criteria to secondary care), hormone replacement therapy (HRT) may be prescribed.
- If the woman is at increased risk of breast cancer (that is, she fulfils the referral criteria to secondary care):
    - Ensure that she has been referred to secondary care for assessment of her breast cancer risk.
    - Inform her of the increase in breast cancer risk with type and duration of HRT, and refer her to secondary care for advice as to whether or not it is appropriate to prescribe HRT.
    - Provide information on non-hormonal and non-pharmacological treatments, such as antidepressants (selective serotonin reuptake inhibitor), vaginal moisturisers and lubricants, cognitive behavioural therapy (CBT), hypnosis, acupuncture, and relaxation techniques, for the management of symptoms.

**Presentation**

- Breast lump is the most common presenting symptom.
- A malignant breast lump is usually painless, but pain can occur.
- Nipple symptoms, including change in shape or nipple bleeding, are recognized symptoms as are skin changes, such as tethering or peau d'orange.

- A diagnosis of breast cancer is generally made using mammography and core biopsy. This is performed in secondary care.

**Prognosis**

- The 5 years survival is 85%, although this figure includes cases detected by screening as well as those identified after symptoms have occurred.

Globally, breast cancer is the most frequently diagnosed malignancy, accounting for over two million cases each year.
It is the leading cause of cancer death in women worldwide

**Factors associated with greater breast cancer risk:**

- Increasing age
- Female sex
- Endogenous estrogen 'contraceptive pill' and hormone replacement therapy
- Early menarche or late menopause
- Nulliparity
- Increasing age at first full-term pregnancy; 1st pregnancy > 30 yrs (twice risk of women having 1st child < 25 yrs)
- Previous history of breast cancer
- Family history of breast cancer
- Genetic factors; BRCA1 60-80%, BRCA2 40-60% - associated with risk of ovarian cancer
- P53 gene mutation
- Exposure to therapeutic ionizing radiation
- Alcohol and smoking
- Obesity
- Tall stature
- Benign proliferative breast lesion

*Breast augmentation is NOT generally associated with increased risk of breast cancer

**Types and classifications:**

Most breast cancers arise from duct tissue followed by lobular tissue, described as **ductal** or **lobular carcinoma** respectively. These can be further subdivided as to whether the cancer hasn't spread beyond the local tissue (described as **carcinoma-in-situ**) or has spread (described as **invasive**). Therefore, common breast cancer types include:

- **Invasive ductal carcinoma**. This is the most common type of breast cancer. To complicate matters further this has recently been renamed 'No Special Type (NST)'. In contrast, lobular carcinoma and other rarer types of breast cancer are classified as 'Special Type'
- **Invasive lobular carcinoma**
- **Ductal carcinoma-in-situ (DCIS)**
- **Lobular carcinoma-in-situ (LCIS)**

**Rarer types of breast cancer are shown in the following list. These are classed as 'Special Type' but as noted previously remember that a relatively common type of breast cancer (lobular) is also Special Type:**

- Medullary breast cancer
- Mucinous (mucoid or colloid) breast cancer
- Tubular breast cancer

- Adenoid cystic carcinoma of the breast
- Metaplastic breast cancer
- Lymphoma of the breast
- Basal type breast cancer
- Phyllodes or cystosarcoma phyllodes
- Papillary breast cancer

Other types of breast cancer include the following (although please note they may be associated with the underlying lesions seen above, rather than completely separate subtypes):

**Paget's disease of the nipple** is an eczematoid change of the nipple associated with an underlying breast malignancy and it is present in 1-2% of patients with breast cancer. In half of these patients, it is associated with an underlying mass lesion and 90% of such patients will have an invasive carcinoma. 30% of patients without a mass lesion will still be found to have an underlying carcinoma. The remainder will have carcinoma in situ.
**Inflammatory breast cancer** where cancerous cells block the lymph drainage resulting in an inflamed appearance of the breast.

**Clinical Features**
**Breast mass** - The "classic" characteristics of a cancerous lesion include a hard, immovable, single dominant lesion with irregular borders
**Locally advanced disease** — signs include axillary adenopathy (suggesting locoregional disease) or skin findings such as erythema, thickening, or dimpling of the overlying skin (peau d'orange), suggesting inflammatory breast cancer
**Metastatic disease** — Symptoms of metastatic breast cancer depend on the organs involved, with the most common sites of involvement being the bone (e.g., back or leg pain), liver (abdominal pain, nausea, jaundice), and lungs (e.g., shortness of breath or cough)

**Clinical Evaluation:**
Through history and physical examination
**Diagnostic Evaluation:**
Investigations; estrogen and progesterone, epidermal growth factors 'HER2', routine blood tests
Imaging; CXR, mammography and ultrasound, MRI
**Biopsy**; Fine-needle aspiration, core-needle biopsy (image-guided), open biopsy, excision biopsy
Surgical biopsy should not be utilized as a diagnostic tool unless percutaneous palpation-guided or image-guided biopsy is not feasible
**Diagnosis:**
The diagnosis of breast cancer is defined by the **presence of malignant epithelial cells (carcinoma)**
Patients with a suspicious mammographic abnormality or a clinically suspicious palpable breast mass typically undergo biopsy regardless of additional imaging findings.
Genetic testing for BRCA1 and BRCA2 mutations should be offered to women aged under 50 years with triple-negative breast cancer (tests negative for oestrogen receptors, progesterone receptors, and excess HER2 protein), including those with no family history of breast or ovarian cancer.
**Management:**
Breast cancer is treated with a multidisciplinary approach involving surgical oncology, radiation oncology, and medical oncology, which has been associated with a reduction in breast cancer mortality
**The tumour, node, metastasis (TNM) staging system** – The TNM staging system for breast cancer is an internationally accepted system used to determine the disease stage.
**Patient stratification** – Patients with a new diagnosis of breast cancer can be stratified as having early or locally advanced disease

**Non-metastatic breast cancer is broadly considered in two categories:**

- **Early stage** – Patients with stage I, IIA, or a subset of stage IIB disease (T2N1)
- **Locally advanced** –Patients with stage IIB disease (T3N0) and patients with stage IIIA to IIIC disease

**The management of breast cancer depends on the staging, tumour type and patient background. It may involve any of the following:**

- Surgery
- Radiotherapy
- Hormone therapy
- Biological therapy
- Chemotherapy

**Surgery:**

The vast majority of patients who have breast cancer diagnosed will be offered surgery
An exception may be a very frail, elderly lady with metastatic disease who may be better managed with hormonal therapy

**Prior to surgery, the presence/absence of axillary lymphadenopathy determines management:**

- Women with no palpable axillary lymphadenopathy at presentation should have a pre-operative axillary ultrasound before their primary surgery
    - if positive then they should have a sentinel node biopsy to assess the nodal burden
- In patients with breast cancer who present with clinically palpable lymphadenopathy, axillary node clearance is indicated at primary surgery
    - this may lead to arm lymphedema and functional arm impairment

Depending on the characteristics of the tumour women either has a wide-local excision or a mastectomy.

Around two-thirds of tumours can be removed with a wide-local excision

| Mastectomy | Wide Local Excision |
|---|---|
| Multifocal tumour | Solitary lesion |
| Central tumour | Peripheral tumour |
| Large lesion in small breast | Small lesion in large breast |
| Ductal carcinoma in situ “DCIS” > 4cm | DCIS < 4cm |
| Diffuse malignant-appearing calcifications on imaging | |
| Prior history of chest radiotherapy | |
| Pregnancy<br>Persistently positive margins despite attempts at re-excision | |

Women should be offered breast reconstruction to achieve a cosmetically suitable result regardless of the type of operation they have. For women who've had a mastectomy this may be done at the initial operation or at a later date

**Radiotherapy:**
Whole breast radiotherapy is recommended after a woman has had a wide-local excision as this may reduce the risk of recurrence by around two-thirds
Post-mastectomy radiation therapy (RT) is indicated for patients at high risk for local recurrence, such as those with cancer involving the deep margins and pathologically involved axillary lymph nodes.
**Hormonal therapy:**
Adjuvant hormonal therapy is offered if tumours are positive for hormone receptors. For many years this was done using **Tamoxifen** for 5 years after diagnosis. Tamoxifen is still used in pre- and peri-menopausal women.
In post-menopausal women, **aromatase inhibitors** such as **Anastrozole** are used for this purpose. This is important as aromatisation accounts for the majority of oestrogen production in post-menopausal women and therefore anastrozole is used for ER +ve breast cancer in this group.
Important side-effects of tamoxifen include an increased risk of endometrial cancer, venous thromboembolism and menopausal symptoms.
**Biological therapy:**
The most common type of biological therapy used for breast cancer is Trastuzumab (Herceptin). It is only useful in the 20-25% of tumours that are HER2 positive.
Trastuzumab cannot be used in patients with a history of heart disorders
**Chemotherapy**
Cytotoxic therapy may be used either prior to surgery ('neoadjuvanant' chemotherapy) to downstage a primary lesion or after surgery depending on the stage of the tumour, for example, if there is axillary node disease - FEC-D is used in this situation.
Screening in Kuwait is offered to women from the age of 40 years by mammogram every year.

## Ultrasound exam

A procedure in which high-energy sound waves are bounced off internal tissues or organs and make echoes. The echoes form a picture of body tissues called a sonogram.

An ultrasound is sometimes used as an additional screening test for women who are at increased risk of developing breast cancer (such as women with dense breasts). It is not known if supplemental screening with ultrasound leads to better health outcomes.

MRI may be used as a screening test for women who have a high risk of breast cancer. Factors that put women at high risk include:

- Certain gene changes, such as changes in the *BRCA1* or *BRCA2* gene.
- A family history of breast cancer in a first-degree relative, such as a mother, daughter, or sister.
- Certain genetic syndromes, such as Li-Fraumeni syndrome or Cowden syndrome.

An MRI is more likely than mammography to find a breast mass (lump) that is not cancer.

Women with dense breasts who have supplemental screening (for example, an MRI) show higher rates of breast cancer detection, but there is limited evidence about whether this leads to better health outcomes.

## ACS recommendations for breast MRI screening as an adjunct to mammography

| **Recommend annual MRI screening (based on high risk of breast cancer and high sensitivity of MRI*)** |
|---|
| *BRCA* mutation |
| First-degree relative of *BRCA* carrier, but untested |
| Lifetime risk >20 to 25% or greater, as defined by BRCAPRO or other models that are largely dependent on family history |
| **Recommend annual MRI screening (based on high risk of breast cancer)** |
| Radiation to chest between age 10 and 30 years |
| Li-Fraumeni syndrome and first-degree relatives |
| Cowden and Bannayan-Riley-Ruvalcaba syndromes and first-degree relatives |
| **Insufficient evidence to recommend for or against MRI screening[Δ]** |
| Lifetime risk 15 to 20%, as defined by BRCAPRO or other models that are largely dependent on family history |
| Lobular carcinoma in situ or atypical lobular hyperplasia |
| Atypical ductal hyperplasia |
| Heterogeneously or extremely dense breast on mammography |
| Women with a personal history of breast cancer, including ductal carcinoma in situ |
| **Recommend against MRI screening (based on expert consensus opinion)** |
| Women at <15% lifetime risk |

ACS: American Cancer Society; MRI: magnetic resonance imaging.
* Evidence from nonrandomized screening trials and observational studies.
Δ Payment should not be a barrier. Screening decisions should be made on a case-by-case basis, as there may be particular factors to support MRI.

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TOPIC: Anti-estrogen drugs

**Selective estrogen Receptor Modulators (SERM):**
Tamoxifen is a SERM which acts as an estrogen receptor antagonist and partial agonist. It is used in the management of estrogen receptor-positive breast cancer.

**Adverse effects:**

- menstrual disturbance: vaginal bleeding, amenorrhea
- hot flushes - 3% of patients stop taking tamoxifen due to climacteric side-effects
- venous thromboembolism
- endometrial cancer

**Aromatase inhibitors:**
Anastrozole and letrozole are aromatase inhibitors that reduces peripheral estrogen synthesis. This is important as aromatization accounts for the majority of estrogen production in postmenopausal women and therefore anastrozole is used for ER +ve breast cancer in this group.

**Adverse effects:**

- osteoporosis
  - NICE recommends a DEXA scan when initiating a patient on aromatase inhibitors for breast cancer
- hot flushes
- arthralgia, myalgia
- insomnia

## TOPIC: CYCLICAL BREAST PAIN

### DEFINITION

- Cyclical breast pain is related to the menstrual cycle, with symptoms usually starting within 2 weeks of, and improving at the onset of, the menstrual period.

### CAUSES

- The exact cause of cyclical breast pain is not fully understood — as pain is related to the menstrual cycle it is thought that hormonal changes affecting the breast tissue are involved.

- Other contributing factors (such as caffeine intake, iodine deficiency, dietary fat intake, and smoking) have been suggested but have not been found to definitively cause symptoms.

### PROGNOSIS

- In up to 20% of cases spontaneous resolution occurs within 3 months or with a hormonal event such as pregnancy or menopause.
- Symptoms may recur in up to 60% of women within 2 years.
- In the absence of accompanying suspicious clinical features (such as a lump or nipple discharge) the likelihood of an associated breast cancer is very low.

### DIAGNOSIS

- **Take a history asking about:**
  - Onset, severity, location, radiation, and duration of pain.
  - Relationship of pain to menstrual cycle, physical activity or recent trauma.
  - Associated features such as skin or nipples changes or a lump.
  - Risk factors for breast cancer including past medical history and family history.
  - The possibility of pregnancy.
  - Smoking, alcohol, caffeine and stress.
- **Carry out an examination to identify signs suggestive of other conditions such as:**
  - Breast cancer
    - for example, an unexplained breast lump or thickening; suspicious skin changes; discharge, retraction, or other concerning nipple changes; axillary lump.
  - Infection
    - for example, localized breast swelling, redness, warmth, and pain; associated systemic symptoms such as fever.
- **Consider a diagnosis of cyclical breast pain when a woman with a normal breast exam describes pain that:**
  - Usually starts during the luteal phase of the cycle (within 2 weeks before menses), increases until menstruation begins, and improves after menses.
  - Is dull, heavy, or aching in nature.
  - Is usually bilateral.
  - May be poorly localized and extend to the axilla.
- **Consider the use of a breast pain diary to aid diagnosis.**
  - Advise the woman to complete this daily for at least 2 cycles to assess the severity and timing of breast pain.
- **Exclude pregnancy.**
- **If the woman is experiencing additional physical and/or psychological premenstrual symptoms, consider a diagnosis of premenstrual syndrome.**

## MANAGENENT

- **Reassure the woman that there is no serious underlying pathology.**
- **Consider the following treatment options:**
  - Wearing a well-fitted bra during the day, more supportive bra during exercise, and a soft support bra at night.
  - Oral paracetamol and/or ibuprofen, or a topical nonsteroidal anti-inflammatory preparation, as required.
- **Ask the woman to keep a pain diary (if she has not already done so)** to evaluate the severity and timing of the pain, and its response to treatment.
- **Consider referring to a breast specialist if the pain is severe enough to affect quality of life or sleep and does not respond to first-line treatment after 3 months.**
  - Specialist treatment options include danazol and tamoxifen.
- **The following treatments for cyclical breast pain are not recommended as evidence is limited:**
  - Evening primrose oil.
  - Flaxseed.
  - Diets low in fat or low in caffeine.
  - Vitamin B6.
  - Vitamin E.
  - Stopping or changing contraceptives.

## TOPIC: MASTITIS AND BREAST ABSCESS

### MASTITIS

- Mastitis is a painful inflammatory condition of the breast. It usually occurs in lactating women ('lactational' or 'puerperal mastitis') but can also occur in non-lactating women ('non-lactational mastitis').

- **Mastitis can be classified as:**
  - **Non-infectious** — breast inflammation due to a non-infectious and/or idiopathic cause.
  - **Infectious** — infection of breast tissue which usually occurs by retrograde spread through a lactiferous duct or a traumatized nipple. Very rarely, the infection occurs through the lymphatics or by haematogenous spread.

### BREAST ABSCESS

- A breast abscess is a localized collection of pus within the breast.

  - A **lactational abscess** is usually located in the peripheral region of the breast, more commonly in the upper and outer quadrant.
  - A **non-lactational abscess** tends to be located in the central/subareolar or lower quadrants of the breast.

### CAUSES

- **In lactating women,** milk stasis is usually the primary cause of mastitis.
  - The accumulated milk causes an inflammatory response which may or may not progress to infection.
  - The most common organism associated with infectious mastitis in breastfeeding women is Staphylococcus aureus, including strains of meticillin-resistant S. aureus (MRSA) if the infection was hospital-acquired.
- **In non-lactating women,** mastitis is usually accompanied by infection, which can be categorized as either central/subareolar or peripheral.
  - Central/subareolar infection is usually secondary to periductal mastitis (a condition where the subareolar ducts are damaged and become infected). Less commonly, it may be caused by duct ectasia, a harmless, age-related breast change.
  - Peripheral non-lactating infection (less common) has been associated with diabetes mellitus, rheumatoid arthritis, trauma, corticosteroid treatment, and granulomatous mastitis, but often there is no underlying cause.
  - Granulomatous mastitis is a rare inflammatory disease of the breast which is thought to be an autoimmune reaction to substances secreted from the mammary ducts. It can be idiopathic or occur in people with certain risk factors, although it is not exactly clear how it develops.
  - The most common organisms associated with infectious mastitis in non-lactating women are S. aureus, enterococci, and anaerobic bacteria (such as Bacteroides spp. and anaerobic streptococci).
- A **breast abscess** is a severe complication of mastitis, although it may occur without apparent preceding mastitis.

### PREDISPOSING FACTORS

### MASTITIS – IN LACTATING WOMEN

- **Milk stasis is the primary cause of mastitis in lactating women.**

- **Predisposing factors include:**
  - Poor infant attachment to the breast, for example due to a short frenulum (tongue-tie) in the infant — this may also lead to nipple damage which can cause pain and provide an entry point for bacteria.
  - Reduced number or duration of feeds, for example due to:
    - Partial bottle feeding, changes in feeding regime (common when the infant first starts to sleep through the night), and rapid weaning from breast milk.
    - Painful breasts.
    - Use of a dummy or bottle — this may also result in poor infant attachment to the breast.
    - Having a preferred breast for feeding, leading to milk accumulation in the other breast.
    - Maternal stress and fatigue.
  - Pressure on the breast, for example from tight clothing or bra, baby sling straps, or prone sleeping position.
- **Other predisposing factors include:**
  - Age
  - Non-lactational predisposing factors, such as smoking, nipple damage, trauma to the breast, an underlying breast abnormality, and immunosuppression.
- **Recurrence of lactational mastitis is common. Predisposing factors include:**
  - Staphylococcus aureus carriage — asymptomatic carriage of S. aureus on a person's skin or mucous membranes (colonization) is implicated in recurrent infections.
  - Previous mastitis
    - this association is thought to be due to inadequate treatment of previous mastitis (which can lead to a recurrence of more severe mastitis or a breast abscess) and/or inadequate management of predisposing factors.

## MASTITIS -IN NON-LACTATING WOMEN

- **Smoking**
  - is a major predisposing factor for periductal mastitis.
  - Nicotine and other toxins found in cigarette smoke accumulate in the breasts causing damage to the ducts (directly or by a localized hypoxic effect).
- **Other predisposing factors**:
  - Nipple damage
    - for example, from nipple piercing or skin conditions such as eczema, infection, or Raynaud's disease — this can cause pain and provide an entry point for bacteria.
  - Trauma to the breast
    - may damage duct and gland tissue.
  - Underlying breast abnormality
    - such as ductal abnormality, cyst, or tumour — can cause persistent poor drainage of part of the breast, leading to breast infection.
  - Immunosuppression
    - women with diabetes mellitus or HIV infection, and those on immunosuppressive therapy, are at increased risk for developing breast infections.
  - Shaving or plucking areolar hair
    - pulling hair from the areola may result in the formation of a boil or an abscess, with potential for more widespread infection.
  - Foreign body

- materials such as silicone and paraffin, which are used for both breast augmentation and reconstruction, may cause a foreign body-type reaction in the breast.

- **Predisposing factors for granulomatous mastitis**
  - include bacterial infections (Corynebacterium spp.), childbirth, oral contraceptive use, trauma, autoimmune disease, hyperprolactinaemia, tuberculosis, sarcoidosis, and diabetes mellitus.
- **Recurrence of non-lactational mastitis is common. Predisposing factors include:**
  - Staphylococcus aureus carriage — asymptomatic carriage of S. aureus on a person's skin or mucous membranes (colonization) is implicated in recurrent infections.
  - Previous mastitis — this association is thought to be due to inadequate treatment of previous mastitis (which can lead to a recurrence of more severe mastitis or a breast abscess) and/or inadequate management of predisposing factors.

## BREAST ABSCESS

**Predisposing factors:**

- Previous mastitis — this association is thought to be due to:
  - Delayed, inadequate, or inappropriate treatment of previous mastitis
    - can lead to complications, including a breast abscess.
  - Sudden cessation of breastfeeding in women with lactational mastitis
    - without effective milk removal, infectious mastitis is likely to progress to an abscess.
- Immunosuppression
  - women with diabetes mellitus or HIV infection, and those on immunosuppressive therapy, are at risk for developing recurrent breast infections.
- Staphylococcus aureus carriage
  - asymptomatic carriage of S. aureus on a person's skin or mucous membranes (colonization) is implicated in recurrent infections.
- Socio-economic status
  - there is some evidence of a higher incidence of breast abscess (especially lactational) in low-income women compared with higher income women.
- Poor hygiene
  - better maternal and infant hygiene reduces the risk of abscess formation.
- Breast abscess may also be more common in women who are primiparous, aged over 30 years, and following post-term delivery.

## PROGNOSIS

**When treated promptly and appropriately, recovery from mastitis or a breast abscess is usually rapid and complete. However, if treatment is delayed, inadequate, or inappropriate, the woman may develop serious complications.**

- Granulomatous mastitis is often treated with antibiotics, corticosteroids, or surgery.

**Recurrence of mastitis or breast abscess is common:**

- The World Health Organization (WHO) found that more than half of women with lactational mastitis had experienced at least one previous episode.
- Abscesses associated with periductal mastitis have a high recurrence rate (up to 28%). This is thought to be because treatment (aspiration or incision) does not usually remove the underlying diseased duct, and many affected women continue to smoke — smoking is the main predisposing factor for this condition.

- Granulomatous mastitis is often recurrent (in up to 78% of cases), even following surgical excision (in around 10% of cases).

## COMPLICATIONS

- Breast abscess
- Mammary duct fistula
  - recurrent episodes of periductal mastitis and infection can result in mammary duct fistula.
- Sepsis
  - breast infections may be associated with bacteraemia, especially in immunocompromised people.
- Scarring
  - mastitis and breast abscesses are unlikely to cause significant breast scarring. However:
    - Surgical intervention other than needle aspiration may cause a post-operative scar.
    - Recurrent mastitis may lead to chronic inflammation and disfigurement of the breast.
    - Pain — up to 30% of women report long-term pain following granulomatous lobular mastitis.
- Additional infections
  - mastitis may be the initiating factor for necrotizing fasciitis, especially in children. In addition, people with Staphylococcus aureus mastitis are at increased risk for subsequent skin infections at extra-mammary sites.
- Death
  - mastitis and breast abscesses can occasionally be fatal if inadequately treated, especially in women who are immunocompromised.

Additional complications in lactating women include:

- Emotional distress due to unplanned early cessation of breastfeeding.
- Inability to breastfeed in the future
  - future lactation may be compromised in up to 10% of women who have had a breast abscess.

## DIAGNOSIS

### MASTITIS

- **Suspect mastitis in a woman who presents with:**
  - A painful breast.
  - Fever and/or general malaise.
  - A tender, red, swollen, and hard area of the breast, usually in a wedge-shaped distribution.
- **Be aware that the symptoms and signs of non-lactational mastitis may mimic breast cancer or a breast abscess:**
  - Periductal mastitis
    - may present with periareolar inflammation (with or without an associated mass), an established abscess, nipple retraction at the site of the diseased duct, central breast pain, and/or greenish discharge from the nipple.
  - Granulomatous mastitis
    - may present with a firm, unilateral breast mass, breast distortion, nipple retraction, skin thickening, axillar adenopathy, ulceration, or a large area (or areas) of infection with multiple simultaneous peripheral abscesses.

        - Note: the clinical features of granulomatous mastitis overlap with those of breast cancer and a biopsy is required to confirm the diagnosis.
- **It is not possible to reliably distinguish between non-infectious and infectious mastitis based on clinical features.** However, features that may raise suspicion of infection include:
    - A nipple fissure that looks infected.
    - Purulent discharge.
    - Influenza-like symptoms and pyrexia lasting for longer than 24 hours.
    - Considerable breast discomfort.
- **For lactating women, features of infection may include:**
    - Symptoms not improving (or worsening) after 12–24 hours despite effective milk removal.
    - Positive breast milk culture.

## BREAST ABSCESS

- **Suspect a breast abscess if the woman has:**
    - A history of recent mastitis, or prior breast abscess.
    - Fever and/or general malaise — these may have subsided if the woman has taken antibiotics for suspected infectious mastitis.
    - A painful, swollen lump in the breast, with redness, heat, and swelling of the overlying skin.
        - On examination, the lump may be fluctuant with skin discolouration.

## INVESTIGATIONS

**Breast milk culture is not routinely required in primary care for women with mastitis.**

- However, in women with lactational mastitis, send a sample of breast milk for microscopy, culture, and antibiotic sensitivity, if:
    - Mastitis is severe or recurrent, or presentation is unusual.
    - Hospital-acquired infection is likely.
    - There is severe deep 'burning' breast pain (indicative of ductal infection).
- Advise the women on how to collect a sample of breast milk. She should:
    - Clean the nipple of the affected breast.
    - Express a small amount of milk by hand and discard it (to avoid skin contamination).
    - Express milk into a sterile container, avoiding touching the inside of the container with the nipple or hands.
- **All women with a breast abscess should be urgently referred to secondary care for confirmation of the diagnosis and management**

## DIFFRENTIAL DIAGNOSIS

**Conditions that cause breast pain and that are associated with lactation include:**

- Full breasts
    - these are common between the second and sixth day after birth, and both breasts are usually affected. The breasts feel hot, heavy, and hard (and are not shiny, oedematous, or red).
- Engorged breasts
    - this either occurs in the first few days after an infant is born (when the breasts overfill with milk [primary engorgement]) or when feeding is less frequent, or the infant's demands have decreased (secondary engorgement). It is usually bilateral, and the breasts are enlarged, swollen, and painful.
- A blocked duct
    - there is a painful lump in the breast, and the skin overlying the lump may be red. There is usually no fever.

- Galactocoele
  - Typically, there is a smooth, rounded, and painless swelling in the breast. Milky fluid is discharged from the nipple when pressed. Systemic symptoms are absent.
- Infection of the mammary ducts
  - there is a deep burning, aching, or shooting pain in the breast that is worse during or just after breastfeeding. There may be accompanying pain radiating down the arm or into the back. There is usually no fever or malaise.

**Conditions that cause breast pain that are not associated with lactation include:**

- Breast cancer
  - this includes ductal cancer (characterized by breast pain and/or bloody discharge), inflammatory breast cancer (characterized by rapid onset of warmth of the breast, diffuse redness, and oedema causing an orange skin [peau d'orange] appearance), and Paget's disease of the nipple (characterized by an itchy, red rash on the nipple, burning sensation in the breast, and bleeding from the nipple).
- Duct ectasia
  - a harmless, age-related breast change which causes the milk duct under the nipple to become blocked or clogged with a thick, sticky substance. It is usually asymptomatic, but there may be discharge from the breast (non-bloody or bloody), a lump behind the nipple (which is just scar tissue or a dilated duct), and/or an inverted nipple.
- Cellulitis
  - acute bacterial infection of the dermis and subcutaneous tissue. It presents with an acute onset of red, painful, hot, swollen, and tender skin, with possible blister or bullae formation. Fever, malaise, nausea, and rigors may accompany or precede the skin changes.
- Fibroadenosis
  - describes a group of benign conditions affecting the breast. Clinical features include breast pain, increase in breast size, and breast nodularity.
- Ruptured breast cyst
  - may present with breast pain and a palpable mass.
- Necrotizing fasciitis of the breast
  - a rare and potentially fatal soft tissue infection of the breast. It can occur after trauma, around foreign bodies in surgical wounds, or can be idiopathic. Clinical features include painful enlargement of the breast, breast ulceration, foul-smelling discharge from the breast, and fever.
- Fat necrosis of the breast — a firm, round lump (or lumps) may be seen on the affected breast. This is usually painless but may feel tender or even painful. The skin around the lump may look red, bruised, or occasionally dimpled. Rarely the nipple of the affected breast may be retracted.

**Conditions that cause nipple pain include:**

- Poor infant attachment
  - this is the most common cause of nipple pain and/or damage.
- Candidal infection of the nipple
  - this often follows antibiotic treatment but may also be a predisposing factor for recurrent mastitis.
- Blanching of the nipple

- this is due to the pressure of suckling, may cause pain, and is related to poor breast-feeding technique.

- Eczema of the nipple
  - presents as a red, itchy rash with a well-demarcated edge.
- Bacterial infection of the nipple
  - may present as a yellow discharge from the nipple, or a sloughy appearance.
- Raynaud's disease of the nipple
  - blanching of the nipple is followed by cyanosis and/or erythema. Pain is severe, debilitating, and throbbing, and breastfeeding is very painful.

## MANAGENENT

### LACTATIONAL MASTITIS

#### When should I arrange immediate admission or referral for a woman with lactational mastitis?

- **Arrange hospital admission if:**
  - There are signs of sepsis (such as tachycardia, fever, and chills).
  - The infection progresses rapidly.
  - The woman is haemodynamically unstable or immunocompromised.
    - If appropriate and depending on the woman's wishes, the infant can be admitted with her to allow continuation of breastfeeding.
- **Arrange an urgent 2-week wait referral** if there is an underlying mass or breast cancer is suspected.
- **Refer urgently to a general surgeon** if a breast abscess is suspected.

#### How should I manage a woman with lactational mastitis in primary care?

**If immediate admission or referral is not indicated:**

- **Reassure the woman** that her breast should return to normal size, shape, and function after treatment.
- **To relieve pain and discomfort:**
  - Prescribe a simple analgesic, such as paracetamol or ibuprofen.
  - Advise the woman to place a warm compress on the breast, or bathe or shower in warm water, to relieve pain and help milk to flow.
- **Advise the woman to continue breastfeeding if possible (including from the affected breast).** If necessary, involve a breastfeeding specialist to assist the woman in improving the infant's attachment to the breast. This will improve milk removal and prevent nipple damage.
  - If breastfeeding is too painful, or the infant refuses to breastfeed from the affected breast, advise the woman to express sufficient milk to match the infant needs until she is able to resume breastfeeding from that breast. Mothers using breast pumps should express only the volume their infant consumes.
  - If the woman does not wish to continue breastfeeding, give advice on stopping breastfeeding.
- **Identify and manage any predisposing factors for mastitis**. For example:
  - Advise the woman to rest and avoid wearing a bra, especially at night.
- **Prescribe an oral antibiotic** if the woman has a nipple fissure that is infected, symptoms have not improved (or are worsening) after 12–24 hours despite effective milk removal, and/or breast milk culture is positive.
  - If breast milk culture results are available, treat with an antibiotic that the organism is sensitive to.
  - If breast milk culture results are not available:
    - Treat empirically with flucloxacillin 500 mg four times a day for 10–14 days.

- If the woman is allergic to penicillin, prescribe either erythromycin 250–500 mg four times a day or clarithromycin 500 mg twice a day for 10–14 days.

- **Advise the woman to seek immediate medical advice** if symptoms fail to settle after 48 hours of antibiotic treatment.
- **Give advice on measures to prevent recurrence.**

## How should I manage treatment failure or recurrence in a woman with lactational mastitis?

**If symptoms fail to settle after 48 hours of first-line antibiotic treatment:**

- Check that the woman has taken the antibiotic correctly.
- Consider the possibility of an alternative diagnosis (such as breast cancer or a breast abscess) and the need for admission or referral.
  - If an abscess is suspected, be aware that malaise and fever may have subsided if antibiotics have been started.
- If an alternative diagnosis is unlikely:
  - Send a sample of breast milk for microscopy, culture, and antibiotic sensitivity (if this has not already been done)
  - Prescribe a second-line antibiotic, co-amoxiclav 500/125 mg three times a day, for 10–14 days; review this choice when breast milk culture results become available. Seek specialist advice if the woman is allergic to penicillin.

**If mastitis recurs, manage as for treatment failure. In addition:**

- Identify and manage any predisposing factors, such as:
  - Nipple damage
  - Staphylococcus aureus carriage — send nasal swabs from both the woman and the infant to identify nasal carriage of S. aureus.
- Ensure that the woman is aware of measures to prevent recurrence, such as having a good breastfeeding technique and maintaining good hygiene.

## What advice should I give to prevent recurrence?

- **Explain that good breastfeeding technique is necessary to prevent mastitis.**
- Advise the woman to:
  - Make sure the infant is attached to the breast correctly.
  - Feed on demand, both in terms of frequency and duration.
  - Avoid missed feeds, especially when the infant starts to sleep through the night.
  - Finish the first breast before offering the other.
  - Breastfeed exclusively for 4–6 months, if possible.
  - Avoid the use of a dummy, which may result in poor attachment to the breast.
  - For future pregnancies, start to breastfeed within an hour of delivery, if possible.
- **Ensure that the woman knows:**
  - How to express milk manually
  - How to check breasts for lumps, redness, and tenderness.
  - How to recognize milk stasis.
  - What to do if milk stasis develops — rest, breastfeed frequently, massage any lumpy areas, and seek medical help if problems do not resolve within 24 hours.
- **If the woman does not wish to continue breastfeeding, advise that she should:**
  - Not stop breastfeeding abruptly.
  - Support the breasts with a comfortable bra.
  - Express enough milk to keep the breasts comfortable.
  - Take paracetamol or ibuprofen if pain occurs.
- **Give advice on hygiene measures, such as:**
  - Thorough and frequent hand washing.

- Ensuring that the breast pump is washed thoroughly with soap and hot water (and air dried) after every use.
- Rinsing the nipple area with water before and after each feed.
- Ensuring that potentially-contaminated nipple ointments or creams are discarded.

## NON-LACTATIONAL MASTITIS

### When should I arrange immediate admission or referral for a woman with non-lactational mastitis?

- **Arrange hospital admission if:**
  - There are signs of sepsis (such as tachycardia, fever, and chills).
  - The infection progresses rapidly.
  - The woman is haemodynamically unstable or immunocompromised.
- **Arrange urgent 2-week wait referral** if there is an underlying mass or breast cancer is suspected.
- **Refer urgently to a general surgeon** if a breast abscess is suspected.

### How should I manage a woman with non-lactational mastitis in primary care?

**If immediate admission or referral is not indicated:**

- **Reassure the woman** that her breast should return to normal size, shape, and function after adequate treatment.
- **To relieve pain and discomfort:**
  - Prescribe a simple analgesic, such as paracetamol or ibuprofen.
  - Advise the woman to place a warm compress on the breast, or bathe or shower in warm water, to relieve pain.
- **Identify and manage any predisposing factors for mastitis.** For example, if there is nipple damage, manage the underlying cause.
- **Prescribe an oral antibiotic for all women with non-lactational mastitis:**
  - Prescribe co-amoxiclav 500/125 mg three times a day for 10–14 days.
  - If the woman is allergic to penicillin, prescribe a combination of erythromycin (250–500 mg four times a day) or clarithromycin (500 mg twice a day) plus metronidazole (400 mg three times a day) for 10–14 days.
- **Advise the woman to seek immediate medical advice** if symptoms worsen or fail to settle after 48 hours of antibiotic treatment. See the section on Treatment failure/recurrence for further information.
- **Give advice on measures to prevent recurrence.**

### How should I manage treatment failure or recurrence in a woman with non-lactational mastitis?

- **If symptoms fail to settle after 48 hours of first-line antibiotic treatment:**
  - Check that the woman has taken the antibiotic correctly.
  - Consider the possibility of an alternative diagnosis (such as breast cancer or a breast abscess) and refer accordingly.
    - If an abscess is suspected, be aware that malaise and fever may have subsided if antibiotics have been started.
  - Also consider the possibility of methicillin-resistant Staphylococcus aureus (MRSA) infection (risk factors include local prevalence and hospital-acquired infection) where first-line treatment is ineffective.
    - If MRSA is suspected, seek local microbiology advice regarding possible investigations and treatments.
  - Consider the need for referral or admission.

- **If mastitis recurs, manage as for treatment failure.** In addition:
  - Check that previous episodes of mastitis were managed appropriately.
  - If an alternative diagnosis is unlikely:
    - Prescribe co-amoxiclav 500/125 mg three times a day for 10–14 days.
    - If the woman is allergic to penicillin, prescribe a combination of erythromycin (250–500 mg four times a day) or clarithromycin (500 mg twice a day) plus metronidazole (400 mg three times a day) for 10–14 days.
    - If antibiotic treatment has failed more than once, seek advice from a local microbiologist regarding culture and sensitivity testing and/or alternative antibiotic treatment, and consider the possibility of an alternative diagnosis.
- **Identify and manage any predisposing factors,** such as:
  - Nipple damage from skin conditions such as psoriasis, eczema, infection, or Raynaud's disease of the nipple
  - S. aureus carriage
  - Ensure that the woman is aware of measures to prevent recurrence, such as stopping smoking and maintaining good hygiene.
  - Consider referring the woman to secondary care (using clinical judgement to determine the urgency) if there is no obvious cause for recurrence (such as continued smoking or nipple piercing).

### What advice should I give to prevent recurrence?

- Advise women who smoke to stop — smoking is a major risk factor for periductal mastitis.
- Give advice on hygiene measures, including thorough and frequent hand washing and removal of nipple rings (due to potential bacterial contamination).

## BREAST ABSCESS MANAGEMENT

- **Refer the woman urgently to a general surgeon for:**
  - Confirmation of the diagnosis (by ultrasound).
  - Drainage of the abscess (by ultrasound-guided needle aspiration or surgical drainage).
  - Culture of fluid from the abscess (which will be used to guide the choice of antibiotic).
- **Advise lactating women to continue breastfeeding if possible (including from the affected breast).**
  - If this is too painful, or the infant refuses to breastfeed from the affected breast, advise the woman to express the milk (by hand or with a breast pump) until she is able to resume breastfeeding from that breast.

TOPIC: Breast Disorders

| Disorder | Features |
|---|---|
| **Fibroadenoma** | Common in women under the age of 30 years<br>Often described as 'breast mice' due as they are discrete, non-tender, highly mobile lumps |
| **Fibroadenosis (fibrocystic disease, benign mammary dysplasia)** | Most common in middle-aged women<br>'Lumpy' breasts which may be painful. Symptoms may worsen prior to menstruation |
| **Breast cancer** | Characteristically a hard, irregular lump. There may be associated nipple inversion or skin tethering<br><br>**Paget's disease of the breast** - intraductal carcinoma associated with a reddening and thickening (may resemble eczematous changes) of the nipple/areola |
| **Mammary duct ectasia** | Dilatation of the large breast ducts<br>Most common around the menopause<br>May present with a tender lump around the areola +/- a green nipple discharge<br>If ruptures may cause local inflammation, sometimes referred to as 'plasma cell mastitis' |
| **Duct papilloma** | Local areas of epithelial proliferation in large mammary ducts<br>Hyperplastic lesions rather than malignant or premalignant<br>May present with blood-stained discharge |
| **Fat necrosis** | More common in obese women with large breasts<br>May follow trivial or unnoticed trauma<br>Initial inflammatory response, the lesion is typical firm and round but may develop into a hard, irregular breast lump<br>Rare and may mimic breast cancer so further investigation is always warranted |
| **Breast abscess** | More common in lactating women<br>Red, hot tender swelling |

| Lesion | Features | Treatment |
|---|---|---|
| Fibroadenoma | • Develop from a whole lobule<br>• Mobile, firm breast lumps<br>• 12% of all breast masses<br>• Over a 2years period up to 30% will get smaller<br>• No increase in risk of malignancy | If >3cm surgical excision is usual, Phyllodes tumours should be widely excised (mastectomy if the lesion is large) |
| Breast cyst | • 7% of all Western females will present with a breast cyst<br>• Usually presents as a smooth discrete lump (may be fluctuant) | Cysts should be aspirated, those which are blood stained or persistently refill should be biopsied or excised |

| Lesion | Features | Treatment |
|---|---|---|
| | • Small increased risk of breast cancer (especially if younger) | |
| Sclerosing adenosis, (radial scars and complex sclerosing lesions) | • Usually presents as a breast lump or breast pain<br>• Causes mammographic changes which may mimic carcinoma<br>• Cause distortion of the distal lobular unit, without hyperplasia (complex lesions will show hyperplasia)<br>• Considered a disorder of involution, no increase in malignancy risk | Lesions should be biopsied, excision is not mandatory |
| Epithelial hyperplasia | • Variable clinical presentation ranging from generalised lumpiness through to discrete lump<br>• Disorder consists of increased cellularity of terminal lobular unit, atypical features may be present<br>• Atypical features and family history of breast cancer confers greatly increased risk of malignancy | If no atypical features then conservative, those with atypical features require either close monitoring or surgical resection |
| Fat necrosis | • Up to 40% cases usually have a traumatic aetiology<br>• Physical features usually mimic carcinoma<br>• Mass may increase in size initially | Imaging and core biopsy |
| Duct papilloma | • Usually present with nipple discharge<br>• Large papillomas may present with a mass<br>• The discharge usually originates from a single duct<br>• No increased risk of malignancy | Microdochectomy |

## TOPIC: Balanitis

- Balanitis describes inflammation of the glans penis, and posthitis describes inflammation of the prepuce (foreskin). It is caused by a range of conditions which can affect the penile skin.
    - In clinical practice, both areas are often affected, and the term 'balanoposthitis' may be used.
- Balanitis may be caused by a range of different conditions affecting the penile skin (which may present similarly and co-exist) including:
    - Non-specific dermatitis with Candida albicans or bacterial infection.
    - Other infections such as anaerobes or sexually transmitted infections.
    - Inflammatory skin conditions such as irritant or allergic contact dermatitis, seborrhoeic dermatitis, psoriasis, lichen sclerosus, lichen planus, or Zoon's balanitis.
    - Mechanical irritation and poor hygiene.
    - Pre-malignant conditions (penile intraepithelial neoplasia).
- It affects about 4% of uncircumcised boys between the ages of 2–5 years.
- Possible complications depend on the underlying cause and may include:
    - Phimosis and paraphimosis.
    - Urinary symptoms due to urethral meatal stenosis and urethral stricture.
    - Sexual dysfunction.
    - Penile intraepithelial neoplasia and malignancy.
- Most episodes will improve following accurate diagnosis and appropriate treatment of the underlying cause.
- A diagnosis of balanitis should be suspected if there is:
    - Penile discomfort and itch, bleeding, skin splitting, urethral discharge, urinary symptoms and/or sexual dysfunction.
    - Redness and swelling of the glans penis and/or foreskin with possible sub-preputial discharge, fissuring, or ulceration.
- Assessment of suspected balanitis should include:
    - Asking about the onset and duration of symptoms; hygiene practices; exposure to potential irritants, allergens and new drugs; underlying skin conditions; sexual history and risk of sexually transmitted infections (STIs); any causes for immunosuppression.
    - Examination of the penile skin, genital area, and extra-genital skin/mucosae to assess for an underlying cause.
    - Arranging sub-preputial swabs for Candida sp and bacterial culture and sensitivity if symptoms are severe, or infection is suspected.
    - Arranging an STI screen if clinically indicated.
    - Arranging blood testing for HbA1c and HIV (if clinically appropriate) if balanitis is severe or persistent.

- Management of suspected balanitis should include:
  - Advising on the importance of good daily hygiene.
  - Advising on management options depending on the likely underlying cause and age of the person, such as a topical corticosteroid or imidazole preparation for non-specific dermatitis; a topical imidazole or oral antifungal and topical corticosteroid preparation for candida infection; a topical or oral antibiotic and topical corticosteroid preparation for bacterial balanitis; and avoiding triggers, a topical corticosteroid preparation and regular emollient for irritant or allergic contact dermatitis.
  - Managing any other underlying condition as clinically indicated.
  - Reviewing the person if symptoms are severe or have not improved with initial treatment and taking sub-preputial swabs.
- Referral to an appropriate specialist should be arranged if there is:
  - Diagnostic uncertainty.
  - Persistent or recurrent balanitis not responding to primary care treatment.
  - Persistent phimosis.
  - Suspected allergic contact dermatitis, lichen sclerosus, or Zoon's balanitis.
  - A suspected pre-malignant or malignant lesion.

## TOPIC: Erectile dysfunction

**Definition**

- Erectile dysfunction is defined as the persistent inability to attain and maintain an erection sufficient to permit satisfactory sexual performance.
    - It is a symptom and not a disease, and may be due to multiple underlying causes.

**Causes**

Erectile dysfunction is caused by various vascular, neuronal, hormonal, and metabolic factors, mediated by endothelial and smooth muscle dysfunction, as normal erectile function relies on arterial dilatation, smooth muscle relaxation, and veno-occlusion within the penile corpora.

- Organic causes include:
    - Vasculogenic (most common) — cardiovascular disease (CVD), hypertension, peripheral arterial disease, hyperlipidaemia, type 1 and 2 diabetes mellitus, metabolic syndrome, smoking, obesity, major pelvic surgery (radical prostatectomy), or radiotherapy (to pelvis or retroperitoneum).
    - Neurogenic (central) — degenerative disorders (such as multiple sclerosis, Parkinson's disease, and multiple system atrophy), stroke, spinal cord trauma or disease, or central nervous system (CNS) tumours.
    - Neurogenic (peripheral) — type 1 and 2 diabetes mellitus, chronic kidney disease, chronic liver disease, polyneuropathy, major surgery of the pelvis or retroperitoneum, or urethral surgery (for example, urethroplasty for urethral stricture).
    - Anatomical or structural — Peyronie's disease, penile cancer, prostate cancer, congenital curvature of the penis, micropenis, hypospadias, epispadias, or phimosis.
    - Endocrine — type 1 and type 2 diabetes mellitus, metabolic syndrome, primary or secondary hypogonadism, hyperprolactinaemia, hyper- or hypothyroidism, Cushing's disease, panhypopituitarism and multiple endocrine disorders, or hypopituitarism following traumatic brain injury.
- Psychogenic causes include:
    - Generalized — for example, due to lack of arousability and disorders of sexual intimacy.
    - Situational — for example, due to partner- or performance-related issues, relationship problems (marital, extramarital, or new relationship), stress, depression, anxiety, post-traumatic stress disorder, or psychosis.
- Drugs that can cause or exacerbate symptoms include:
    - Antihypertensives — beta-blockers, verapamil, methyldopa, and clonidine.
    - Diuretics — spironolactone and thiazides.
    - Antidepressants — tricyclics, monoamine oxidase inhibitors (MAOIs), selective serotonin reuptake inhibitors (SSRIs), lithium, and venlafaxine.
    - Antiarrhythmic drugs — digoxin and amiodarone.
    - Anticholinergics — pregabalin, gabapentin, and duloxetine.
    - Antiepileptics — carbamazepine, topiramate, gabapentin, and pregabalin.
    - Antipsychotics/tranquilizers — chlorpromazine, haloperidol, and phenothiazines.
    - Hormones and hormone-modifying drugs — anti-androgens (such as cyproterone acetate); gonadotrophin-releasing hormone agonists (such as leuprorelin, goserelin);

corticosteroids; 5-alpha reductase inhibitors (such as finasteride), oestrogens, and progesterone.
    - Histamine ($H_2$)-antagonists — cimetidine and ranitidine.
    - Cytotoxic drugs — cyclophosphamide and methotrexate.
    - Recreational drugs — alcohol, heroin, cocaine, cannabis, methadone, anabolic steroids, and opiates.

**Risk factors**

- The risk factors for erectile dysfunction are similar to those for cardiovascular disease (CVD) and involve endothelial dysfunction and small vessel atherosclerosis, such as:
    - Increasing age.
    - Obesity — associated with androgen deficiency.
    - Cardiovascular disease.
    - Hypertension.
    - Dyslipidaemia.
    - Diabetes mellitus (types 1 and 2) — type 2 diabetes may be associated with androgen deficiency and may be the first symptom of diabetes in up to 20% of men.
    - Metabolic syndrome including non-alcoholic fatty liver disease (NAFLD) — associated with androgen deficiency.
    - Obstructive sleep apnoea.
    - Benign prostatic hypertrophy and lower urinary tract symptoms (LUTS).
    - Lifestyle factors:
        - Lack of exercise and sedentary lifestyle.
        - Smoking and alcohol.
        - Bicycle-riding for more than 3 hours a week (may cause a neuropraxia and/or vasculogenic erectile dysfunction due to compression of the pudendal nerve and arteries).

**Prognosis:**

- Depending on the likely underlying cause of erectile dysfunction, some cases can be cured, and others may be managed successfully with medical or surgical treatment.
    - Psychogenic erectile dysfunction, post-traumatic arteriogenic erectile dysfunction in young men, and erectile dysfunction due to hormonal causes (for example, hypogonadism or hyperprolactinaemia) may be cured with specialist treatment.
    - Drug-induced erectile dysfunction can be cured in some cases if the offending drug is stopped, if a clear temporal relationship is established between starting a drug and the onset of erectile dysfunction.
    - Erectile dysfunction resulting from radical prostatectomy may be reversible depending on pre-operative erectile function, the nerve-sparing nature of the procedure, and how soon after the procedure drugs such as phosphodiesterase-5 (PDE-5) inhibitors are started.
    - Expert opinion in a review article states that 60–65% of men with erectile dysfunction can successfully complete sexual intercourse with phosphodiesterase -5 (PDE-5) inhibitor drug treatment.

        - These drugs can be used long-term if there are no contraindications or significant adverse effects, and there is no clear evidence in the literature of tachyphylaxis or drug tolerance. Men may; however, become less responsive to drug treatment over time, due to worsening of underlying endothelial dysfunction and small vessel atherosclerosis.

**Complications:**
Erectile dysfunction may result in various psychosocial and physical health complications.

- Anxiety:
    - Lack of sexual confidence and low self-esteem may lead to avoidance or abstinence from sexual intimacy.
    - Secondary performance anxiety may develop due to an underlying organic cause, which perpetuates symptoms of erectile dysfunction.
- Depression:
- Interpersonal and relationship difficulties including impact on the person's partner.
- Increased risk of cardiovascular disease (CVD), stroke, peripheral arterial disease, and all-cause mortality.

**Diagnosis**
If a man presents with erectile dysfunction, assess with their partner if possible and appropriate, to determine any underlying cause and their cardiac risk of sexual activity.

- Ask about psychosexual factors:
    - Sexual orientation, gender identity, and past and current sexual relationships.
    - Sexual function, including onset and duration of erectile problems; quality of erections (morning, spontaneous, self-stimulated, and/or partner-related); and problems with sexual desire, arousal, ejaculation, and orgasm.
        - An organic cause is suggested by a gradual onset of symptoms, lack of tumescence, and low-to-normal libido.
        - A psychogenic cause is suggested by sudden onset of symptoms, low libido, and good quality spontaneous or self-stimulated erections.
        - Consider using a validated patient questionnaire, such as the 15-item International Index of Erectile Function (IIEF) or the shortened 5-item version, the Sexual Health Inventory for Men (SHIM/IIEF-5), to assess the presence, severity, and impact of erectile dysfunction.
    - Any issues around sexual aversion or pain, cultural or religious beliefs, and any issues for sexual partner(s) including sexual desire, satisfaction, or pain.
    - Any contributory life events including recent pregnancy or childbirth, relationship dynamics, intimacy difficulties, previous sexual trauma or abuse, anxiety, and/or depression.
    - Any complications, including impact on emotional wellbeing, relationships, and daily functioning.
- Ask about other medical history:
    - History or symptoms of associated conditions such as cardiovascular disease, hypertension, heart failure, peripheral arterial disease, or stroke.
        - Assess the cardiac risk of sexual activity, and advise and/or arrange referral accordingly.

    - Surgical history including pelvic or retroperitoneal surgery and coronary revascularisation procedures (to assess cardiac risk of sexual activity).
    - Symptoms suggesting a possible organic cause, such as hypogonadism (reduced energy and/or reduced libido); lower urinary tract symptoms (LUTS); and deviation of the erection during tumescence or painful erection (may suggest Peyronie's disease).
    - Previous episodes of erectile dysfunction, treatments including those bought over-the-counter and online, and response to treatment.
    - Any risk factors including smoking history and alcohol intake; prescribed or recreational drugs; cycling for more than 3 hours per week; and exercise tolerance (to assess cardiac risk of sexual activity).
- Perform a focused physical examination to identify any underlying organic cause(s).
    - Measure blood pressure, heart rate, waist circumference, and body mass index (BMI).
    - Check for gynaecomastia, sparse body hair, and reduced muscle mass to assess for testosterone deficiency.
    - Consider examination of the external genitalia to assess for:
        - Signs of hypogonadism, such as testicular atrophy and alterations in secondary sexual characteristics.
        - Penile foreskin conditions, such as phimosis or lichen sclerosus.
        - Penile structural abnormalities, including Peyronie's disease (causes abnormal angulation of the erect penis), hypospadias, phimosis, and penile cancer.
        - Testicular abnormalities if there are symptoms of testicular lump, swelling, or pain.
    - Consider a digital rectal examination (DRE) to assess for an abnormality of the prostate:
        - If there are obstructive LUTS (such as reduced flow and/or intermittent urinary stream) or prolonged secondary ejaculation (due to an enlarged prostate obstructing the flow of ejaculate).
        - In men with a history of prostate cancer and/or over 50 years of age.
- For all men, arrange blood tests to assess for an underlying cause and cardiovascular risk:
    - Serum HbA1c or fasting blood glucose to diagnose diabetes or assess blood sugar control (if confirmed diabetes).
    - Serum lipid profile.
        - Calculate the 10-year cardiovascular risk and manage accordingly. information.
    - A fasting serum total testosterone (taken between 9–11 am) to assess for testosterone deficiency and hypogonadism.
        - If the free testosterone level is low or borderline, arrange a repeat serum testosterone, together with follicle-stimulating hormone (FSH), luteinizing hormone (LH), sex hormone binding globulin, and prolactin levels.
- Consider arranging additional investigations, depending on the likely underlying cause and clinical judgement, such as:

    - Serum prostate-specific antigen (PSA) level to assess for prostate cancer, depending on clinical judgement.
    - Thyroid function tests.
    - Renal function tests.

**Cardiac risk stratification**

Men with erectile dysfunction can be stratified into low-, intermediate-, or high-risk cardiovascular categories, depending on cardiovascular risk factors and co-morbidities.

- Low risk
    - Asymptomatic; able to perform exercise of modest intensity without symptoms; less than 3 risk factors for coronary artery disease (CAD, excluding sex).
    - Mild, stable angina.
    - Uncomplicated previous myocardial infarction (MI).
    - Previous successful coronary revascularization.
    - Reduced ejection fracture heart failure (New York Heart Association [NYHA] class I and II).
    - Controlled hypertension.
    - Mild valvular heart disease (VHD).
- Intermediate risk
    - Three or more risk factors for CAD (excluding sex).
    - Moderate stable angina.
    - Recent MI (within the last 2–6 weeks).
    - Reduced ejection fracture heart failure (NYHA class III).
    - Peripheral arterial disease, stroke disease.
- High risk
    - Unstable or refractory angina.
    - Recent MI (within the last 2 weeks).
    - Reduced ejection fracture heart failure (NYHA class IV).
    - Uncontrolled hypertension.
    - High-risk arrhythmia (exercise-induced ventricular tachycardia, implanted internal cardioverter defibrillator with frequent shocks, and poorly controlled atrial fibrillation).
    - Hypertrophic obstructive or other cardiomyopathies.
    - Moderate-to-severe VHD.

**Management**
**When should I refer a man with erectile dysfunction?**

If a man presents with erectile dysfunction following initial assessment:

- Arrange emergency hospital admission if there is evidence of priapism (painful prolonged erection for more than 4 hours — may be a rare adverse effect of specialist treatments for erectile dysfunction).
- Arrange referral to a urology specialist if:

    - The man is young or has a lifelong history of difficulty in obtaining or maintaining an erection (possible primary erectile dysfunction).
    - There is a history of pelvic, perineal, or genital trauma.
    - There is a penile structural abnormality or abnormal testicular examination.
- Arrange referral to an endocrinologist if:
    - A diagnosis of hypogonadism is suspected (low serum testosterone level) or there are other abnormalities of testosterone, follicle-stimulating hormone (FSH), luteinizing hormone (LH), or prolactin levels.
    - Testosterone replacement is being considered.
- Arrange referral to a cardiologist or seek specialist advice if:
    - The man is at high cardiac risk (or intermediate cardiac risk, depending on clinical judgement), where sexual activity may be unsafe or phosphodiesterase-5 (PDE-5) inhibitor use is contraindicated.
        - Advise stopping all sexual activity until specialist assessment, and a decision has been made that further cardiac assessment and management is needed, or it is safe to resume sexual activity.
- Arrange referral to mental health services, psychosexual or relationship counselling, depending on clinical judgement, if:
    - There is a suspected psychogenic or relationship cause for symptoms.
    - There is a complex or severe mental health condition which cannot be managed in primary care.

**How should I manage a man with erectile dysfunction in primary care?**

Offer management of erectile dysfunction symptoms in primary care following initial assessment.

- Arrange hospital admission or specialist referral if clinically indicated.
- Advise about lifestyle modification that may help, including:
    - Weight loss, if indicated, and regular exercise.
    - Smoking cessation, if indicated.
    - Alcohol reduction to recommended levels, if indicated.
    - Stopping all sexual activity until specialist cardiology assessment, if a man is at high cardiac risk from sexual activity (or intermediate risk, depending on clinical judgement).
    - Stopping cycling for a trial period, if a man regularly cycles for more than 3 hours a week. If this is not possible, advise to use a properly fitted bicycle seat and ride with the seat in a suitable position.
- Optimize management of any reversible or modifiable risk factors or conditions, such as diabetes, hypertension, and dyslipidaemia.
- If a drug cause is suspected, consider stopping medication if the onset of erectile dysfunction correlated with starting a specific drug and a causative relationship seems likely, depending on clinical judgement.
    - Be aware that for most men, management of any underlying condition with appropriate drug treatment is more likely to be beneficial than changes to potentially contributory drugs.
- Do not recommend the use of unlicensed herbal preparations or complementary medicines.

    - Advise these may contain pharmacological substances which may be contraindicated or interact with other prescribed medication(s).
- For men not at high cardiac risk of sexual activity, consider prescribing drug treatment with a phosphodiesterase-5 (PDE-5) inhibitor, irrespective of the suspected underlying cause(s). Drug choice depends on the frequency of intercourse, speed of onset, duration of action, dosing regimen, personal experience and preference, adverse effect profile, cost, and local prescribing guidelines.
    - Options include sildenafil (Viagra®), tadalafil (Cialis®), vardenafil (Levitra®), and avanafil (Spedra®).
- Arrange follow-up 6–8 weeks after starting treatment, to assess its efficacy, tolerability, and patient satisfaction.

**How should I follow up with a man with erectile dysfunction?**

When reviewing a man with erectile dysfunction following initial management in primary care:

- If initial treatment is effective, reinforce lifestyle modification advice and arrange review depending on clinical judgement.
- If initial treatment is ineffective or unsatisfactory, reassess for underlying cause(s) and risk factor(s) and manage appropriately, including arranging specialist referral if clinically indicated.
    - Be aware that hypogonadism and a low testosterone level may result in a reduced response or non-response to phosphodiesterase-5 (PDE-5) inhibitor drug treatment.
- If the man is using PDE-5 inhibitor drug treatment, ensure it is taken correctly and advise on possible dose and drug regimen changes.
    - Advise about the importance of dose timing and the fact drug treatment requires sexual stimulation in order to facilitate erection.
    - Consider increasing to the maximum dose depending on symptom response and adverse effects. Advise to try each PDE-5 inhibitor 4–8 times at the maximum tolerated dose before switching to an alternative drug.
    - Suggest a trial of at least two different PDE-5 inhibitors taken sequentially before being classed as a 'non-responder'.
    - Consider increasing the dose frequency, such as switching to once daily (rather than 'on-demand') dosing if taking tadalafil, depending on clinical judgement.
- If PDE-5 inhibitor drug or other treatment is ineffective, not tolerated, or contraindicated, offer referral to a urology specialist.

    - Specialist treatment options may include vacuum erection assistance devices (VEDs), alprostadil penile intracavernous injections (Caverject®, Viridal®), medicated urethral system for erection (MUSE®), vacuum erection assistance devices (VEDs), vascular surgery/angioplasty, or penile prostheses.

**Prescribing information:**
**Phosphodiesterase-5 (PDE-5) inhibitors**
**Contraindications and cautions**

- Do not prescribe a phosphodiesterase-5 (PDE-5) inhibitor to a man with:
    - Unstable angina or angina occurring during sexual intercourse.

    - Regular or intermittent use of nitrates in any form — risk of severe, life-threatening hypotension. The manufacturer advises that prescribers should avoid concomitant prescribing.
    - Hypotension (systolic blood pressure below 90 mmHg).
    - History of non-arteritic anterior ischaemic optic neuropathy (NAION).
    - Recent history of myocardial infarction (within 90 days for tadalafil).
    - Recent history of stroke.
    - Heart failure (mild to severe, tadalafil, and avanafil).
    - Uncontrolled arrhythmias (tadalafil and avanafil).
    - Uncontrolled hypertension (tadalafil and avanafil).
    - Severe hepatic impairment (sildenafil, vardenafil film-coated tablets, and avanafil); moderate-to-severe (vardenafil orodispersible tablets).
    - Renal impairment (creatinine clearance less than 30 mL/min, avanafil).
    - A hereditary degenerative retinal disorder, such as retinitis pigmentosa (sildenafil, vardenafil, and avanafil).
- Prescribe a PDE-5 inhibitor with caution to a man with:
    - Cardiovascular disease.
    - Left ventricular outflow obstruction (for example aortic stenosis and idiopathic hypertrophic subaortic stenosis).
    - Anatomical deformity of the penis (for example angulation, cavernosal fibrosis, or Peyronie's disease).
    - A predisposition to priapism (for example sickle-cell disease, multiple myeloma, or leukaemia).
    - Active peptic ulceration or bleeding disorders (sildenafil, vardenafil, and avanafil).
    - Hepatic impairment (sildenafil — consider initial dose reduction if mild-to-moderate impairment); severe hepatic impairment (tadalafil); mild-to-moderate (caution if vardenafil film-coated tablets or avanafil); mild (caution if vardenafil orodispersible tablets).
    - Renal impairment (creatinine clearance less than 30 mL/min, sildenafil and vardenafil — consider initial dose reduction); severe renal impairment (tadalafil — avoid regular once-daily dosing and consider dose reduction for intermittent use).
    - Susceptibility to prolongation of QT interval, or older age (vardenafil).
    - Autonomic dysfunction.
- Prescribe sildenafil with caution to men taking sacubitril/valsartan — a single dose of sildenafil in people with hypertension significantly reduces blood pressure.

**Adverse effects**
**Possible adverse effects of phosphodiesterase-5 (PDE-5) inhibitors include:**

- Most common — headache, flushing, dyspepsia, nasal congestion (sildenafil, tadalafil, avanafil, vardenafil), back pain, and myalgia (with tadalafil)
- Common — alopecia, anaemia, anxiety, cough, diarrhoea, dizziness, fluid retention, gastrointestinal discomfort, increased risk of infection, insomnia, nausea, night sweats, pain, skin reactions, tremor, and vision disorders (sildenafil).

- Rare — optic neuropathy, retinal occlusion (sildenafil, tadalafil, vardenafil — discontinue if sudden visual impairment), and priapism (persistent erection).

**Drug interactions**

Possible drug interactions with phosphodiesterase-5 (PDE-5) inhibitors include:

- Nitrates — concurrent use of PDE-5 inhibitors and organic nitrates (such as isosorbide mononitrate or isosorbide dinitrate), nicorandil, or amyl nitrate ('poppers') are absolutely contraindicated due to the risk of hypotension.
    - If a PDE-5 inhibitor is taken and the man develops chest pain, organic nitrates should not be used for at least 24 hours for sildenafil and possibly vardenafil (half-life 4 hours), 48 hours for tadalafil (due to its longer half-life of 17.5 hours), and at least 12 hours for avanafil (half-life 6–17 hours).
- Alpha-blockers — concurrent use of PDE-5 inhibitors and alpha-blockers can increase the risk of postural hypotension as both drugs are vasodilators.
    - PDE-5 inhibitors should not be used by men taking non-selective alpha blockers (such as doxazosin, indoramin, terazosin, or prazosin) unless they have finished alpha-blocker dose titration and are on a stable dose. The manufacturers also state that:
        - Sildenafil should be used with caution in men taking an alpha-blocker (especially doxazosin). Hypotension is more likely to occur within four hours following treatment with an alpha-blocker. A sildenafil starting dose of 25 mg is recommended.
        - Vardenafil should be initiated at the lowest starting dose of 5 mg (if the man has been stabilised on his alpha-blocker treatment). Vardenafil may be given at any time with tamsulosin or alfuzosin but with other alpha-blockers, a time separation of dosing should be considered. In men already taking an optimized dose of vardenafil, alpha-blocker treatment should be initiated at the lowest dose.
        - Tadalafil is not recommended in men taking doxazosin. For all other alpha-blockers, tadalafil should be used with caution; treatment should be initiated at the minimum dose and progressively adjusted.
        - Avanafil should be initiated at the lowest dose of 50 mg. Conversely in men already taking an optimal dose of avanafil, alpha-blocker treatment should be initiated at the lowest dose.
- Cytochrome P450 (CYP) 3A4 and 2C9 inhibitors — may reduce the clearance of PDE-5 inhibitors, as PDE-5 inhibitors are metabolized by this enzyme.
    - Sildenafil
        - With ritonavir — should be avoided if possible, otherwise a maximum dose of 25 mg of sildenafil within 48 hours must not be exceeded.
        - With all other CYP3A4 inhibitors (such as ketoconazole, itraconazole, erythromycin, cimetidine, and grapefruit juice) — a sildenafil starting dose of 25 mg should be considered.
    - Vardenafil
        - HIV protease inhibitors (such as ritonavir and indinavir; very potent inhibitors of CYP3A4) — co-administration is contraindicated.

        - Potent CYP3A4 inhibitors (such as clarithromycin and oral ketoconazole and itraconazole) — co-administration is contraindicated in men older than 75 years and should be avoided in all other men.
        - Moderate CYP3A4 inhibitors (such as erythromycin) — vardenafil dose adjustment may be necessary.
        - Grapefruit — co-administration should be avoided as concomitant intake of grapefruit or grapefruit juice is expected to increase the plasma concentrations of vardenafil.
    - Tadalafil
        - with CYP3A4 should be avoided if possible, otherwise tadalafil should be initiated at the lowest recommended starting dose.
    - Avanafil
        - Very potent and potent CYP3A4 inhibitors (such as itraconazole, clarithromycin, saquinavir, and indinavir) — co-administration is contraindicated.
        - Moderate CYP3A4 inhibitors (such as erythromycin) — a maximum dose of 100 mg of avanafil within 48 hours must not be exceeded.
        - Other CYP3A4 inhibitors — co-administration with caution.
        - Grapefruit — the man should be advised to avoid grapefruit juice within 24 hours prior to taking avanafil.
- CYP3A4 inducers (such as rifampin, phenobarbital, phenytoin, and carbamazepine) — may enhance the clearance of PDE-5 inhibitors.
    - should be avoided if possible, otherwise higher doses of the PDE-5 inhibitor may be required.

**Dosing regimen**

Phosphodiesterase-5 (PDE-5) inhibitor drugs are usually taken intermittently as needed. The drug tadalafil may be prescribed once daily in specific circumstances, depending on clinical judgement.

- Sildenafil
    - Dosage
        - Advise taking 50 mg approximately one hour before sexual activity (or longer if taken with food). Based on efficacy and tolerability, the dose may be increased to 100 mg or decreased to 25 mg, to be taken as a single dose. The maximum recommended dose is 100 mg. The maximum recommended dosing is once per 24 hours.
    - Renal impairment
        - If the creatinine clearance is less than 30 mL/min, use an initial dose of 25 mg. This can be increased to 50 mg, then 100 mg based on efficacy and tolerability.
    - Hepatic impairment
        - In men with mild-to-moderate hepatic impairment, consider a starting dose of 25 mg. This can be increased to 50 mg, then 100 mg based on efficacy and tolerability.
        - Avoid in men with severe hepatic impairment.
- Tadalafil

- Dosage
  - Advise taking 10 mg (with or without food) taken at least 30 minutes prior to sexual activity. Based on efficacy and tolerability, this can be increased to the maximum dose of 20 mg. The maximum dose frequency is once per 24 hours.
  - For men who prefer spontaneous (rather than planned) sexual activity, or who anticipate frequent sexual activity (at least twice a week), consider prescribing tadalafil 5 mg once daily, reduced to 2.5 mg once daily, dose adjusted according to response.
- Renal impairment
  - If severe impairment, use a maximum dose of 10 mg for intermittent use. Avoid regular once-daily dosing regimen.
- Hepatic impairment
  - A maximum daily dose of 10 mg once daily should be considered for intermittent use.
  - Use with caution in severe hepatic impairment and in regular once-daily dosing regimens.

- Vardenafil (film-coated and orodispersible formulations available)
  - Dosage
    - For film-coated tablets — 10 mg (with or without food) taken at least 25–60 minutes before sexual activity (or longer if taken with a high fat meal). Based on efficacy and tolerability, this can be increased to the maximum recommended daily dose of 20 mg or decreased to 5 mg. The maximum recommended dosing frequency is once per 24 hours.
    - For orodispersible tablets — 10 mg (with or without food) taken at least 25–60 minutes before sexual activity (not affected by fatty meals). The maximum recommended dose is 10 mg a day. The maximum recommended dosing frequency is once per 24 hours.
  - Renal impairment
    - For film-coated tablets — if severe renal impairment (creatinine clearance less than 30mL/min), a starting dose of 5 mg should be considered. This can be increased to 10 mg, then 20 mg based on tolerability and efficacy.
    - For orodispersible tablets — if severe renal impairment (creatinine clearance less than 30 mL/min), a starting dose of vardenafil 5 mg film-coated tablets should be considered. Based on tolerability and efficacy, the dose may be increased to vardenafil 10 mg and 20 mg film-coated tablets, or vardenafil 10 mg orodispersible tablets. Do not use orodispersible tablets in men with end-stage kidney disease.
  - Hepatic impairment
    - For film-coated tablets — a starting dose of 5 mg should be considered in men with mild-to-moderate hepatic impairment. Based on tolerability and efficacy, the dose may subsequently be increased. The maximum dose recommended in moderate hepatic impairment is 10 mg.
    - For orodispersible tablets — in mild hepatic impairment, use a starting dose of vardenafil 5 mg film-coated tablets. Based on tolerability and efficacy, the dose may be increased to vardenafil 10 mg and 20 mg film-coated tablets, or

vardenafil 10 mg orodispersible tablets. Do not use orodispersible tablets as a starting dose in men with mild hepatic impairment and in men with moderate or severe hepatic impairment.

- Elderly men (aged 65 years and older)
    - A dose increase to the maximum of 20 mg should be used cautiously, depending on individual tolerability.

- Avanafil
    - Dosage
        - The recommended dose is 100 mg (with or without food) taken as needed approximately 15–30 minutes before sexual activity (or longer if taken with food). Based on individual efficacy and tolerability, the dose may be increased to a maximum dose of 200 mg or decreased to 50 mg. The maximum recommended dosing frequency is once per 24 hours.
    - Renal impairment
        - If the creatinine clearance is less than 30 mL/min, do not use.
    - Hepatic impairment
        - If severe hepatic impairment, do not use.
        - In men with mild-to-moderate hepatic impairment, use the lowest effective initial dose and adjust according to response.
    - Elderly men (aged 65 years and older)
        - No dose adjustment is needed, however, there are limited data for its use in men aged 70 years and older.

## TOPIC: Premature Ejaculation

**Definition:**

- Premature ejaculation which always or nearly always occurs prior to or within about one minute of vaginal penetration from the first sexual experience (lifelong premature ejaculation), OR, a clinically significant reduction in latency time, often to about three minutes or less (acquired premature ejaculation); and
- The inability to delay ejaculation on all or nearly all vaginal penetrations; and
- Negative personal consequences, such as distress, bother, frustration and/or the avoidance of sexual intimacy.

This definition incorporates both lifelong and acquired premature ejaculation but only applies to vaginal intercourse, as the committee felt there was not the evidence to extend the definition to other situations or groups.

**Risk Factors:**

- Genetic predisposition.
- Obesity.
- Poor general health.
- Emotional problems and stress.
- History of traumatic sexual experiences.
- Medication and recreational drugs. Possible causes include amfetamine, cocaine and dopaminergic medication.
- Prostatitis.
- Neurological causes - eg, multiple sclerosis, peripheral neuropathies.
- Thyroid disorder.
- Varicocele.
- Erectile dysfunction.

**Assessment:**

**History should include:**

- An estimate of intravaginal ejaculation latency time.
- Perceived control.
- Degree of distress.
- Impact on sexual activity and relationship and quality of life.
- Two questionnaires are available for the assessment of premature ejaculation - the Premature Ejaculation Diagnostic Tool (PEDT) and the Arabic Index of Premature Ejaculation (AIPE).

**Examination and investigations:**

Examination may be indicated to exclude endocrine, urological or neurological conditions which may be relevant. Further investigation is not routinely indicated, unless suggested by findings on history and/or examination.

## Management:

Premature ejaculation management should be tailored to the needs of the individual. Premature ejaculation may be more of an issue in some relationships than in others and patient expectation should be explored. Psychosexual counselling may be sufficient.

### General advice

- More frequent sex (or masturbation): premature ejaculation is more likely if there is a longer gap between sexual intercourse.
- Using a condom may decrease sensation.
- Sex with the woman on top reduces the likelihood of premature ejaculation.
- Squeeze and stop-go techniques: stimulating the penis almost to the point of ejaculation and then stopping. These techniques are often effective but may take a few months to produce any benefit and relapse is common.
- Behavioural treatments are useful for secondary premature ejaculation but are not recommended first-line for lifelong premature ejaculation. They are time-intensive and require commitment from the partner.
- Where present, treat erectile dysfunction and/or prostatitis first.
- Reviews suggest that pharmacological therapy is most effective, particularly for lifelong premature ejaculation.

### Pharmacological therapy

- Selective serotonin reuptake inhibitor (SSRI) antidepressants are commonly used off-label but need to be taken daily, as it takes one to two weeks for the effect to be achieved. Paroxetine, fluoxetine, citalopram, escitalopram and sertraline have all been shown to be effective. The serotonin-noradrenaline reuptake inhibitor (SNRI) duloxetine is also effective. On-demand treatment with clomipramine may be a suitable alternative.
- Dapoxetine - a short-acting SSRI which has been specifically developed for the treatment of premature ejaculation - has been shown to be effective and is approved for use in the UK. It is taken as required, one to three hours before anticipated sexual activity. It is not recommended for men over the age of 65.
- Anaesthetic creams may be effective. Options with evidence of efficacy are EMLA® cream and lidocaine gel. Aerosol sprays are proving popular and novel preparations are being developed but are not available on the NHS. Topical preparations may be the preferred therapy for some patients
- Tramadol has been found to have beneficial effect in the treatment of premature ejaculation and may be used "on-demand". The most common adverse effect was sleep-disturbance. Due to its potential for addiction, it is not routinely recommended.
- Phosphodiesterase type 5 (PDE5) inhibitors may be helpful in some cases, although the evidence for this is limited. There is some evidence that a combination of sildenafil with SSRI is better than SSRI monotherapy. It may be that the PDE5 inhibitor allows a second erection within a short space of time with subsequent improvement in intravaginal ejaculatory latency time (IELT)

### Psychosexual therapy:

The evidence base for the effectiveness of psychological interventions is limited for premature ejaculation treatment. Psychological therapy needs compliance from the partner and is time-consuming,

expensive and less effective than pharmacotherapy, and its efficiency decreases with time. It is however the first-line therapy in patients with subjective premature ejaculation or where there are underlying psychological problems. Psychotherapy is effective in managing psychological distress. Methods like meditation/relaxation, hypnotherapy and neuro-biofeedback help improve ejaculatory control.

### Surgery

One study reported that a short frenulum was found in 43% of individuals affected by lifelong premature ejaculation. Frenulectomy was effective in relieving the problem and the authors recommended excluding short frenulum in all patients with lifelong premature ejaculation. This option is not reflected in any of the available guidelines.

### Complications:

Premature ejaculation may have a significant adverse effect on both self-confidence and the relationship. One study reported that premature ejaculation can lead to sexual dissatisfaction, a feeling that something is missing from the relationship and an impaired sense of intimacy. If the premature ejaculation remains untreated it can lead to increased irritability, interpersonal difficulties and deepening of an emotional divide. Studies report as well as the detrimental effect on sexual function and relationships, premature ejaculation may lead to reduced quality of life, mental distress, anxiety and depression

## TOPIC: Haematospermia

**Definition**

- Haematospermia is the presence of blood in seminal fluid.
    - Blood may be visible or identified incidentally (for example in semen analysis during fertility investigations).
    - The blood appears bright red (with or without clots) when bleeding has occurred recently and darker red/brown if older.
- Haematospermia can occur as a single episode, or repeatedly over time.

**Causes**

- Haematospermia may be associated with pathology of the prostate gland, seminal tract (seminal vesicles, vasa deferentia, and ejaculatory ducts), verumontanum, urethra, urinary bladder, epididymis, or testes.
- The most common causes are:
    - Iatrogenic for example following a recent urological procedure such as prostate biopsy or radiotherapy, cystoscopy or vasectomy.
    - Urogenital infection (the most common cause in men under 40 years of age) including:
        - Sexually transmitted infections such as chlamydia, gonorrhoea and herpes simplex.
        - Organisms associated with urinary tract infection such as Escherichia coli, Proteus mirabilis and Enterobacter.
        - Rarer infective cause such as tuberculosis, and schistosomiasis.
- Other causes include:
    - Trauma for example, coital trauma or injury of the pelvis, perineum or genitals.
    - Malignant tumours of the prostate, bladder, urethra, testes/epididymis, or seminal vesicles.
    - Prostatic conditions such as acute/chronic prostatitis, benign prostatic hyperplasia, prostatic calculi, and prostatic cysts.
    - Testicular or epididymal conditions such as orchitis or epididymitis.
    - Seminal vesicle/ejaculatory duct conditions such as cysts or calculi, seminal vesiculitis, or ductal obstruction. Improved imaging techniques have confirmed that one of the main sites of bleeding is the seminal vesicles.
    - Systemic disorders such as severe uncontrolled hypertension, bleeding disorders, lymphoma, leukaemia, amyloidosis and severe liver disease.
    - Medications such as anticoagulants.
    - Vascular malformations (rare).
- Most cases of haematospermia have a benign cause.
- In men younger than 40 years of age:
    - The most common cause is infection (urinary tract infection or sexually transmitted infection).

- In men of 40 years of age or older:
    - Malignancy is a more common cause of haematospermia than in men under 40 years old.

**Prognosis**

- Most cases of haematospermia are benign and self-limiting.
- Recurrent or persistent haematospermia, and/or specific associated symptoms increase the chance of malignancy as an underlying pathology.
- The majority of men referred to urology for further assessment do not have an underlying cancer.

**How should I assess a person with haematospermia?**

- Note the person's age. Men older than 40 years of age with haematospermia are at increased risk of a serious underlying cause.
- Ask about:
    - The duration and frequency of haematospermia.
        - Prolonged and/or recurrent episodes increase the risk of a serious underlying cause.
    - The colour of the semen.
        - Bright red (with or without clots) indicates fresh bleeding, while older blood may appear brown.
    - Associated symptoms.
        - Symptoms that can indicate a urinary tract infection include dysuria, frequency, urgency, visible haematuria, and pain in the abdomen, scrotum, pelvis, or perineum.
        - Perineal, penile, rectal or suprapubic pain associated with symptoms of a urinary tract infection may indicate acute or chronic prostatitis.
        - Painless haematuria may indicate an underlying urological cancer.
        - The presence of urethral discharge may indicate a sexually transmitted infection.
        - Urinary hesitancy, frequency, or dribbling may indicate benign prostatic hypertrophy or prostate cancer.
        - Unilateral testicular enlargement, swelling, change in shape or discomfort may indicate a testicular cancer.
    - Recent urological investigations or interventions.
    - Past medical history including hypertension, prolonged or excessive bleeding, easy bruising from minor injuries or surgical procedures, or personal or family history of coagulopathy.
    - Sexual history including risk of sexually transmitted infections.
    - Travel history, particularly to areas where tuberculosis and schistosomiasis are endemic.
    - Medication including anticoagulants and antiplatelets.
- Examine the person:

    - Measure blood pressure for signs of uncontrolled hypertension and temperature to check for fever.
    - Palpate the abdomen (to identify masses such as hepatosplenomegaly, which could indicate an underlying haematological, liver, or infectious disease) and local lymph nodes (looking for lymphadenopathy).
    - Examine the urethral meatus looking for trauma, discharge and skin lesions. Palpate the testes to detect lumps, scrotal tenderness, or swelling.
    - Consider performing a digital rectal examination to detect prostatic enlargement, tenderness, fluctuance, firmness, or nodules.
        - Use clinical judgement to determine whether this is necessary, taking into account factors such as the person's age, family history of prostate cancer (particularly in first degree relatives), and ethnicity (incidence of prostate cancer is higher in men of black ethnicity).
    - Look for signs of easy bruising or bleeding tendencies, such as petechiae and large bruises in the absence of trauma
- Carry out a urinalysis and send a mid-stream urine sample for culture and analysis in all people with haematospermia.
- Measure prostate specific antigen (PSA) level in all men aged over 40 years and men of any age with signs and symptoms, or a family history of prostate cancer.
- Consider other investigations guided by clinical findings, including:
    - A full blood count, coagulation screen, and renal and liver function tests.
    - Semen microscopy, culture and sensitivity if infection (for example tuberculosis or schistosomiasis) is suspected.
    - Scrotal ultrasound if there is testicular swelling or discomfort.

**How should I manage a person with haematospermia?**

- If the person is under 40 years old, and no underlying cause is identified by initial assessments and investigations:
    - Reassure them that a single episode is likely to be benign and self-limiting.
    - Ask them to return if the problem recurs.
- Refer the person to a genitourinary medicine (GUM) clinic if a sexually transmitted infection is suspected.
    - This will facilitate identification of the infectious agent, and enable appropriate contact tracing.
- If a urinary tract infection is suspected,
    - Treat with antibiotics based on the results of culture and sensitivity tests, and according to local prescribing guidelines.
- If the person has hypertension, treat appropriately.
- If acute or chronic prostatitis is suspected:
- Refer to a urologist:
    - Men of any age with signs and symptoms suggestive of prostate cancer (such as elevated PSA levels or suspicious findings on digital rectal examination), and men or boys with signs or symptoms suggestive of testicular cancer or other urological malignancies.

        - Refer using a suspected cancer pathway referral (for an appointment within 2 weeks).
    - All men aged over 40 years with no identifiable cause for haematospermia found in primary care.
    - Men and boys of any age:
        - Who have recurrent or persistent haematospermia, where no identifiable cause has been found in primary care.
        - Who continue to experience haematospermia despite treatment of a suspected underlying cause identified in primary care.
        - Where initial investigations have suggested that the underlying cause of haematospermia may be cysts or calculi of the prostate or seminal vesicles.
- Refer to an appropriate specialist:
    - If the person has signs and symptoms suggestive of a haematological malignancy such as lymphoma or leukaemia.
        - Refer using a suspected cancer pathway referral (for an appointment within 2 weeks).
    - If an acquired bleeding disorder, tuberculosis, or schistosomiasis is suspected.
- Reassure the person that most causes of haematospermia have no effect on fertility. The exception to this is sexually transmitted infections such chlamydia or gonorrhoea.
- Reassure men who have had a recent prostate procedure that any associated haematospermia should resolve within three to four weeks.

## TOPIC: Priapism

Priapism is defined as a persistent penile erection lasting longer than 4 hours which is not associated with sexual stimulation.

**Classification:**

- Ischemic priapism is typically due to impaired vasorelaxation and therefore reduced vascular outflow resulting in congestion and trapping of de-oxygenated blood within the corpus cavernosa.
- Non-ischemic priapism is due to high arterial inflow, typically due to fistula formation often either as the result of congenital or traumatic mechanisms.

**Causes**

- Idiopathic
- Sickle cell disease or other haemoglobinopathies
- Erectile dysfunction medication (e.g. Sildenafil and other PDE-5 inhibitors), this also includes intra-cavernosal injected therapies.
- Other drugs both prescribed (anti-hypertensives, anticoagulants, antidepressants etc.) and recreational (specifically cocaine, cannabis and ecstasy).
- Trauma

**Patients typically present acutely with:**

- A persistent erection lasting over 4 hours
- Pain localised to the penis
- Patients may, more rarely, present with either a non-painful erection or an erection that is not fully rigid: these are both suggestive of non-ischaemic priapism.
- History of trauma to the genital or perineal region: also suggestive of non-ischemic priapism.

Investigations:

- Cavernosal blood gas analysis to differentiate between ischaemic and non-ischaemic: *in ischaemic priapism pO2 and pH would be reduced whilst pCO2 would be increased.*
- Doppler or duplex ultrasonography: this can be used as an alternative to blood gas analysis to assess for blood flow within the penis.
- A full blood count and toxicology screen can be used to assess for an underlying cause of the priapism.
- Diagnosis of priapism is largely clinical, with investigations helping to categorise into ischemic and non-ischemic as well as assessing for the underlying cause.

**Management:**

- Ischemic priapism is a medical emergency and delayed treatment can lead to permanent tissue damage and long-term erectile dysfunction.

- If the priapism has lasted longer than 4 hours, the first-line treatment is aspiration of blood from the cavernosa, this is often combined with injection of a saline flush to help clear viscous blood that has pooled.
- If aspiration and injection fail, then intra-cavernosal injection of a vasoconstrictive agent such as phenylephrine is used and repeated at 5-minute intervals.
- If medical therapy fails then surgical options can be considered.

- Non-ischemic priapism is not a medical emergency and is normally suitable for observation as a first-line option.

## TOPIC: Vasectomy

Vasectomy is a permanent surgical procedure for male sterilisation via excision a segment from vas deferens.

**Operation:**

- Simple operation, usually peformed as outpatient, usually performed under local anesthesia

**complications:**

- Bruising, infection, sperm granuloma, hematoma (most common side effect)
- chronic testicular pain (affects between 5-30% men)

**Failure rate:**

- 1 in 2000, might be due to technical errors, recanalization or unprotected intercourse prior to the confirmation of azoospermima

**Vasectomy Reversal:**

- The success rate of vasectomy reversal is up to 55%, if done within 10 years, and approximately 25% after more than 10 years
- It is important for the patient to understand that this is a permanent procedure even if there was a chance to reverse it

**Post-vasectomy follow-up:**

- It is important to inform the patient that vasectomy does not provide immediate sterilisation
- semen analysis needs to be performed twice following a vasectomy before a man can have unprotected sex (usually at 12 weeks post-vasectomy and preferably with at least 20 ejaculations since vasectomy)
- Vasectomy does not protect from STI and proper protection is needed

**Post-operation care:**

- Should not drive immediately post-operatively, hence a transportation should be arranged
- Scrotal support (at least 48 hours)
- Intermittent ice application (at least 48 hours)
- Rest and gentle activity (24 hours)
- Restrain from heavy activity such as lifting (at least 24 hours)
- Avoid sexual activity for at least a week
- Use an additional method for contraception until azoospermia is confirmed

**POST-Vasectomy Pain Syndrome:**

- Some patient might complain of dull ache in the testes that worsens with ejaculation
- The causes of the syndrome are chronic congestive epididymitis, nerve entrapment or sperm granuloma
- It should be managed by a urologist
- First line of treatment is NSAIDs and warm baths
- Second line of treatment is local nerve block or steroid injection

## TOPIC: Circumcision

Circumcision has been performed in a variety of cultures and mainly people of the Jewish and Islamic faith who undergo circumcision for religious/cultural reasons.

**Prior to circumcision**

- Exclude hypospadias as the foreskin may be used in surgical repair.

**Procedure**

- Circumcision may be performed under a local or general anesthesia (mainly adults).

**The medical benefits of routine circumcision remain controversial some evidence has emerged that it:**

- reduces the risk of penile cancer
- reduces the risk of UTI
- reduces the risk of acquiring sexually transmitted infections including HIV

**Medical indications for circumcision**

- phimosis
- recurrent balanitis
- balanitis xerotica obliterans
- paraphimosis

## TOPIC: Lacerations

- A laceration is a deep cut or tear in the skin and/or underlying tissue, commonly caused by blunt trauma (such as a fall or collision), incision by a sharp object, or mammalian bite. Infection, is the most common complication of a laceration. However, there may be injuries to deeper structures, such as the nerves, blood vessels, muscles, bones, and/or tendons. The risk of infection is high in people with a laceration contaminated with soil, faeces, body fluids, or pus. Other risk factors for infection include diabetes, increasing age, stellate shape or jagged wound margins, and wound length of more than 5 cm.
- Assessment of a person with a laceration involves:
    - Obtaining a detailed history, including the time and cause of the injury, as well as the person's past medical history, allergies, and tetanus vaccine history.
    - Examining the size, shape, and location of the wound.
    - Assessing for injuries to deeper structures.
    - Assessing for signs or symptoms of infection.
- Primary care management of a laceration involves:
    - Cleaning, closing, and dressing wounds at low risk of infection. Infected wounds or wounds at high risk of infection should be cleaned and dressed initially; the wound can be closed after 2–5 days if there are no signs or symptoms of infection.
    - Considering the need for antibiotics to treat, or reduce the risk of, infection. A swab of the wound should be taken before starting antibiotic treatment.
    - Considering the need for tetanus prophylaxis for susceptible wounds, such as those exposed to soil or other external contaminants.
    - Giving appropriate information and advice to the person, including that they should take paracetamol or ibuprofen for pain relief; keep the wound clean and dry to reduce the risk of infection; and seek medical attention if they develop signs or symptoms of infection.
    - Arranging appropriate follow-up, for example, to close a wound or remove wound closure.
- Hospital admission should be arranged if the person:
    - Has a severe infection or is systemically unwell.
    - Has signs or symptoms of tetanus (generalized rigidity and spasm of skeletal muscles, including lockjaw).
- Referral to A&E is recommended if:
    - There is possible vascular, nerve, tendon, or bone damage.
    - It is a facial laceration, particularly if it crosses the margins of the lips, nose, or ears.
    - It is a laceration of the palm of the hand with any sign of infection.
    - There is associated cellulitis over a joint.
    - There is significant contamination of the wound, it is a high-risk tetanus prone wound
    - There is a possible foreign body remaining in the wound after cleaning, including all injuries caused by glass.
    - The laceration is complex, widely gaping, or extensively devitalized.
    - There is significant tissue loss or inability to primarily close wound edges.
    - Local anaesthesia is unsuitable, for example, in children or people with needle phobia.

## Local anesthetic (LA)

Lidocaine is the most widely used LA. It has a rapid onset of action and anesthesia lasts for around 1 hour.

- the maximum safe dose is 3mg/kg. The BNF states 200mg (or 500mg if given in solutions containing adrenaline), which equates to 3mg/kg for a 66kg patient. This is the equivalent of 20ml of 1% solution or 10ml of 2% solution
- lidocaine is available pre-mixed with adrenaline. This increases the duration of action of lidocaine and reduces blood loss secondary to vasoconstriction. It must never be used near extremities due to the risk of ischemia

## Suture material

| Non-absorbable | Absorbable |
|---|---|
| Silk<br>Novafil<br>Prolene<br>Ethilon | Vicryl<br>Dexon<br>PDS |

Non-absorbable sutures are normally removed after 7-14 days, depending on the location. Absorbable sutures normally disappear after 7-10 days. Removal times for non-absorbable sutures are shown below:

| Area | Removal time (days) |
|---|---|
| Face | 3 - 5 |
| Scalp, limbs, chest | 7 - 10 |
| Hand, foot, back | 10 - 14 |

TOPIC: Leg ulcer – venous

**Definition**

- A leg ulcer is a break in the skin below the knee which has not healed within 2 weeks.
- A venous leg ulcer occurs in the presence of venous disease and is the most common type of leg ulcer, accounting for 60–80% of cases. It typically occurs in the gaiter area of the leg (from the ankle to mid-calf).
    - Other causes of leg ulceration and/or delayed wound healing include arterial disease, diabetes, rheumatoid arthritis, vasculitis, sickle cell disease, malignancy, and drugs (such as nicorandil, corticosteroids, and nonsteroidal anti-inflammatory drugs).

**Causes and risk factors**

- Venous leg ulceration is caused by sustained venous hypertension, which results from chronic venous insufficiency due to venous valve incompetence or an impaired calf muscle pump.
    - In the normal venous system, venous pressure decreases with exercise as a result of the action of the calf muscle pump. When the calf muscles relax, the valves in the perforating veins (which connect the superficial veins to the deep veins) prevent reflux of blood and the pressure remains low.
    - If there is reflux of blood due to valve damage or calf muscle pump impairment, the venous pressure remains high, resulting in enlarged veins, oedema, and venous skin changes (hyperpigmentation, venous eczema, lipodermatosclerosis, and atrophie blanche). As the condition of the skin and subcutaneous tissue worsens, it becomes increasingly vulnerable to ulceration.
- Risk factors for developing venous leg ulcers include:
    - Increasing age, obesity, immobility, previous ulcer, female sex
    - Limited range of ankle function, multiple pregnancies, sedentary lifestyle.
    - Personal or family history of varicose veins.
    - Personal history of deep vein thrombosis.
    - Arteriovenous fistula, prolonged standing
    - History of leg fracture or trauma.

**Complications**

- Complications of venous leg ulcers include:
    - Chronic pain.
    - Impaired mobility (often due to pain).
    - Infection, including osteomyelitis and septicaemia.
    - Allergic contact dermatitis (caused by allergens in creams and dressings, such as preservatives, emulsifiers, latex, resins, lanolin, and topical steroids, antibiotics, and antiseptics).
    - Malignant transformation in the ulcer bed (Marjolin's ulcer).
    - Sinus formation and fistula (uncommon).
    - Negative impacts on quality of life and daily functioning — time off work, psychological distress, loss of independence, and social isolation are common.

- Poor prognostic factors include:
    - Wound duration of more than 1 year — recurrence rate in these ulcers is more than 70%.
    - Larger wounds.
    - History of surgical treatment of varicose veins.
    - Ankle brachial pressure index (ABPI) of less than 0.8.
    - Impaired calf muscle pump — this reduces ulcer healing rates.
    - Fibrin in more than 50% of wound surface.
    - Being from a lower socioeconomic group — ulcers take longer to heal and recurrence rates are higher in people from lower socioeconomic groups.
    - Poor compliance with treatment and lifestyle advice.

**How should I assess a person with a suspected venous leg ulcer?**

- Take a history.
    - Ask about:
        - The duration of the ulcer.
        - Symptoms of venous insufficiency, such as pain, heaviness, aching, swelling, and itching of the affected leg. Symptoms are typically worse at the end of the day and are relieved by leg elevation.
        - Risk factors for venous leg ulcers, such as immobility or a history of deep vein thrombosis.
        - Other possible causes of ulceration and/or delayed wound healing, including comorbidities (such as diabetes and rheumatoid arthritis) and the use of certain medications (such as nicorandil, corticosteroids, nonsteroidal anti-inflammatory drugs, and anticoagulants).
        - How symptoms are affecting the person's daily functioning, quality of life, and overall health.
        - A history of allergy, for example to latex or lanolin. This is useful for when the choice of appropriate dressing is being considered.
- Examine the person.
    - Assess the ulcer. Record the following details about the ulcer (take photographs if possible and appropriate) to aid diagnosis, guide referral, and help assess ulcer healing:
        - Site — venous leg ulcers typically occur in the gaiter area of the leg (ankle to mid-calf). Assess and note the position of the ulcer (medial, lateral, anterior, posterior, or a combination).
        - Wound edge — venous leg ulcers typically have gently-sloping, irregular edges. 'Punched out', rolled, or everted edges should raise the suspicion of an alternative diagnosis.
        - Size — assess the size of the wound at first presentation and regularly thereafter. Trace out the outline of the wound margin on to transparent acetate sheets and estimate the surface area.
        - Depth — take approximate measurements of the greatest depth of the ulcer to assess progress, noting any exposed underlying structures (bone or tendon).

        - Appearance of the wound bed — look for granulation tissue (an indicator of wound healing), and slough (dead tissue, usually cream or yellow in colour) or necrotic tissue, which may need to be removed to allow healing.
        - Amount of exudate — excessive exudate may be due to wound infection or gross oedema in the wound area. The amount of exudate will help determine which dressing is needed.
    - Assess for signs of infection — look for cellulitis (characterized by pain, warmth, swelling, and erythema of the infected area), fever, increased pain, rapid extension of the area of ulceration, malodour, and increased exudate.
    - Examine both legs for signs of venous insufficiency, including:
        - Pitting oedema — rule out non-venous causes of oedema, such as heart failure and chronic kidney disease.
        - Skin changes — look for hyperpigmentation (brown-red discolouration caused by haemosiderin deposition), venous eczema (itchy, red, scaly, and/or flaky skin which may have blisters and crusts on the surface), lipodermatosclerosis (painful, hardened, tight skin), and atrophie blanche (star-shaped, ivory-white, depressed, atrophic scars with surrounding pigmentation).
    - Examine the person (lying and standing) for evidence of varicose veins.
    - Check joint mobility, particularly that of the ankle, which is an important component of calf muscle pump function.
    - Assess for peripheral arterial disease, which may coexist with venous ulceration.
        - Look for hair loss, pallor, coldness on palpation, and dusky appearance on lowering the leg.
        - Assess capillary refill — greater than 4 seconds is suggestive of arterial insufficiency.
        - Check peripheral pulses (femoral, popliteal, foot) and the abdomen for signs of an abdominal aneurysm.
- Arrange investigations.
    - Arrange a Doppler assessment of both legs to determine the ankle brachial pressure index (ABPI) to exclude arterial insufficiency.
    - Consider the need for additional investigations, such as:
        - Full blood count — presence of anaemia may delay healing; high white blood cell count and platelet count may indicate infection.
        - Erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP) — markers for inflammation and infection.
        - Urea and creatinine — high urea levels may indicate dehydration, which may impair healing.
        - Albumin — low albumin may be associated with protein loss and malnutrition, which may delay healing.
        - HbA1c — to assess for diabetes mellitus.
        - Other investigations to rule out other possible causes of ulceration and delayed wound healing.

- Bacteriological swabs should only be taken where there is clinical evidence of infection.

**What else might it be?**

- Exclude the following conditions in the initial assessment of a person with a suspected venous leg ulcer:
  - Peripheral arterial disease — ulcers due to arterial disease often appear well demarcated and have a 'punched out' appearance. There may be signs of arterial compromise, such as pallor, loss of hair, nail dystrophy, coldness, and diminished capillary refill. A history of intermittent claudication, cardiovascular disease, or stroke may indicate the presence of arterial disease.
  - Rheumatoid arthritis — rheumatoid ulcers may be venous, arterial, or vasculitic. Vasculitic ulcers can occur on the calf or dorsum of the foot and are typically deep and well demarcated with a 'punched out' appearance. People with vasculitic ulcers will have clinical features of established disease which may be associated with systemic vasculitis.
  - Systemic vasculitis — people with systemic vasculitis often develop multiple leg ulcers that are necrotic and deep. There is usually an atypical distribution and evidence of vasculitic lesions elsewhere (for example nail-fold infarcts and splinter haemorrhages). Associated conditions include systemic lupus erythematosus, scleroderma, polyarteritis nodosa, and granulomatosis with polyangitis (previously known as Wegener's granulomatosis).
  - Diabetes mellitus — diabetic ulcers may be venous, arterial, or neuropathic. Neuropathic ulcers typically develop under calluses or over pressure points, such as metatarsal heads, sole of foot, and balls of toes. They are often irregular and correspond to the shape of the pressure point that has become exposed. The edge and base are often clean and may be deep, with exposure of bone and tendon. People with diabetes may also have diabetic bullae or necrobiosis lipoidica, both of which can ulcerate.
  - Malignancy — malignant ulcers often occur in sun exposed areas. The ulcer edge may be rolled (basal cell carcinoma) or everted (squamous cell carcinoma). Malignancy should be considered if the ulcer does not respond to medical treatment or if the appearance is unusual.
- Other types of lower limb ulcers include:
  - Pressure ulcers — caused by unrelieved pressure over bony prominences, such as the malleolus or the heel. Prolonged compression of the tissues, along with friction and shear, results in local tissue ischemia, necrosis, and ulcer formation.
  - Drug/treatment-induced ulcers — certain drugs/treatments can:
    - Cause ulceration, for example nicorandil, hydroxycarbamide, and radiotherapy.
    - Delay wound healing, for example hydroxycarbamide, immunosuppressive drugs, antipsychotics, beta-blockers, nicorandil, corticosteroids, nonsteroidal anti-inflammatory drugs (NSAIDs), anticoagulants, and radiotherapy.
    - Increase oedema, for example calcium channel blockers (such as amlodipine) and NSAIDs.
    - Affect surrounding skin, for example systemic and topical corticosteroids, and radiotherapy.

    - Others — tropical ulcers, tuberculosis ulcers, syphilitic ulcers, and ulcers caused by sarcoidosis, pyoderma gangrenosum, livedoid vasculopathy, and sickle cell disease.

**Overview of management of a person with a venous leg ulcer**

- Assess the need for immediate referral.
- If immediate referral is not indicated:
    - Ensure that compression therapy is started immediately if appropriate.
        - The ulcer should be cleaned and dressed appropriately before compression is applied.
        - Compression should only be applied by trained healthcare professionals and in accordance with the manufacturer's instructions.
    - Consider prescribing pentoxifylline depending on local prescribing policies, to aid ulcer healing.
        - Pentoxifylline is an effective adjunct to compression bandaging for treating venous leg ulcers (off-label indication) and may be effective in the absence of compression.
    - Assess for (and manage):
        - Associated conditions, such as oedema and venous eczema.
        - Complications of the ulcer, such as pain and infection.
    - Provide information and lifestyle advice to promote ulcer healing and reduce the risk of recurrence, and sources of further information.
    - Review the person appropriately.
- After a venous leg ulcer has healed, offer below-knee graduated compression stockings to prevent recurrence of ulceration.

**When should I refer a person with a venous leg ulcer?**

- Refer to a specialist if:
    - There is diagnostic uncertainty — refer for specialist vascular assessment (to a leg ulcer clinic, tissue viability clinic, or a vascular specialist depending on clinical judgement and availability) or to dermatology.
    - The ulcer is rapidly deteriorating or has an atypical location (outside the gaiter area) and/or appearance — refer to dermatology for a possible skin biopsy.
    - There is a suspected alternative cause of ulceration, such as:
        - Arterial or mixed venous/arterial disease.
        - Malignancy — refer to dermatology using the suspected cancer pathway referral (for an appointment within 2 weeks).
        - Diabetes mellitus — refer to the diabetes ulcer clinic.
        - Rheumatoid arthritis or systemic vasculitis — refer to rheumatology.
    - The person has poor ankle mobility, reduced joint function, or a history of falls.
    - There is suspected iliac vein stenosis — these people may need investigation by CT venography and intravenous ultrasound.
- After compression therapy is initiated, arrange specialist referral if:
    - There is a complication related to the ulcer or the treatment, including:

        - Suspected contact dermatitis — refer to dermatology for consideration of patch testing.
        - Osteomyelitis (suggested by fever, disproportionate pain, local malodour, non-healing sinus, and oedema around wound) — arrange urgent hospital admission for intravenous antibiotics.
        - Sepsis (suggested by tachycardia, fever, and chills) — arrange urgent hospital admission for intravenous antibiotics.
        - Necrotising fasciitis (suggested by disproportionate pain and tenderness, dusky-violaceous areas along with erythema, crepitus, blisters, and necrosis) — arrange urgent hospital admission for intravenous antibiotics.
        - Uncontrolled pain — refer to a specialist pain team.
    - There is delayed or no healing after 2 weeks of compression therapy — refer to a vascular specialist or dermatologist depending on clinical judgement.
- If the ulcer is recurrent, refer to a vascular specialist for consideration for superficial venous surgery.

**How should a venous leg ulcer be cleaned and dressed?**

A healthcare professional with expertise in wound management (such as a district nurse or tissue viability nurse) should be involved in the cleaning and dressing of a venous leg ulcer.

- Ensure that the ulcerated leg is washed normally in tap water and carefully dried.
    - Strict aseptic technique is not required, as the aim is not to remove surface bacteria but rather to avoid cross infection from contamination.
    - Debridement of slough or necrotic tissue from the wound surface may facilitate wound healing.
- Ensure that simple non-adherent dressings are used in the management of venous leg ulcers.
    - The choice of dressing should be based on factors such as the properties of the wound (location, size, depth, exudate level, presence of infection), the stage of the healing process, the condition of the surrounding skin, the person's preference or tolerance (including allergies and concurrent medical conditions), and cost.
        - The Wound management section of the British National Formulary (BNF) provides information on the types and properties of different dressings, and a table that suggests choices of primary dressing depending on the nature of the wound.
        - Antimicrobial dressings (for example silver, iodine, or honey dressings) should not be used routinely.
        - There is no evidence to support regular application of topical antimicrobials or antiseptics.
    - The frequency of dressing changes depends on the exudate volume and the stage of ulcer healing.
        - Once compression therapy has begun, dressing changes should lessen as exudate reduces. Dressing changes may only be needed weekly; the frequency should be tailored so that the dressing is neither oversaturated nor dry and adherent to the wound when it is removed.
        - Dressings are typically used continually until the wound has healed. Occasionally, they may be used after healing to protect fragile scar tissue.

    - The minimum quantity of dressings necessary to meet the person's needs should be provided to avoid wastage and stockpiling.
- If there is pain at dressing changes:
    - Ensure an appropriate dressing is used and that the frequency of dressing changes is minimized.
    - Advise the person to take a simple analgesia (such as paracetamol with or without codeine) at least 30 minutes before a dressing change, where possible.
    - Manage neuropathic pain if present.

**How should compression therapy be used to treat a venous leg ulcer?**

Compression therapy should be applied by a healthcare professional trained in venous ulcer management (such as a district nurse or tissue viability nurse). Before starting compression therapy, the ankle brachial pressure index (ABPI) in both legs should be measured to exclude arterial insufficiency.

- Ensure that compression therapy is started immediately if appropriate.
    - The Scottish Intercollegiate Guidelines Network (SIGN) recommends high compression multilayer compression bandaging for the treatment of venous leg ulcers.
        - The person should be offered the strongest compression that they can tolerate, and assessed for skin complications within 24–48 hours of initiation of compression.
        - Multilayer compression bandaging systems are commercially available as kits. Compression is achieved by the combined effects of two or three extensible bandages applied over a layer of orthopaedic wadding and a wound contact dressing.
        - Different kits are available comprising slightly different components for different ankle sizes.
    - Evidence suggests that compression hosiery is as effective as bandaging in certain people.
- Encourage compliance with compression therapy by:
    - Ensuring that the person understands the reasons for, and the benefits of, compression therapy.
    - Managing associated conditions, such as oedema and venous eczema.
    - Managing complications of the ulcer, such as pain and infection.
    - Addressing the person's concerns and offering options that suit their preferences. When deciding on the type of compression to use, the following should be considered:
        - The person's preference, lifestyle, and likely concordance.
        - The required frequency of application.
        - The size and shape of leg.
- Give general lifestyle advice to promote ulcer healing, and provide sources of further information.

**How should I prescribe pentoxifylline?**

- Consider prescribing pentoxifylline to increase microcirculatory blood flow and improve ulcer healing.

    - Pentoxifylline is an effective adjunct to compression therapy for treating venous leg ulcers (off-label indication) and may be effective in the absence of compression.
    - The recommended dosage is 400 mg three times daily for up to 6 months, depending on local prescribing policies.

**How should I manage infection in a person with venous leg ulcer?**

If there are symptoms or signs of infection, such as cellulitis (characterized by pain, warmth, swelling, and erythema of the infected area), fever, increased pain, rapid extension of the area of ulceration, malodour, and increased exudate:

- Consider taking a swab (from deep within the ulcer after cleaning) for microbiological testing.
- Prescribe an oral antibiotic.
    - For most people, prescribe flucloxacillin 500 mg four times daily for 7 days.
    - If the person is penicillin-allergic or flucloxacillin is unsuitable, consider one of the following options:
        - Clarithromycin 500 mg twice a day for 7 days.
        - Doxycycline 200 mg on first day, then 100 mg once a day for 6 days (7-day course in total).
        - Erythromycin 500 mg four times daily for 7 days (in pregnant women).
    - Routine long-term use of topical antiseptics and antimicrobials is not recommended.
- Advise the person to seek medical help if symptoms of the infection worsen rapidly or significantly at any time, or do not start to improve within 2-3 days.
- Compression therapy can be continued if tolerated. Otherwise, it can be temporarily reduced or stopped to ease pain and then returned to usual compression as soon as possible.
    - Uncontrolled exudate and oedema will prolong the cellulitis, so the aim is to return to therapeutic compression as soon as possible.
- Reassess the person if the infection worsens rapidly or significantly at any time, does not start to improve within 2–3 days, or the person becomes systemically very unwell or has severe pain out of proportion to the infection.
    - When reassessing the person, take account of:
        - Any symptoms or signs suggesting a more serious illness or condition, such as sepsis (tachycardia, fever, and chills), necrotising fasciitis (disproportionate pain and tenderness, dusky-violaceous areas along with erythema, crepitus, blisters, and necrosis), osteomyelitis (fever, disproportionate pain, local malodour, non-healing sinus, and oedema around wound), or lymphangitis (fever, chills, swollen lymph glands, malaise, and aches).
        - Other comorbidities, such as diabetes or immunosuppression.
        - Possible complications, including allergic contact dermatitis (which may cause infection).
        - Whether uncontrolled oedema or poor compliance with compression therapy contributed to the cellulitis, and whether any changes to compression therapy are required.
        - Previous antibiotic use, which may have led to resistant bacteria.

    - If the infection is worsening or not improving, check swab results and change the antibiotic based on swab sensitivities, using a narrow-spectrum antibiotic if possible. Consider one of the following options (guided by microbiological results):
        - Co-amoxiclav 500/125 mg three times a day for 7 days, or
        - Co-trimoxazole 960 mg twice a day for 7 days (in people with penicillin allergy).
    - If the infection is sensitive to the first-line antibiotic but only slowly responding, review the person after 7 days and consider continuing the antibiotic for a further 7 days.
- Arrange urgent hospital referral if the person has any symptoms or signs suggesting a more serious illness or condition, such as sepsis, necrotising fasciitis, or osteomyelitis.
- Consider referring or seeking specialist advice if the person:
    - Has a high risk of complications.
    - Has lymphangitis.
    - Has spreading infection that is not responding to oral antibiotics.
    - Cannot take oral antibiotics (where appropriate, explore locally available options for giving intravenous antibiotics at home or in the community, rather than in hospital).

**How should I manage pain in a person with a venous leg ulcer?**

- Determine the nature of the pain to help exclude an underlying cause.
    - Ask about the location, severity, duration, frequency (and when it occurs), triggers and relievers, and the impact of the pain on the person's quality of life and functional ability.
        - Venous leg ulcers can sometimes cause constant or intermittent pain.
        - Severe or worsening pain may indicate a complication, such as an infection.
        - Pain can also be caused by arterial disease, diabetic neuropathy, vascular structures (superficial, deep phlebitis), pitting oedema, lipodermatosclerosis, and contact dermatitis.
        - Changing dressings, debridement, and firm bandages can also cause pain.
    - Consider using a validated pain tool, such as the Visual Analogue Scale (VAS), to assess pain.
- Advise the person:
    - To take a simple analgesic, such as paracetamol with or without codeine, as needed.
        - Nonsteroidal anti-inflammatory drugs (NSAIDs) are not recommended for pain control because they impair wound healing and may worsen leg oedema.
    - That leg elevation may help with the pain associated with oedema.

**How should I manage oedema in a person with venous leg ulcer?**

- Check compliance with compression therapy.
- Advise the person to:
    - Elevate their legs (above hip level) for 30 minutes three to four times a day.
    - Consider placing pillows under their feet and legs while sleeping.
- Exclude other causes of oedema, such as:

    - Medication, for example, calcium-channel blockers and oral corticosteroids.
    - Comorbidities, such as heart failure and chronic kidney disease.
- If oedema is persistent or worsening, refer for specialist assessment.
    - Do not prescribe a diuretic for persistent or worsening oedema.

**How should I manage eczema in a person with venous leg ulcer?**

- If there is worsening venous eczema, assess for signs of infection, such as cellulitis (characterized by pain, warmth, swelling, and erythema of the infected area), fever, increased pain, rapid extension of the area of ulceration, malodour, and increased exudate.
    - If there are signs of infection.
    - If there are no signs of an infection:
        - Advise regular application of an emollient, and treat flares with a topical corticosteroid.
        - If compression bandaging is being used, consider replacing bandages more frequently than once weekly to allow application of topical treatment.
- If there is no clinical improvement or allergic contact dermatitis is suspected, refer the person to dermatology for consideration of patch testing.
    - Advise the person to avoid any allergens subsequently identified.
    - Common allergens that may be present in creams and dressings used for venous leg ulcer management include preservatives, emulsifiers, latex, resins, lanolin, and topical steroids, antibiotics, and antiseptics.

**How should I follow up a person with venous leg ulcer?**

Ideally, people with uncomplicated ulcers should be reviewed at least weekly for the first 2 weeks after initiation of primary care treatment. If the ulcer is healing, this period can be extended depending on clinical judgement.

- During the review:
    - Inspect and compare the ulcer with the initial assessment to see if there is evidence of healing, suggested by reduced ulcer size, development of healthy pink granulation tissue, reduced amounts of exudate, and improved symptoms of pain and oedema.
    - Check compliance with compression therapy — poor compliance may be due to heat, discomfort, and the impractical nature of the bandaging.
    - Check compliance with lifestyle advice.
    - Assess for (and manage):
        - Associated conditions, such as oedema and venous eczema.
        - Complications related to the ulcer, including pain and infection.
        - Complications related to the dressings applied, including skin maceration (indicated by a marked cut-off appearance) and allergic contact dermatitis. Maceration is usually caused by the inability of the dressing to manage exudate. More frequent dressing changes, or a change in dressing type, should be considered. The surrounding skin should also be protected with an emollient.
        - Complications related to compression bandaging, including pressure damage and arterial insufficiency. Compression bandages should be removed immediately if the person experiences a change in foot colour or

temperature, or increased pain. Seek further medical advice if there is no improvement after removing the bandages.

- Refer to a vascular specialist or dermatologist (depending on clinical judgement) if there is delayed or no healing after 2 weeks of compression therapy.
- There is no consensus among guidelines on the frequency of repeat ankle-brachial pressure index (ABPI) measurements.
    - Repeat ABPI measurements should ideally be done at the first signs of ulcer deterioration, ulcer recurrence, sudden increase in ulcer size or pain, or change in foot colour or temperature.

**How should I prevent recurrence of venous leg ulcer?**

- After a venous leg ulcer has healed, offer below-knee graduated compression stockings to prevent recurrence.
    - Advise the person that compression hosiery has been proven to reduce the rate of ulcer recurrence and that it is likely that compression will be required indefinitely.
- Give general lifestyle advice to reduce the risk of ulcer recurrence, and provide sources of further information.
- For people with chronic venous leg ulcer and superficial venous reflux, consider referring for superficial venous surgery to prevent recurrence.

**What information and advice should I give a person with a venous leg ulcer?**

- To promote healing of the ulcer and reduce the risk of recurrence:
    - Encourage the person to be compliant with compression therapy. Advise that:
        - Compression therapy has been proven to promotes venous leg ulcer healing and reduce the risk of recurrence.
        - If there are any difficulties, they should contact the district/tissue viability nurse before deciding to stop using them.
        - They should remove the compression immediately and seek urgent medical advice if they experience features of arterial disease, such as numbness, tingling, pain, or dusky toes.
    - Advise the person to:
        - Keep mobile with regular walking (to exercise calf muscle pump function).
        - Elevate the legs when immobile.
        - Avoid trauma and wear appropriate (well-fitting) footwear.
        - Use an emollient frequently, and avoid products that may contain any identified sensitizing agents (such as lanolin and preservatives).
        - Examine their legs regularly for broken skin, blisters, swelling, or redness.
        - Adopt a healthy lifestyle. They should:
            - Lose weight (if overweight or obese).
            - Eat a balanced diet (malnutrition impairs ulcer healing).
            - Drink alcohol within recommended levels.
            - Stop smoking (if applicable).

TOPIC: Pressure ulcers

- A pressure ulcer is defined as localized damage to the skin and/or underlying tissue, as a result of pressure or pressure in combination with shear. They usually occur over a bony prominence but may also be related to a medical device or other objects.
- Pressure ulcers can range in severity, from patches of discoloured skin to extensive wounds filled with necrotic tissue and involving fascia, muscle, and bone.
- Pressure ulcers most commonly occur over bony prominences but can develop on any part of the body, including mucosal surfaces.

**Causes**

Pressure ulcers are caused by pressure, or pressure in combination with shear forces:

- The duration and intensity of pressure are important: tissue damage may occur as a result of relatively short exposures to intense pressure or prolonged exposure to lower levels of pressure. Pressure forces are distributed throughout the soft tissue, the extent depending on the mechanical properties of both the soft tissues and any external support surface or device.
- Shear forces, typically generated when people slide down in a bed or chair, may cause stretching and tearing of small blood vessels and contribute to pressure-induced skin damage.

Friction and moisture can also contribute to the development of pressure ulcers:

- Friction can cause superficial injuries, including skin tears, which could facilitate the transmission of pressure to the deeper tissues.
- Moisture decreases the stiffness of the stratum corneum and increases the coefficient of friction so that skin is more adherent to the contact surface, resulting in greater shear forces being transmitted to deeper tissues. Additionally, moisture can cause moisture-associated dermatitis, which could also facilitate the transmission of pressure to the deeper tissues

Pressure ulcers can occur due to the forces of a person's body weight or as a result of externally exerted forces, such as those applied by a medical device and/or other objects.

When a person is immobilized for extended periods, for example, in the supine position, the pressure and shear forces cause tissue deformation, inflammatory oedema, and ischaemia, leading to pressure ulceration in bony anatomical sites, such as the sacrum, ischium, elbows, and heels.

Medical device-related pressure ulcers result from the use of devices designed and applied for diagnostic or therapeutic purposes (such as continuous positive airway pressure masks, nasogastric tubes, cervical collars, or splints). Localized forces from the device deform/damage the underlying skin and soft tissues, and the resultant pressure ulcer generally conforms to the pattern or shape of the device. Non-medical devices (such as bed clutter and furniture) can also result in pressure ulcers when they remain in contact with skin and tissues.

The tolerance of soft tissue for sustained deformations differs by tissue type and may also be affected by age, health status, microclimate, comorbidities, and conditions of the soft tissues. People with conditions that cause inadequate blood flow to the skin and soft tissues (such as peripheral vascular disease, heart failure, diabetes, and sepsis) are at increased risk of developing pressure ulcers.

**Risk factors:**

- Anyone is potentially at risk of developing a pressure ulcer. However, they are more likely to occur in the presence of the following risk factors:
  - Significantly limited mobility, for example, due to spinal cord injury.
  - Significant loss of sensation, for example, due to neurological impairment.
  - Significant cognitive impairment.
  - Poor posture and/or deformity.
  - Inability to independently reposition.
  - Nutritional deficiency.
  - A previous or current pressure ulcer.

- General skin status and skin moisture.
- Conditions that cause inadequate blood flow to the skin and soft tissues, for example, hypotension, diabetes, heart failure, and peripheral vascular disease.
- Older age, largely due to age-related problems, such as hip fractures, incontinence, dry skin, chronic systemic conditions, and terminal illness.
- Regular use of non-pressure-relieving support surfaces, such as chairs or beds, that are not specifically designed to provide pressure relief.
- Regular use of medical or other devices.

**Prognosis**

The prognosis of a pressure ulcer is determined by the severity of the wound and the presence of ongoing risk factors.

- In people with superficial damage, the prognosis is reasonable if appropriate local wound care is provided.
- In seriously ill people with extensive ulcers, the prognosis is poor, especially because they may be unable to tolerate or accept the intensive treatment required to close their wound surgically.

**Complications**

- Pain and distress.
- Infection (for example cellulitis or osteomyelitis), which can lead to systemic infection or sepsis.
- Longer hospital stays (an average of 5–8 days per pressure ulcer).
- Increased mortality and morbidity.
- Reduced quality of life.

**Diagnosis**

**Risk assessment:**

**How should I assess a person's risk of developing a pressure ulcer?**

• Anyone is potentially at risk of developing a pressure ulcer.

Carry out an assessment of pressure ulcer risk for:

- All people being admitted to secondary care or to care homes in which NHS care is provided.
- All people receiving NHS care in other settings (such as primary and community care, and emergency departments) if they have a risk factor for developing pressure ulcers, such as significantly limited mobility or neurological impairment.

Consider carrying out an assessment of pressure ulcer risk for all people in their own homes or in social/nursing care settings.

Consider using a validated risk assessment scale to support clinical judgement when assessing pressure ulcer risk.

**The recommended risk assessment scales include:**

- For adults: Braden Risk Assessment tool, Water-low score, Norton risk assessment scale, or the PURPOSE-T (Pressure Ulcer Risk Primary or Secondary Evaluation Tool).
- For children: the Braden Q Scale.

**People are considered to be:**

▪ At risk of developing a pressure ulcer if they are deemed to be at risk based on clinical judgement and/or the use of a risk assessment tool.

▪ At high risk of developing a pressure ulcer if multiple risk factors (for example significantly limited mobility, nutritional deficiency, inability to reposition themselves, or significant cognitive impairment) are identified during the risk assessment, with or without a validated risk assessment tool. People with a history of pressure ulcers or a current pressure ulcer are also considered to be at high risk.

• Document the findings of the risk assessment.

• Reassess pressure ulcer risk, particularly if there is a change in clinical status (for example after surgery, on worsening of an underlying condition, or with a change in mobility).

**Diagnosis and assessment:**

**How should I diagnose and assess a person with a pressure ulcer?**

Pressure ulcers should be diagnosed and assessed by healthcare professionals with the necessary skills and competencies.

Diagnose a pressure ulcer if a person with risk factors develops evidence of skin damage over a bony prominence.

The diagnosis of a pressure ulcer is supported by the presence of one of the following:

- An area of non-blanchable erythema. Be aware that non-blanchable erythema may present as colour changes or discolouration, particularly in darker skin tones or types.
- Marked localized skin changes.
- A wound of varying severity on an anatomical site that is known (or suspected) to have previously been exposed to significant unrelieved pressure.

Document:

- The surface area of the pressure ulcer, preferably using a validated measurement technique (such as a photograph).
- An estimate of the depth of the pressure ulcer and the presence of undermining. Do not routinely measure the volume of a pressure ulcer.

Categorize the pressure ulcer using a validated classification tool, such as the NPUAP–EPUAP Pressure Ulcer Classification System (published by the National Pressure Ulcer Advisory Panel and the European Pressure Ulcer Advisory Panel).

The NPUAP-EPUAP system classifies pressure ulcers into 6 categories/stages:

- ▪ Category/Stage I: non-blanchable erythema.
- ▪ Category/Stage II: partial thickness skin loss.
- ▪ Category/Stage III: full-thickness skin loss.
- ▪ Category/Stage IV: full-thickness tissue loss.
- ▪ Unstageable: depth unknown.
- ▪ Suspected deep tissue injury: depth unknown.

Use the classification to guide ongoing preventative strategies and management.

Repeat and document the classification each time the ulcer is assessed.

• Investigations are not used to make the diagnosis of pressure ulcers, However, they may be required if the diagnosis is uncertain or to exclude complications and/or differential diagnoses of pressure ulcers.

Wounds should only be swabbed if infection is suspected.

Consider or suspect abuse or neglect (particularly malnourishment, lack of repositioning, or poor continence management) as a contributory factor to (or cause of) pressure ulcers in children and adults.

Child abuse is common, can present in all care settings (including in the child's home), and may coexist with pressure ulcers.

The Department of Health and Social Care has issued a protocol for safeguarding adults at risk of pressure ulcers, which aims to help practitioners and managers across health and care organizations to provide caring and quick responses to people at risk of developing pressure ulcers.

**Differential diagnosis:**

- Diabetic foot ulcer.
- Moisture lesions.
- Osteomyelitis.
- Dermatological lesions, such as pyoderma gangrenosum and bullous pemphigoid.
- Malignancy.
- Peripheral arterial disease.

- Venous ulcers.

**Management**

**Scenario: Prevention of pressure ulcers**

**From birth onwards.**

**How should I prevent pressure ulcers in adults?**

Adults assessed as being at risk of pressure ulcers should be managed by healthcare professionals with the necessary skills and competencies.

Offer a skin assessment by a trained healthcare professional.
This should consider any pain or discomfort reported by the person, and the skin should be checked for:

- Skin integrity in areas of pressure.
- Colour changes or discolouration. Be aware that non-blanchable erythema may present as colour changes or discolouration, particularly in darker skin tones or types.
- Variations in heat, firmness, and moisture (for example due to incontinence, oedema, and dry or inflamed skin).

**Take into account:**

- The outcome of the risk assessment.
- The outcome of the skin assessment.
- Other comorbidities.
- The person's preferences.

Consider the person's mobility and their ability to reposition themself.

▪ Encourage the person to change their position at least every 6 hours if they have been assessed as being at risk of developing a pressure ulcer, or at least every 4 hours if assessed as being at high risk of developing a pressure ulcer.

▪ If they are unable to reposition themselves, they should be offered help to do so, using appropriate equipment if needed.

o Consider the need for pressure relief, including additional pressure relief at specific at-risk body sites (such as the heel).

▪ Recommend a high-specification foam mattress for people assessed as being at high risk of developing pressure ulcers.

▪ Consider the seating needs of people who are sitting for prolonged periods. Consider a high-specification foam or equivalent pressure redistributing cushion for people who use a wheelchair or who sit for prolonged periods.

▪ If the person is at high risk of developing a heel pressure ulcer, discuss (and document) a strategy to offload heel pressure.

• Consider prescribing a barrier preparation to prevent skin damage in people who are at high risk of developing a moisture lesion or incontinence-associated dermatitis (such as those with incontinence or oedema).

o If the skin is dry or inflamed, an emollient should be used.

• Do not offer:

- Skin massage or rubbing to prevent pressure ulcers.
- Subcutaneous or intravenous fluids specifically to prevent pressure ulcers in adults whose hydration status is adequate.
- Nutritional supplements specifically to prevent pressure ulcers in adults whose nutritional intake is adequate.

**Heel pressure ulcers**

• The heel is a common site for pressure ulcers to develop.

o Many interventions aim to reduce either the intensity or the duration of pressure, or the friction or sheer forces.
o Heel-specific pressure-reducing devices can be categorized as:
▪ Constant low-pressure (CLP) devices, such as gel or foam heel pads/cups and booties, that aim to distribute the pressure over a larger surface area.
▪ Offloading devices, such as pillows, wedges, or splints, that prevent contact between the heel and the bed.
▪ Low friction devices, such as dressings or booties, that reduce friction and sheer when the person moves their foot.
▪ Devices with combinations of functions, for example, prophylactic dressing or devices that reduce pressure and/or friction and shear, such as multilayer heel dressings and medical-grade sheepskin.
A clinical guideline by the European Pressure Ulcer Advisory Panel (EPUAP), National Pressure Injury Advisory Panel (NPIAP), and Pan Pacific Pressure Injury Alliance (PPPIA) makes the following recommendations:

- Assess the vascular/perfusion status of the lower limbs, heels, and feet when performing a skin and tissue assessment, and as part of a risk assessment.
- For individuals at risk of heel pressure injuries and/or with Category/Stage I or II pressure injuries:
  - Elevate the heels using a specifically designed heel suspension device or a pillow/ foam cushion.
  - Offload the heel completely in such a way as to distribute the weight of the leg along the calf without placing pressure on the Achilles tendon and the popliteal vein.

For individuals with a Category/Stage III or greater heel pressure injury:

- Elevate the heels using a specifically designed heel suspension device, offloading the heel completely in such a way as to distribute the weight of the leg along the calf without placing pressure on the Achilles tendon and the popliteal vein.
- Use a prophylactic dressing as an adjunct to heel offloading and other strategies to prevent heel pressure injuries.
  - Parafricta® garments (Bootees and Undergarments) are designed specifically to reduce friction and associated shear, thereby reducing the potential for pressure ulcer and friction lesion development. Bootees protect the heel and ankle; Undergarments protect the sacrum, buttocks, and hips.

o The NICE medical technology guidance Parafricta Bootees and Undergarments to reduce skin breakdown in people with or at risk of pressure ulcers concluded that although the garments are a promising technology, with the potential to reduce skin damage and the incidence and severity of pressure ulcers in both hospitals and the community, more evidence about their clinical benefits is needed to support the case for more widespread, routine adoption.

**Scenario: Management of pressure ulcers**

**From birth onwards.**

**How should I manage an adult with a pressure ulcer?**

Adults with pressure ulcers should be assessed and managed by healthcare professionals with the necessary skills and competencies.

In addition to the recommendations for prevention:

• Perform a nutritional risk assessment.
o If the person is assessed as being at risk of nutritional deficiency, refer to a dietitian to provide:
▪ Advice on how to follow a balanced diet to maintain an adequate nutritional status, taking into account energy, protein, and micronutrient requirements.
▪ Nutritional supplements (for people with nutritional deficiencies).
o Do not offer:

▪ Nutritional supplements to treat pressure ulcers in adults whose nutritional intake is adequate.
▪ Subcutaneous or intravenous fluids to treat pressure ulcers in adults whose hydration status is adequate.

o If debridement is indicated:
- Autolytic debridement should be used, with an appropriate dressing to support it.
- Sharp debridement should be considered if autolytic debridement is likely to take longer and prolong healing time, and should only be performed by a trained and competent practitioner.
- Enzymatic debridement should not be routinely offered.
- Larval (maggot) therapy should not be routinely offered. However, it can be considered if sharp debridement is contraindicated or if there is associated vascular insufficiency.

Consider the need for antibiotic treatment.

o Offer systemic antibiotics only if there is clinical evidence of systemic sepsis, spreading cellulitis, or underlying osteomyelitis.

o Discuss with microbiology to ensure that the chosen systemic antibiotic is effective against specific local pathogens.

o Do not offer systemic antibiotics:
- Specially to heal a pressure ulcer.
- Based only on positive wound cultures, without clinical evidence of infection.

o Do not routinely use topical antiseptics or antimicrobials to treat a pressure ulcer.

• Recommend an appropriate wound dressing, the choice of dressing should take into account:
- The person's pain and tolerance level.
- The position of the ulcer.
- The amount of exudate.
- Frequency of dressing change.

o Consider using a dressing that promotes a warm, moist, wound-healing environment to treat Category II, III, and IV pressure ulcers.

o Do not offer gauze dressings to treat a pressure ulcer.

**• Do not:**
- Offer electrotherapy or hyperbaric oxygen therapy to treat pressure ulcers.
- Routinely offer negative pressure wound therapy to treat pressure ulcers, unless it is necessary to reduce the number of dressing changes (for example in a wound with a large amount of exudate).

TOPIC: Superficial vein thrombosis (superficial thrombophlebitis)

**Definition**

Superficial vein thrombosis (also known as superficial thrombophlebitis) results from thrombus formation in a superficial vein, with associated inflammation in the tissue surrounding the vein:

- It most commonly occurs in the saphenous vein and its tributaries of the lower limbs (60–80% of cases). Superficial veins in other areas of the body (for example the upper limbs, neck, abdominal wall, breasts, or arms) can also be affected but are outside the scope of this CKS topic.
- Thrombus formation may be idiopathic or associated with one or more risk factors.
- Migratory thrombophlebitis is thrombophlebitis that recurs at varying sites but most commonly in the leg. It is often associated with a serious underlying cause, such as cancer.
- Trousseau syndrome is a rare condition characterized by recurrent and migratory superficial vein thrombosis, frequently in unusual sites, such as the arm or chest (also called Mondor's disease when involving chest wall veins). It is most often associated with gliomas and adenocarcinomas of the gastrointestinal tract (stomach, pancreas, and colon), lungs, breasts, ovaries, and prostate. It can also be seen with Behcet's and Buerger's disease

**Causes and risks factors**

- Superficial vein thrombosis is most often associated with prothrombotic conditions characterized by one or more of the components of the Virchow triad: vessel wall damage, venous stasis, and hypercoagulability of blood.
- Thrombus formation in the vein is likely to be the result of interplay between the damaged blood vessel, the coagulation cascade, the fibrinolytic system, platelet activation, and activation of the inflammatory process. However, the specific relation between these processes is unclear.
- Varicose veins are the most common risk factors for superficial vein thrombosis of the lower legs.
- Their dilatation and tortuosity predispose to stasis, inflammation, and thrombosis.
- About 80% of people with superficial vein thrombosis of the lower limbs have pre-existing varicose veins with or without chronic venous insufficiency.

Other risk factors include:

- o Thrombophilia — abnormalities of coagulation are associated with superficial vein thrombosis, especially in the absence of varicose veins and in people with superficial vein thrombosis involving the greater saphenous vein trunk. The main coagulation abnormalities associated with superficial vein thrombosis are factor V Leiden mutation (16%), prothrombin 20210 mutation (10%), and deficiencies in antithrombin III, protein C, or protein S (10%).
- o Autoimmune disease — autoimmune diseases, particularly Behcet's and Buerger's disease, are associated with superficial vein thrombosis. About 53% of people with Behcet's disease will experience an episode of superficial vein thrombosis usually within 5 years of the diagnosis of Behcet's disease (rarely superficial vein thrombosis precedes the diagnosis). Superficial vein thrombosis (more commonly migratory thrombophlebitis) is thought to occur in 27–50% of people with Buerger's disease
- o Previous history of superficial vein thrombosis — this is likey to be an important predictor of future episodes, especially in people with persistent risk factors, such as varicose veins.
- o Intravenous (IV) injections or cannulation — superficial vein thrombosis commonly occurs after vessel wall damage caused by IV injection or infusion (especially if an irritant substance was infused), mechanical injuries (for example during cannulation or hemodynamic procedures), or sclerotherapy.
- o Cancer — evidence from small retrospective cohort studies suggests that 10–15% of people with superficial vein thrombosis may have a diagnosis of cancer. Migratory thrombophlebitis is often associated with a serious underlying cause, such as cancer.
- o Pregnancy and up to 6 weeks after delivery — pregnancy-related changes, such as increase in procoagulant factors and reduced fibrinolytic activity, may explain the increased risk during pregnancy, particularly in the puerperium. Venous dilation (and the resulting venous stasis),

particularly in the third trimester, is also a risk factor for superficial vein thrombosis during pregnancy.
o Female sex — superficial vein thrombosis is more common in women than men, possibly because of the increased prevalence of varicose veins during pregnancy.
o Old-age.
o Prolonged immobilization — prolonged immobilization (for example due to long-haul flight or hospitalization) may predisposes to venous stasis and/or hypercoagulability of blood.
o Oral contraceptives or hormone replacement therapy (HRT)
o Obesity

**Complications**

• Deep vein thrombosis (DVT) or pulmonary embolism (PE) may develop in association with superficial vein thrombosis.

the risk of DVT is greater in people with superficial vein thrombosis if:

- The superficial thrombus extends near where the affected superficial vein joins the deep veins, for example where the long saphenous vein joins the femoral vein at the groin. Superficial vein thrombosis of the long saphenous vein within 3 cm of the saphenofemoral junction is considered to be equivalent to a DVT.
- The superficial thrombus is 5 cm or greater in length.
- The person has reduced mobility.
- The superficial thrombus is not associated with varicose veins.
- There is a past history of DVT or PE.
- The person has active cancer.

**Other possible complications include:**

- Infection — septic (infected) thrombophlebitis may occur, usually when superficial vein thrombosis is the result of vein instrumentation or cannulation. Although infected thrombophlebitis most commonly presents as a localized infection, it may spread (especially in this group of people) and become life threatening.
- Skin changes — hyperpigmentation and/or a persistent firm subcutaneous nodule may occur at the affected site.
- Varicose veins — can develop if the valves are damaged during recanalization of the thrombosed vein.

**Prognosis**

• Uncomplicated superficial vein thrombosis is generally considered to be a benign and self-limiting condition, and symptoms generally subside in 1–2 weeks, although hardness of the vein may persist for longer (several weeks to months).
• If there is concurrent venous thromboembolism (deep vein thrombosis or pulmonary embolism), the course may be more prolonged and treatment will be required.
• The risk of recurrence of superficial vein thrombosis ranges between 1.6–12.2% in treated cases and 3.3–36.7% in untreated cases, depending on the presence or absence of underlying risk factors, such as varicose veins and malignancy. If superficial vein thrombosis is associated with varicose veins, it has a high likelihood of recurrence unless excision of the varicose vein is performed.

**Diagnosis**

**Take a history.**

Ask about symptoms experienced, including the duration and severity.

▪ Superficial vein thrombosis is generally characterized by pain, tenderness, itching, and reddening of the skin, and/or hardening of the surrounding tissue. Pigmentation changes of the overlying skin are often observed.
▪ Symptoms typically develop over hours to days and resolve in days to weeks, although hardness of the vein may persist for longer (several weeks to months).

- Identify risk factors for superficial vein thrombosis, such as varicose veins, prolonged immobilization, and a history of previous episodes.

**Examine the person.**

- Superficial vein thrombosis most often occurs in the lower limbs (in the saphenous vein or its tributaries), and varicose veins are the most common risk factor.
- Examine the legs for varicose veins and other signs of venous insufficiency. Examine the person standing and lying, and document the location and severity of any varicose veins.
- In people with pre-existing varicose veins, there is a tender 'worm-like' mass deep to the skin. The overlying skin is warm and erythematous, and there is often oedema of the surrounding area without generalized swelling of the whole limb.
- In people without varicose veins, there is often a palpable cord, sometimes nodular, associated with warmth, tenderness, and erythema along the course of a non-varicose superficial vein.
- Assess for signs of deep vein thrombosis (DVT) or pulmonary embolism (PE), which may occur in association with superficial vein thrombosis.

▪ DVT is distinguished from superficial vein thrombosis by generalized pain and swelling of the leg that is not localized to a thrombosed vein.

- Features of PE include dyspnoea, chest pain, and syncope.
- Identify people with factors that put them at increased risk of developing DVT or PE, such as reduced mobility or a superficial thrombus that extends near where an affected superficial vein joins a deep vein (for example where the long saphenous vein joins the femoral vein at the groin).
- Assess for other complications of superficial vein thrombosis, such as infection.
- Suspect infected superficial vein thrombosis if there are features of cellulitis (acute onset of red, painful, hot, swollen, and tender skin) or an abscess, especially if the person has a fever.
- Suspect migratory superficial thrombophlebitis if a person has repeated thromboses developing in superficial veins, at varying (or multiple) sites but most commonly in the leg.
- Consider differential diagnoses in people with other atypical presentations of superficial vein thrombosis.

**Differential diagnosis**

**Differential diagnoses of superficial vein thrombosis include:**

- Deep vein thrombosis (DVT) — presents with pain and swelling, tenderness, redness and warmth of the skin, and vein distension in one leg.
- Cellulitis — erythema is usually localized and confluent, with associated tenderness and generalized swelling. There may be associated lymphangitis or tender lymphadenopathy, which is not characteristic of superficial vein thrombosis. A palpable cord is not present with cellulitis. Fever, chill, and myalgia is more common with cellulitis.
- Chronic venous insufficiency — usually chronic leg swelling with associated ectatic veins, varicose veins, and skin changes that may include hyperpigmentation, venous eczema, lipodermatosclerosis, and venous ulcers.
- Insect bite or stings — there is usually a recent history of an insect bite or sting. Swelling and redness usually extend over a large area and do not follow the course of a vein. There may be associated pruritus.
- Lymphangitis — presents with red streaks from the infected area to the groin or armpit (may be faint or obvious), warmth, enlarged and tender lymph nodes (usually in the elbow, armpit, or groin), and a raised border around the affected area. High fever, chills, and moderate pain and swelling may also occur.
- Tendonitis — usually pain, mild swelling, and warmth in the area of a tendon. Pain is worse during and after activity, and the tendon and joint area can become stiff as the swelling impinges on the movement of the joint.
- Vasculitic conditions (such as erythema nodosum and polyarteritis nodosum) — characterized by inflammation within or around blood vessels, with or without necrosis. Presentation is variable and includes skin signs, such as palpable purpura (often painful) usually on the lower leg/buttock, and systemic signs, such as fever, night sweats, malaise, weight loss, and arthralgia.

**Scenario: Management of superficial vein thrombosis**

**From age 16 years onwards.**

**How should I manage a person with superficial vein thrombosis?**

- If deep vein thrombosis (DVT) or pulmonary embolism (PE) is suspected
- If the person has varicose veins or a venous leg ulcer:
- If a serious underlying cause of superficial vein thrombosis, such as cancer, is suspected:
- Refer appropriately for specialist assessment.

**For all other people with superficial vein thrombosis**:
Treat pain with a simple analgesic.

- Prescribe an oral nonsteroidal anti-inflammatory drug (NSAID) and/or paracetamol.
- Consider a topical NSAID for mild and limited superficial vein thrombosis.
- Continue treatment until pain settles (usually within 1–2 weeks, although the thrombosed vein may be palpable and tender for several weeks to months).

Give self-care advice to relieve symptoms and reduce the risk of complications. Advise that:

- A warm, moist towel or flannel applied to the affected limb may provide symptomatic relief.
- Keeping the leg elevated when sitting may improve venous blood flow and reduce swelling.
- They should continue using the affected limb and remain mobile in order to reduce the risk of DVT.

o Be aware that antibiotics are not indicated unless there are signs of infection (such as high fever and malaise).

▪ If there are signs of infection, consider arranging hospital admission, especially if the person is at risk of complications because they are frail, elderly, immunocompromised, or have an indwelling intravenous catheter.

o Consider referral to a vascular service for venous duplex scanning to help guide treatment and the need for further specialist intervention.

- Low molecular weight heparin (LMWH) and fondaparinux have been shown to reduce the risk of superficial vein thrombosis extension and recurrence.
- Fondaparinux has also been shown to significantly reduce the risk of symptomatic venous thromboembolism and may be considered in people at increased risk of a DVT or PE, for example those with superficial thrombus near the saphenofemoral junction.

o Offer compression stockings (after excluding arterial insufficiency).

**For people with recurrent superficial vein thrombosis with no obvious cause or risk factor, consider investigating for a coagulation disorder.**

TOPIC: Varicose veins

**Definition**

• Varicose veins are dilated, tortuous, superficial veins that are most commonly found in the legs. They are often visible and palpable, and are an indication of superficial lower extremity venous insufficiency.

- For most people, varicose veins are mainly a cosmetic concern; however, they can cause symptoms in some people.
- Varicose veins may become more severe over time and can lead to complications.

**Causes and risk factors**

Varicose veins are an indication of superficial lower extremity venous insufficiency.

o In, the normal venous system, blood from superficial venous capillaries is directed upward via one-way valves into superficial veins. These in turn drain via the perforator veins, which pass through a muscular fascial layer, into the deep veins.

o Varicose veins are thought to be caused by incompetent valves in the affected vein, which result in reflux of blood and increased pressure in the vein distally. Unlike deep veins, which have thick walls and are confined by fascia, superficial veins are unable to withstand the high pressure and eventually become dilated and tortuous.

o Other pathological factors, such as weakness or degeneration of the vein wall, may also be involved in the development of varicose veins.

o The, increase in venous pressure in the leg is usually particularly marked if there is also incompetence in deep veins.

**Risk factors for developing varicose veins include:**

- Increasing age
- Family history of varicose veins
- Female sex
- Pregnancy
- Obesity — being overweight or obese can increase the risk of developing varicose veins. The impact of body weight on the development of varicose veins appears to be more significant in women than in men.
- Prolonged standing or sitting
- History of deep vein thrombosis (DVT) — DVT may cause valvular damage and dysfunction in the deep veins, leading to increased pressure with subsequent distension and varicose vein formation

**Complications**

- Bleeding, especially when the varicose veins are large, traumatized, or located over bony prominences.
- Superficial vein thrombosis (characterized by the appearance of hard, painful veins).
- Deep vein thrombosis.
- Changes in skin pigmentation.
- Skin ulceration (occurs in 3–6% of people with varicose veins).
- Depression (due to cosmetic skin changes).
- Decreased quality of life.

**Prognosis**

• Without appropriate management, varicose veins may progress in severity and extent and complications, such as skin changes and ulceration, may occur.

**Diagnosis**

**Take a history.**

Ask about symptoms experienced, if any.

▪ Pain, aching, itching, and swelling of the affected leg are commonly reported but may be due to other causes. Discomfort after prolonged standing, relief with leg elevation, and discomfort over the affected veins suggest that the symptoms are due to varicose veins.
▪ Restless legs and leg cramps (usually nocturnal) may also be present.

Ask about previous interventions on the veins (including varicose vein surgery) as well as a history of trauma, deep vein thrombosis, ulcerations, or bleeding from varicose veins.

Identify risk factors for varicose veins, such as prolonged standing or a family history of varicose veins.

Identify why the person is concerned about their varicose veins. For many people, it is because of the cosmetic appearance, or worries that they will get worse or cause complications.

**Examine the legs.**

Examine the skin visually and by palpation (with the person standing and lying) for irregularities and bulges consistent with varicose veins. Note the extent, size, and location of the dilated veins to help exclude telangiectasias and reticular veins.

▪ Varicose veins are 3 mm or more in diameter in the upright position.
▪ Telangiectasias (also known as spider or thread veins) are intradermal veins that are less than 1 mm in diameter.
▪ Reticular veins are permanently dilated intradermal veins that range between 1–3 mm in diameter. They may be tortuous but are usually asymptomatic.

**Assess for any complications, including:**

▪ Skin changes — look for hyperpigmentation (brown-red discolouration caused by haemosiderin deposition), venous eczema (itchy, red, scaly, and/or flaky skin which may have blisters and crusts on the surface), lipodermatosclerosis (painful, hardened, tight skin), and atrophie blanche (star-shaped, ivory-white, depressed, atrophic scars with surrounding pigmentation).
▪ Areas of active or healed ulceration — venous leg ulcers typically occur in the gaiter area of the leg (from the ankle to mid-calf).
▪ Superficial vein thrombosis — characterized by the appearance of hard, painful veins.

**Management**

**From age 18 years onwards.**

If varicose veins are bleeding, offer first aid and admit to a vascular service.

- If there is doubt about whether admission is required, seek immediate specialist advice from a vascular surgeon.

Explain that varicose veins have faulty valves that allow reversed venous blood flow, leading to pooling of blood and venous distension.

Advise on the possible risk factors.

Reassure the person that complications, such as skin changes and leg ulcers, are uncommon.

Give self-care advice to relieve symptoms and reduce the risk of complications. Advise the person to:

- Lose weight (if overweight or obese).
- Engage in light to moderate physical activity.
- Avoid factors that exacerbate symptoms, such as sitting or standing for long periods of time.
- Elevate the legs when possible.

The National Institute for Health and Care Excellence (NICE) recommends referral to a vascular service in the following cases for consideration of interventional treatments:

- Primary or recurrent varicose veins associated with lower limb symptoms, typically pain, aching, discomfort, swelling, heaviness, and itching.
- Lower-limb skin changes, such as pigmentation or eczema, thought to be caused by chronic venous insufficiency.
- Superficial vein thrombosis (characterized by the appearance of hard, painful veins) and suspected venous incompetence.
- An active venous leg ulcer (break in the skin below the knee that has not healed within 2 weeks).
- A healed venous leg ulcer.

If referral is not indicated, offer compression stockings (after excluding arterial insufficiency).

If the person is a pregnant woman:

- Reassure her that varicose veins are common in pregnancy, are not harmful to the baby, and often improve considerably after pregnancy.
- Offer treatment with compression stockings. Advise that these may improve the symptoms but will not prevent varicose veins from emerging.
- Be aware that interventional treatments are unlikely to be an option in pregnant women, except in exceptional circumstances.

Advise the person to seek further medical help if:

- Veins are hard or painful.
- There are skin changes (pigmentation, venous eczema, lipodermatosclerosis, and atrophie blanche).
- They develop a venous ulcer (break in the skin below the knee that has not healed within 2 weeks).
- There is bleeding from the varicose veins.

**Interventional treatments for varicose veins**

- Duplex ultrasound provides both anatomical and haemodynamic data on the functioning of the superficial and deep venous systems.
  - People with varicose veins who are seen by a vascular service should be assessed with duplex ultrasound to confirm the diagnosis of varicose veins and the extent of truncal reflux, and to plan treatment for suspected primary or recurrent varicose veins.
- Interventional treatments can be done under local or general anaesthesia and include:
  - Endothermal ablation — using energy either from high-frequency radio waves (radiofrequency ablation) or lasers (endovenous laser treatment) to seal the affected veins.
  - Foam sclerotherapy — injection of an irritant foam into the vein, resulting in an inflammatory response that causes closure of the vein.
  - Surgery — involves ligation (tying off) and stripping (removal) of the affected vein.

## TOPIC: OBESITY

Obesity is a chronic, often progressive metabolic disease, associated with an increase in mortality and morbidity, that is increasing in prevalence in adults.
The National Institute for Health and Care Excellence (NICE) recommends the use of body mass index (BMI) to assess overweight and obese individuals. It advises the measurement of waist circumference to supplement this in individuals with a BMI under 35 kg/m.

### BMI

**BMI = body weight (in kg) ÷ height squared (in meters)**

**Classification of body mass index**

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| Underweight – BMI <18.5 kg/m$^2$ |
| Normal weight – BMI ≥18.5 to 24.9 kg/m$^2$ |
| Overweight – BMI ≥25 to 29.9 kg/m$^2$ |
| Obesity – BMI ≥30 kg/m$^2$ |
| ▪ Obesity class 1 – BMI 30 to 34.9 kg/m$^2$ |
| ▪ Obesity class 2 – BMI 35 to 39.9 kg/m$^2$ |
| ▪ Obesity class 3 – BMI ≥40 kg/m$^2$ |

**Waist circumference**

Waist circumference in men:
- <94 cm is defined as low risk.
- 94 to 102 cm is defined as high risk.
- >102 cm is defined as very high risk.

Waist circumference in women:
- <80 cm is defined as low risk.
- 80 to 88 cm is defined as high risk.
- >88 cm is defined as high risk.

Define the degree of central adiposity based on waist-to-height ratio as follows:
- Healthy central adiposity — 0.4 to 0.49, indicating no increased health risks.
- Increased central adiposity — 0.5 to 0.59, indicating increased health risks.
- High central adiposity — 0.6 or more, indicating further increased health risks.

### CAUSES

| **Iatrogenic causes** |
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| Drugs (antipsychotics, antidepressants, anticonvulsants, lithium, corticosteroids, BB, OCP, sulfonylureas, thiazolidinediones) |
| Hypothalamic surgery |

| **Dietary obesity** |
| --- |
| Infant feeding practices |
| Progressive hyperplastic obesity |
| Frequency of eating |
| High-fat diets |
| Overeating |
| **Neuroendocrine obesities** |
| Hypothalamic obesity |
| Hypothyroidism |
| Seasonal affective disorder |
| Cushing's syndrome |
| Polycystic ovary syndrome |
| Hypogonadism |
| Growth hormone deficiency |
| Pseudohypoparathyroidism |
| **Social and behavioral factors** |
| Socioeconomic status, less formal education |
| Ethnicity |
| Psychological factors, sleep deprivation |
| Restrained eaters |
| Night eating syndrome |
| Binge eating |
| **Sedentary lifestyle** |
| Enforced inactivity (postoperative) |

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| Aging (55-65) |
| **Genetic (dysmorphic) obesities** |
| Autosomal recessive traits |
| Autosomal dominant traits |
| X-linked traits |
| Chromosomal abnormalities |
| **Other** |
| Low birth weight, prior pregnancy |

**RISKS OF OBESITY**

COPD, asthma, Obstructive sleep apnea, obesity hypoventilation syndrome, DVT, breast cancer, carcinoma of endometrium, polycystic ovary syndrome, stress incontinence, fertility impairment, fatty liver, metabolic syndrome, type 2 DM, osteoarthritis, GERD, depression, skin changes (striae, acanthosis nigricans, hirsutism)

**MANAGEMENT**

Realistic targets should be set. This includes weight loss targets of **0.5-1 kg (1-2 lb) per week**, healthier eating and increased physical activity.

**DIET MODIFICATION**

-Diets should be undertaken with expert support and intensive follow-up
-NICE guidelines recommend diets with a **600kcal deficit**, or low-fat diets.
-Low-calorie diets **(800-1600 kcal per day)** may be considered, but NICE points out they may be nutritionally incomplete.
-Very low-calorie diets (< 800 kcal per day) should not be used routinely. They should not be undertaken for more than 12 weeks.

**PHYSICAL ACTIVITY**

-**30 minutes** of **moderate-intensity activity** (60% of maximum heart rate '220-age in years'), either as one session or in bouts of 10 minutes, on at least **five days a week**.
-To prevent obesity, most people would need to do 45-60 minutes of moderate-intensity exercise every day, particularly if calorie intake is not adjusted
-For those who have been obese and lost weight, 60-90 minutes per day are advised to avoid relapse.

**BEHAVIORAL INTERVENTION**

- Self-awareness of behaviour and progress.
- Stimulus control.
- Goal setting.
- Slowing of the rate of eating.
- Exploring and involving social support.
- Problem solving.
- Assertiveness.
- Cognitive restructuring (modifying thoughts).
- Reinforcing changes.
- Relapse prevention strategies.
- Strategies for dealing with weight regain.

**PHARMOCOLGICAL OPTIONS**

- **Obesity + migraine or epilepsy = use topiramate**
- **Obesity + smoking = use bupropion and naloxone**
- **Obesity + prediabetic = GLP1**
- **Obesity + diabetic =GLP1**
- **Obesity + diabetic + needle phobia = use Semaglutide oral or SGLT2 inhibitor**

Orlistat (pancreatic lipase inhibitor)

- BMI of 28 kg/m^2 or more with associated risk factors, or
- BMI of 30 kg/m^2 or more
- Only be continued if weight loss > 5% at 3 months
- normally used for < 1 year

Liraglutide "Saxenda" (GLP-1 receptor antagonist)

- BMI of 27- 30 kg/m^2 or more with associated risk factors, or
- BMI of 30 kg/m^2 or more
- Stopped after 12 weeks if patients have not lost at least 5% of initial body weight

**BARIATRIC SURGERY**

- BMI 40 kg/m2 or more, or between 35 kg/m2 and 40 kg/m2 with other significant disease (for example, type 2 diabetes or high blood pressure) that could be improved if they lost weight.

Primarily restrictive operations

- laparoscopic-adjustable gastric banding (LAGB)
  - it is normally the first-line intervention in patients with a BMI of 30-39kg/m^2
  - produces less weight loss than malabsorptive or mixed procedures but as it has fewer complications
- sleeve gastrectomy
  - stomach is reduced to about 15% of its original size
- intragastric balloon
  - the balloon can be left in the stomach for a maximum of 6 months

Primarily malabsorptive operations

- biliopancreatic diversion with duodenal switch
  - usually reserved for very obese patients (e.g. BMI > 60 kg/m^2)

Mixed operations

- Roux-en-Y gastric bypass surgery
  - is both restrictive and malabsorptive in action

Always most common complication post gastric bypass is gall stone that's why usually remove it with intervention

TOPIC: Pilonidal sinus disease:

**Definition**

• Pilonidal sinus disease is an inflammatory skin condition that usually occurs in the midline of the natal cleft (although there have been few reported cases involving other parts of the body, such as on the scalp, axilla, groin, and between the fingers).

The term 'pilonidal' originates from the Latin words 'pilo' (hair) and 'nidus' (nest), implying a nest of hair.

• Pilonidal sinus may be asymptomatic, or present as an acute abscess (with or without cellulitis) or a chronic cyst and/or sinus with persistent drainage and/or extensive subcutaneous tracts.

**Risk factors**

• Based on the pathophysiology, the risk of pilonidal sinus disease may depend on hair type, the force of insertion created within the natal cleft, and the vulnerability of the skin of the natal cleft to hair insertion . Therefore, the risk may be increased in people with:

- Coarse hair.
- Hirsutism.
- Deep natal cleft.
- Macerated or broken skin.

• Other risk factors for the development of pilonidal sinus disease include:

- Male gender (2–4 times more likely than in women).
- Age 15–40 years (the mean age at presentation has been reported as 21 years in males and 19 years in females. It is uncommon before puberty and after the age of 45 years).
- Being white (more common than in black people, due to different hair characteristics and growth patterns).
- Obesity.
- Poor hygiene.
- Lifestyle or occupation which involves sitting for prolonged periods (more than 6 hours a day).
- Repetitive trauma to the sacrococcygeal region.
- Family history of pilonidal sinus disease.

**Prognosis**

- Pilonidal sinus disease is rarely seen in people aged over 45 years of age, as the natural history of the disease is that it 'burns itself out' over time.
- Active pilonidal sinus disease can progress to chronic or recurrent disease, and approximately 50% of acute pilonidal abscesses may develop into chronic discharging pilonidal sinus disease despite treatment of the acute abscess.
- Pilonidal sinus disease often recurs after surgical treatment, and the recurrence rate depends on the specific surgical method used.
- Simple incision and drainage is associated with an estimated 15–40% recurrence rate, likely attributed to failure to address the underlying debris, epithelization, granulation tissue, and sinus tracts.
- Following wide excisional surgery for more extensive chronic disease, recurrence is less likely to occur if the wound is left open to heal by secondary intention compared with any type of primary wound closure, although off-midline closure (closure on either side of the middle of the natal cleft) have been shown to be statistically superior to midline closure with regard to healing time, infection at the surgical site, and recurrence rate

**Complications**

Complications of pilonidal sinus disease are a considerable cause of morbidity, resulting in loss of productivity in otherwise healthy people.

They include.

- Cellulitis.
- Pilonidal abscess (requiring urgent surgical drainage).
- Sepsis.

- Chronic pain.
- Chronic non-healing wounds.
- Altered body image and self-esteem, and social embarrassment.
- Effects on personal relationships.

**Diagnosis**

**How should I diagnose pilonidal sinus disease?**

Ask about:

- Symptoms experienced and the duration.
- Recurrence and previous treatments for pilonidal sinus disease.
- The presence of suppurative lesions in areas other than the natal cleft (may suggest hidradenitis suppurativa).

Assess for risk factors of pilonidal sinus disease, including a family history and lifestyles and occupations that may involve prolonged sitting and/or repeated chafing of the natal cleft. Also consider other risk factors, such as male sex and age between 16–40 years.

Examine the person — to help confirm the diagnosis of pilonidal sinus disease and to exclude differential diagnoses, such as fistulae and hidradenitis suppurativa. Consider the need for a digital rectal examination.

People with pilonidal sinus disease will almost always have characteristic midline pits in the natal cleft.

An asymptomatic pilonidal sinus is characterized by one or more non-tender pits, or lump, in the natal cleft, which may have an emergent hair.

An acute pilonidal abscess is characterized by a painful, tender lump which may be fluctuant and have purulent discharge. There may be accompanying cellulitis. Rarely, an acute abscess may be associated with fever.

A discharging pilonidal sinus is characterized by chronic pain, discharge of pus or blood from the lateral opening, and recurrent abscess formation and infection. This may be secondary to chronic disease or recurrent disease post-operatively.

Investigations are not routinely required for the diagnosis of pilonidal sinus disease, but may be considered if there is doubt about the diagnosis or to exclude differential diagnoses, such as Crohn's disease and infectious processes (such as tuberculosis and syphilis).

**Differential diagnosis**

• The differential diagnosis of pilonidal sinus disease includes:

- A boil — an infection of the hair follicle where there is purulent extension into the subcutaneous tissue, in which a small abscess form. It is a deep-seated inflammatory nodule.
- Hidradenitis suppurativa — may occur in the perineal/ perianal region, but also in the areolar area of the breasts and frequently in the axillae.
- Ano-rectal fistula and fissure — a tear or ulcer in the lining of the anal canal, immediately within the anal margin.
- Perianal abscess — abscess develops in the perianal region, and the skin of the natal cleft is normal.
- Infected sebaceous cyst — there may be a history of a small preceding skin lesion. Infection of the cyst is characterized by swelling and pain. In about 50% of cases a punctum will be visible.
- Sacral osteomyelitis — infection of the underlying sacral bone may also cause pain and tenderness, but the overlying skin should be normal.
- Crohn's disease — a chronic, relapsing-remitting, non-infectious inflammatory disease of the gastrointestinal tract. May present with perianal pain or tenderness, anal or perianal skin tag, fissure, fistula, or abscess.
- Infectious processes, including tuberculosis, syphilis, and actinomycosis.

**Management**

**From age 16 years onwards.**

**How should I manage asymptomatic pilonidal sinus disease?**

If a person presents with asymptomatic pilonidal sinus disease:

- Advise a 'watch and wait' approach, and reassure that treatment is not necessary.
- Advise on meticulous perianal hygiene with regular baths or showers.
- Provide patient information on pilonidal sinus disease.

**How should I manage acute pilonidal abscess?**

If a person presents with an acute pilonidal abscess:

- Arrange for an urgent same-day incision and drainage.
- Incision and drainage may sometimes be performed in primary care if the lesion is small and superficial, and the expertise and facilities are available. Otherwise, the person should be referred to a colorectal or general surgical unit according to local protocol.
- Offer paracetamol for pain and/or fever.

If the response is insufficient, also offer a nonsteroidal anti-inflammatory drug (NSAID), such as ibuprofen or naproxen (unless not tolerated or contraindicated)

If there is associated cellulitis, consider prescribing antibiotic treatment.

**Following surgical incision and drainage:**

- Advise the person on long-term measures that may reduce the risk of recurrence, including:
- Meticulous perianal hygiene with regular baths or showers.
- Buttock hair removal techniques, such as laser treatment, waxing, shaving, and depilatory creams (there is some evidence of a lower recurrence rate after laser hair removal compared with shaving and depilatory creams).
- Provide patient information on pilonidal sinus disease.

**How should I manage discharging pilonidal sinus disease?**

• If a person presents with discharging pilonidal sinus disease:

- Refer to a colorectal or general surgical unit for surgical excision (urgency dependent on clinical judgement).
- Surgical excision is the standard treatment for chronic pilonidal cyst and sinus disease. The surgery performed is generally one of two procedures: excision of infected tissue with primary closure (including various flap techniques) or excision with healing by secondary intention (including marsupialization).
- The choice of technique depends on the characteristics of the pilonidal sinus, the experience and expertise of the surgeon, and the person's wishes.
- Offer paracetamol for pain and/or fever.
- ▪ If the response is insufficient, also offer a nonsteroidal anti-inflammatory drug (NSAID), such as ibuprofen or naproxen (unless not tolerated or contraindicated).
- Provide patient information on pilonidal sinus disease.

TOPIC: Anorectal disorders

| **Haemorrhoids** | **Location: 3, 7, 11 o'clock position**<br>**Internal or external**<br>**Treatment: Conservative, Rubber band ligation, Haemorrhoidectomy** |
|---|---|
| **Anal fissure** | Typically presents with painful rectal bleeding<br>Location: midline 6 (posterior midline 90%) & 12 o'clock position. Distal to the dentate line<br>Chronic fissure > 6/52: triad: Ulcer, sentinel pile, enlarged anal papillae |
| **Proctitis** | Causes: Crohn's, ulcerative colitis, *Clostridioides difficile* |
| **Ano rectal abscess** | E.coli, staph aureus<br>Positions: Perianal, Ischiorectal, Pelvirectal, Intersphincteric |
| **Anal fistula** | Usually due to previous ano-rectal abscess<br>Intersphincteric, transsphincteric, suprasphincteric, and extrasphincteric.<br>Goodsalls rule determines location |
| **Rectal prolapse** | Associated with childbirth and rectal intussusception. May be internal or external |
| **Pruritus ani** | Extremely common. In children is often related to worms, in adults may be idiopathic or related to other causes such as haemorrhoids. |
| **Anal neoplasm** | Squamous cell carcinoma commonest unlike adenocarcinoma in rectum |
| **Solitary rectal ulcer** | Associated with chronic straining and constipation. Histology shows mucosal thickening, lamina propria replaced with collagen and smooth muscle (fibromuscular obliteration) |

TOPIC: Diseases Affecting the Vertebral Column

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|---|---|
| **Ankylosing spondylitis** | • Chronic inflammatory disorder affecting the axial skeleton<br>• Sacro-ilitis is a usually visible in plain films<br>• Up to 20% of those who are HLA B27 positive will develop the condition<br>• Affected articulations develop bony or fibrous changes<br>• Typical spinal features include loss of the lumbar lordosis and progressive kyphosis of the cervico-thoracic spine |
| **Scheuermann's disease** | • Epiphysitis of the vertebral joints is the main pathological process<br>• Predominantly affects adolescents<br>• Symptoms include back pain and stiffness<br>• X-ray changes include epiphyseal plate disturbance and anterior wedging<br>• Clinical features include progressive kyphosis (at least 3 vertebrae must be involved)<br>• Minor cases may be managed with physiotherapy and analgesia, more severe cases may require bracing or surgical stabilisation |
| **Scoliosis** | • Consists of curvature of the spine in the coronal plane<br>• Divisible into structural and non-structural, the latter being commonest in adolescent females who develop minor postural changes only. Postural scoliosis will typically disappear on manoeuvres such as bending forwards<br>• Structural scoliosis affects > 1 vertebral body and is divisible into idiopathic, congenital and neuromuscular in origin. It is not correctable by alterations in posture<br>• Within structural scoliosis, idiopathic is the most common type<br>• Severe, or progressive structural disease is often managed surgically with bilateral rod stabilisation of the spine |
| **Spina bifida** | • Non fusion of the vertebral arches during embryonic development<br>• Three categories; myelomeningocele, spina bifida occulta and meningocele<br>• Myelomeningocele is the most severe type with associated neurological defects that may persist in spite of anatomical closure of the defect |

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|---|---|
| | • Up to 10% of the population may have spina bifida occulta, in this condition the skin and tissues (but not bones) may develop over the distal cord. The site may be identifiable by a birth mark or hair patch<br>• The incidence of the condition is reduced by use of folic acid supplements during pregnancy |
| **Spondylolysis** | • Congenital or acquired deficiency of the pars interarticularis of the neural arch of a particular vertebral body, usually affects L4/ L5<br>• May be asymptomatic and affects up to 5% of the population<br>• Spondylolysis is the commonest cause of spondylolisthesis in children<br>• Asymptomatic cases do not require treatment |
| **Spondylolisthesis** | • This occurs when one vertebra is displaced relative to its immediate inferior vertebral body<br>• May occur as a result of stress fracture or spondylolysis<br>• Traumatic cases may show the classic 'Scotty Dog' appearance on plain films<br>• Treatment depends upon the extent of deformity and associated neurological symptoms; minor cases may be actively monitored. Individuals with radicular symptoms or signs will usually require spinal decompression and stabilisation |

TOPIC: Benign prostatic hyperplasia:

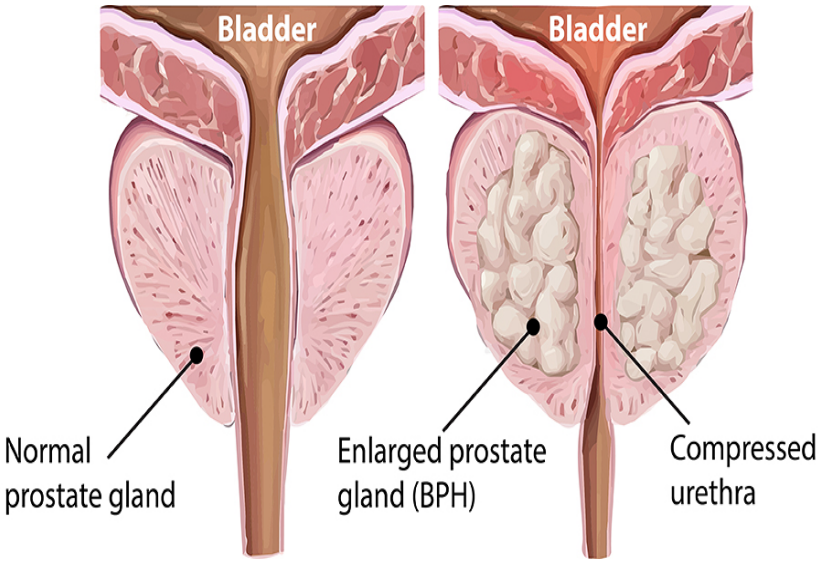
(BPH) is an increase in size of the prostate gland without malignancy present and it is so common as to be normal with advancing age.

**Risk factors:**

- age
  - around 50% of 50-year-old men will have evidence of BPH and 30% will have symptoms
  - around 80% of 80-year-old men have evidence of BPH
- ethnicity: black > white > Asian

**BPH typically presents with lower urinary tract symptoms (LUTS), which may be categorised into:**

- voiding symptoms (obstructive):
  - weak or intermittent urinary flow
  - straining
  - hesitancy
  - terminal dribbling
  - incomplete emptying
- storage symptoms (irritative)
  - urgency
  - frequency
  - urgency incontinence
  - nocturia
- post-micturition
  - dribbling
- complications
  - urinary tract infection
  - retention
  - obstructive uropathy




**Assessment:**

- Check urine by dipstick and send

  MSU for microscopy and culture.

- **Routine blood tests include:**
  - U&E and creatinine: particularly if chronic retention is suspected, FBC, LFTs Abnormal LFTs may indicate other disease. Isolated elevation of alkaline phosphatase can occur:
    - If the prostate is malignant and has metastasised to bone.
    - In an elderly person, where it may represent undiagnosed Paget's disease of bone.
- **PSA:** should be done if there are any obstructive symptoms, or if the patient is worried about prostate cancer PSA is elevated with a large, benign prostate. Experts consider that gentle DRE is unlikely to raise the PSA result. A combination of clinical examination and PSA levels may be a better way of attempting to differentiate between a benign and malignant prostate gland, although evidence to support this may be lacking.
  **It should be recognised that normal PSA values change with age:**

| PSA Cut-off Values | |
|---|---|
| **Age (years)** | **PSA Cut-off** |
| 40-49 | 2.5 micrograms/litre or higher |
| 50-59 | 3.5 micrograms/litre or higher |
| 60-69 | 4.5 micrograms/litre or higher |
| 70-79 | 6.5 micrograms/litre or higher |
| There are no age-specific reference limits for men older than 80 years of age. Whether you are referred will depend on your discussion with your doctor. | |

- urinary frequency-volume chart
  - should be done for at least 3 days
- International Prostate Symptom Score (IPSS)
  - tool for classifying the severity of lower urinary tract symptoms (LUTS) and assessing the impact of LUTS on quality of life
  - Score 20–35: severely symptomatic
  - Score 8–19: moderately symptomatic
  - Score 0–7: mildly symptomatic
- **Imaging**

Imaging may also be necessary if there is any suggestion of urinary tract obstruction.
Ultrasound examination of the prostate may be used prior to surgery or if it will assist choice of medical therapy.

**Examination:**

- **Examination of the abdomen** includes checking for a palpable bladder. This may indicate chronic outflow obstruction or a neurogenic bladder. To exclude the latter, an enquiry about motor or sensory loss along with checking knee and ankle jerks and plantar responses should suffice. Any further abnormalities require a full neurological history and examination.
- **Digital rectal examination** (DRE) includes noting the tone of the anal sphincter and the pelvic floor. It may be poor with a neurogenic bladder. The size of the prostate is assessed. Urologists report their findings in terms of the size of the prostate, a normal gland in a young adult weighing about 20 g. A useful guide for those less familiar with prostates is that a finger's breadth represents about 15 to 20 g and so a gland that is three fingers in breadth across is 45 to 60 g. Symptoms are unusual below two fingers in breadth. It is also important to note the texture and contour of the gland. It should be firm but not hard, and smooth without nodules. The median sulcus should be clearly defined. A gland that is hard rather than firm, nodular and lacks a clear median sulcus suggests carcinoma of prostate.
- If an elderly person is likely to need surgery, a brief check of the cardiovascular and respiratory systems is also needed.

**Indications for referral:**

In September 2020 and more recently in January 2021, NICE provided updated guidance on suspected cancer recognition and referral. There are no specific changes in these versions that relate to prostate cancer

**NICE recommends referral for the following:**

- Acute retention of urine (admit immediately).
- Acute kidney injury (admit immediately).
- Visible haematuria (to be seen in two weeks).
- Suspicion of prostate cancer based on the finding of a nodular or firm prostate, or a raised PSA level, or both (to be seen in two weeks).

- Culture-negative dysuria (to be seen in two weeks).
- Chronic urinary retention with overflow or night-time incontinence (to be seen in two weeks).
- Recurrent UTI.
- Microscopic haematuria.
- Failure to respond to treatment in primary care with poor quality of life as assessed by the I-PSS.

**Differential diagnosis:**

- Bladder tumours, bladder stones, bladder trauma, chronic pelvic pain, chronic prostatitis
- Detrusor instability, interstitial cystitis, neurogenic bladder, prostate cancer, radiation cystitis
- Urethral strictures, UTI

**Management options:**

- watchful waiting
- alpha-1 antagonists e.g. tamsulosin, alfuzosin
    - decrease smooth muscle tone of the prostate and bladder
    - considered first-line: NICE recommend if moderate-to-severe voiding symptoms (IPSS ≥ 8)
    - improve symptoms in around 70% of men
    - adverse effects: dizziness, postural hypotension, dry mouth, depression
- 5 alpha-reductase inhibitors e.g. finasteride
    - block the conversion of testosterone to dihydrotestosterone (DHT), which is known to induce BPH
    - indicated if the patient has a significantly enlarged prostate and is considered to be at high risk of progression
    - unlike alpha-1 antagonists causes a reduction in prostate volume and hence may slow disease progression. This however takes time and symptoms may not improve for 6 months
    - may also decrease PSA concentrations by up to 50%
    - adverse effects: erectile dysfunction, reduced libido, ejaculation problems, gynaecomastia
- the use of combination therapy (alpha-1 antagonist + 5 alpha-reductase inhibitor) was supported by the Medical Therapy of Prostatic Symptoms (MTOPS) trial and is also supported by NICE: '*If the man has bothersome moderate-to-severe voiding symptoms and prostatic enlargement*'
- if there is a mixture of storage symptoms and voiding symptoms that persist after treatment with an alpha-blocker alone, then an antimuscarinic (anticholinergic) drug such as tolterodine or darifenacin may be tried
- surgery
    - transurethral resection of prostate (TURP)

**Complications:**

Bladder outlet obstruction can result in:

- Urinary retention: this may be precipitated by anticholinergic drugs, including tricyclic antidepressants, opiates and diuretics.
- Recurrent UTI, especially with incomplete emptying.
- Impaired kidney function: progression to chronic kidney disease is much rarer now.
- Bladder calculi (may present as ongoing LUTS or recurrent infections).
- Haematuria - may be microscopic or macroscopic.

BPH is not considered to be a precursor of prostate cancer. It is likely that although BPH may not make prostate cancer more likely to occur, it may increase the chance of diagnosing an incidental cancer.

TOPIC: Prostate cancer

**Definition**

- Prostate cancer is a malignant tumour of the prostate. Almost, all cancers of the prostate (95%) are adenocarcinomas (cancers of glandular cells). Prostate cancer is multifocal — the different foci may be caused by different genetic mutations, which can differ greatly in growth rate and ability to metastasize. Most prostate cancers are indolent and grow slowly; a minority are aggressive in their tendency to invade local structures or to metastasize to remote tissues. Localized prostate cancer is confined within the capsule and seldom causes symptoms. Locally advanced prostate cancer extends beyond the capsule of the prostate and is often asymptomatic when diagnosed. Metastatic prostate cancer most frequently affects the bones, where it causes pain and fragility fractures.

**Causes**

• The aetiology of prostate cancer is unknown. However, a number of risk factors have been associated with development of the disease.

**Risk factors**

Risk factors for developing prostate cancer include:

**Increasing age**

Age is one of the strongest risk factors for prostate cancer. People are at higher risk if they are aged 50 years or older. Autopsy studies show that by the age of 80 years, 70% of men will have evidence of prostate cancer.

**Black ethnicity**

There is a higher incidence of, and mortality from, prostate cancer in men of black ethnicity than in white men. The reason for this is unknown.

**Family history of prostate cancer and genetics**

People are at higher risk if they have a close relative, for example a brother or father, who has had prostate cancer.

**Body weight and body mass index (BMI)**

the relationship between BMI and prostate cancer remains unclear.

**Prognosis**

Prostate cancer is curable, and the prognosis depends on the initial stage at diagnosis:

- Prostate cancer has a 5 years survival rate of over 95% when diagnosed at stage 1–3, compared with other cancers. However, for the 1 in 5 people diagnosed with stage 4 prostate cancer (metastatic), the 5 years survival rate is just 49%.

**What is the Gleason score?**

The Gleason scoring system is an internationally recognized grading system, based on examination of prostate tissue. It estimates the grade of prostate cancer according to its architectural differentiation.

When the prostate is biopsied, 10–12 cores are taken from different parts of the gland (5 or 6 per side). The two most common tumour patterns are analyzed and graded from 1–5:

- Grade 1: small, uniform glands with minimal nuclear changes.
- Grade 2: medium-sized acinii, separated by stromal tissue but more closely arranged.
- Grade 3: marked variation in glandular size and organization and infiltration of stromal and neighbouring tissues.
- Grade 4: marked atypical cytology with extensive infiltration.
- Grade 5: sheets of undifferentiated cells.

The Gleason score is the sum of these two grades. For example, if the grade given to the most common growth pattern is 4 and the grade given to the second most common growth pattern is 4, the total Gleason score is 8 (4 + 4).

A Gleason score of 2 is the most well differentiated tumour, and 10 is the most poorly differentiated. Lower scores are associated with a better prognosis than higher scores. Gleason scores under 6 are rarely recorded.

Low-grade tumour: Gleason score ≤6.
Intermediate-grade tumour: Gleason score 7.
High-grade tumour: Gleason score 8–10.

**Complications**

The complications of prostate cancer include:

**Local invasion**

- Prostate cancer can be locally invasive and spread to the seminal vesicles, base of the bladder, urethral sphincter, or side wall of the pelvis.

**Distant metastases**

- Metastasizing prostate cancer most commonly spreads to the bones, where it can cause pain, pathological fractures, or spinal cord compression.

**Lower urinary tract symptoms (LUTS)**

- Early prostate cancer does not usually cause LUTS.
- By the time prostate cancer causes LUTS, it may be advanced and incurable.

**Diagnosis**

How should I assess a person with suspected prostate cancer?

Suspect prostate cancer in men who have any of the following symptoms that are unexplained:

- Lower back or bone pain.
- Lethargy.
- Erectile dysfunction.
- Visible haematuria.
- Anorexia/weight loss.
- Lower urinary tract symptoms (LUTS), such as nocturia, frequency, urgency, hesitancy, retention, terminal dribbling, and/or overactive bladder — these symptoms are common in older men and early prostate cancer will not usually produce these symptoms. However, locally advanced prostate cancer may cause obstructive LUTS.

**Lower back pain, bone pain, weight loss (possible symptoms of advanced prostate cancer).**

Offer a digital rectal examination (DRE).

- DRE allows assessment of the prostate for signs of prostate cancer (a hard gland, sometimes with palpable nodules) or benign enlargement (smooth, firm, enlarged gland).
- Note that a gland that feels normal does not exclude a tumour.

Consider prostate-specific antigen (PSA) testing.

- Before offering PSA testing, provide appropriate information and advice to enable the person to make an informed choice about testing. This should include information on the benefits and limitations of PSA testing, in addition to relevant decision aids and online sources of information.
- If appropriate, arrange testing.
- A normal PSA level ranges from 0–4 nanograms/mL. However, the upper level of normal may vary according to age and race.
- PSA testing should not be offered to asymptomatic people.

Refer the person using a suspected cancer pathway referral (for an appointment within 2 weeks) if their prostate feels malignant on DRE.

Consider referring a person with possible symptoms of prostate cancer using a suspected cancer pathway referral (for an appointment within 2 weeks) if their PSA level is above the threshold for their age in Table 3. Take into account the person's preferences and any comorbidities when making the decision.

Secondary care assessment for people with suspected prostate cancer may include:

- A prostate biopsy — to confirm or exclude the diagnosis.
- Imaging — to assess the stage of the prostate cancer.

**Prostate-specific antigen (PSA) testing**

Prostate-specific antigen (PSA) is a protein produced by the prostate gland. It is secreted by prostate epithelial cells into prostatic fluid, where its function is to liquefy semen and thus allow spermatozoa to move more freely. Although PSA is secreted into prostatic fluid and semen, small amounts are present in blood. Men with prostate cancer tend to have higher levels of PSA in their blood — as a result of altered prostate architecture in prostate cancer, more PSA leaks out increasing the levels in the blood. However, blood PSA is an inaccurate marker for prostate cancer because cancer can be present without increased PSA levels, and PSA levels may be increased by conditions other than cancer (such as benign prostatic enlargement, prostatitis, and urinary tract infection). Also, PSA levels tend to increase naturally with age. The aim of PSA testing is to detect localized prostate cancer when treatment can be offered that may cure cancer or extend life. It is not usually recommended for asymptomatic men with less than 10 years' life expectancy. Most men will have a PSA level less than 3 nanograms/mL. About 3 in 4 men with a raised PSA level (3 nanograms/mL or higher) will not have cancer. Around 15% of men with a normal PSA do have cancer. A PSA test will not distinguish between aggressive tumours (which are at an early stage but will develop quickly) and those which are not. Further tests may provide valuable information.

• **PSA testing should be:**
- Considered in men with suspected prostate cancer.
- Offered to men older than 50 years of age who request a PSA test.

• **Before a PSA test, people should not have**:
- An active urinary infection or within previous 6 weeks.
- Ejaculated in previous 48 hours.
- Exercised vigorously, for example cycling, in the previous 48 hours.
- Had a urological intervention such as prostate biopsy in previous 6 weeks.

• **After taking blood for PSA testing**:
The specimen must reach the laboratory within 16 hours. Information and advice on prostate-specific antigen (PSA) testing. Before offering prostate-specific antigen (PSA) testing, appropriate information should be provided to enable the person make an informed choice about testing.

• **Benefits of PSA testing include:**
- Early detection — PSA testing may lead to prostate cancer being detected before symptoms develop.
- Early treatment — detecting prostate cancer early before symptoms develop may extend life or facilitate a complete cure.

• **Limitations and risks of PSA testing include:**
- False-negative PSA tests — about 15% of people with a normal PSA level (less than 3 nanograms/mL) may have prostate cancer, and 2% will have high-grade cancer. However, it is not known what proportion of these cancers become clinically evident.
- False-positive PSA tests — about 75% of people with a raised PSA level (3 nanograms/mL or higher) have a negative prostate biopsy.
- Unnecessary investigation — a false positive PSA test may lead to invasive investigations, such as prostate biopsy, and there may be adverse effects (for example bleeding or infection).
- Unnecessary treatment — slow-growing tumours are common and may not cause any symptoms or shorten life. Some tested men may therefore face unnecessary diagnosis (overdiagnosis) of prostate cancer, as well as associated anxiety, medical tests, and treatments with adverse effects.

• The Public Health England (PHE) patient information sheet PSA testing and prostate cancer: advice for well men aged 50 and over summarizes the potential benefits and risks of PSA testing.

**What should I advise about screening for prostate cancer?**

• Screening for prostate cancer is currently not recommended in the UK. This is because:
- The prostate-specific antigen (PSA) test is not appropriate as a screening tool as it is neither sensitive nor specific for this purpose. It would generate too many false positives and false negatives. Resulting in potential unnecessary and invasive tests and/or unnecessary treatment.

- There is no single treatment that is quantifiably superior in terms of outcomes for patients with early-stage prostate cancer.
- It is unclear how PSA screening might impact prostate cancer outcomes, specifically death due to prostate cancer. It is still unclear if other tests, such as MRI scanning (with or without PSA), may be more accurate.

**Management**

**From age 30 years onwards (Male).**

Treatment for prostate cancer will be initiated and managed in secondary care. Follow up and monitoring, as well as early recognition and initial management of complications of the condition and adverse effects of its treatment can be carried out in primary care under shared care arrangements. Management options in secondary care depend on the man's prognostic risk.

This is determined by taking into account:

- The clinical staging of the disease.
- Results of the prostate-specific antigen (PSA) test.
- The Gleason score.
- The person's age, comorbidities, and preference.

Treatment options **include:**

- Watchful waiting.
- Active surveillance.
- Radical treatments, such as radical prostatectomy, external beam radiotherapy, and brachytherapy.
- Adjunctive and palliative treatments, such as hormone therapy, chemotherapy, or bisphosphonates.

Watchful waiting is part of a strategy for 'controlling' rather than 'curing' prostate cancer and is aimed at people with localized prostate cancer who are either not suitable for, or do not ever wish to receive, curative treatment, and instead involves the deferred use of hormone therapy.

Watchful waiting avoids the use of surgery or radiation, but implies that curative treatment will not be available; people on watchful waiting who require treatment would receive long-term hormone therapy to control their cancer. A significant number of people on watchful waiting follow up need no treatment at all during the rest of their lives.

Watchful waiting is often suitable for older men, men with significant comorbidities, or those with slowly progressing tumours who are likely to die of other causes and not suffer significant morbidity from their prostate cancer. It is an option for men in any prognostic risk group.

Active surveillance is part of a 'curative' strategy and is aimed at people with localized prostate cancer for whom radical treatments are suitable, keeping them within a 'window of curability' whereby only those whose tumours are showing signs of progressing, or those with a preference for intervention are considered for radical treatment.

Active surveillance may avoid or delay the need for radiotherapy or surgery.

the difference from watchful waiting is that repeating the prostate biopsy at intervals depending on age and prostate-specific antigen (PSA) levels allows the man's prognostic risk category to reassessed. The aim of active surveillance is to safely reduce the risk of over treatment, as treatment is offered only when the risk increases.

Radical treatments include radical prostatectomy, external beam radiotherapy, and brachytherapy.

Radical prostatectomy is the surgical removal of the entire prostate gland and lymph nodes. This can be done by an open approach or by keyhole technique (laparoscopic or robotically assisted laparoscopic prostatectomy).

External beam radiotherapy (EBRT) is radiotherapy given by using ionizing radiation (for example, high-energy X-rays) produced in a machine and directed at the tumour from outside the person.

Brachytherapy is a type of radiotherapy in which the radiation is given using either permanently implanted radioactive seeds (low dose rate) or temporarily inserted radioactive sources (high dose rate) directly into the prostate.

Adjunctive and palliative treatments include hormone therapy, chemotherapy, and bisphosphonates.

Hormonal treatments are the treatment of cancer by removing and/or blocking the effects of hormones which stimulate the growth of prostate cancer cells. Hormonal treatments used for prostate cancer include:

- Androgen deprivation — a treatment that lowers testosterone levels, such as surgery (bilateral orchidectomy) or treatment with luteinizing hormone-releasing hormone (LHRH) agonists (such as goserelin, leuprorelin, triptorelin), or antagonists (such as degarelix).
- Androgen blockade — the use of drugs that bind to and block the hormone receptors of cancer cells (for example, cyproterone acetate), thus preventing androgens from stimulating cancer growth.

Chemotherapy is an option for men with hormone-relapsed metastatic disease. For example:

- Docetaxel in combination with prednisolone is licensed for hormone-resistant metastatic prostate cancer.
- Cabazitaxel in combination with prednisone or prednisolone is an option for treating metastatic hormone-relapsed prostate cancer in people whose disease has progressed during or after docetaxel.
- A corticosteroid, such as dexamethasone, is an option as third-line therapy after androgen withdrawal and anti-androgen therapy for people with hormone-refractory prostate cancer.

Bisphosphonates are calcium-regulated drugs which inhibit bone resorption, used in the treatment of hypercalcemia, osteoporosis, and bone pain.

Bisphosphonates should be offered to people who are having androgen deprivation therapy and have osteoporosis. They can also be considered for pain relief in people with hormone-refractory prostate cancer when other treatments (including analgesics and palliative radiotherapy) have failed.

Bisphosphonates should not be routinely offered to prevent osteoporosis in people with prostate cancer taking androgen deprivation therapy. They should not be offered to prevent bone metastases or to reduce their complications.

**How should I follow up and monitor a person with prostate cancer?**

People with prostate cancer should be followed up in primary care in line with locally agreed protocols.

For people who are being managed with watchful waiting:

- Measure prostate-specific antigen (PSA) levels at least once a year.
- Digital rectal examination (DRE) is not recommended on a routine basis while the PSA level remains at baseline values.
- People with localized prostate cancer who have chosen watchful waiting and who have evidence of significant disease progression (that is, rapidly rising PSA level or bone pain) should have their situation reviewed by a urological cancer specialist.
- Follow up people with prostate cancer who have chosen a watchful waiting regimen with no curative intent in primary care only if protocols for this have been agreed between the local urological cancer multidisciplinary team (MDT) and the relevant primary care organization(s).

**For people who are being managed with active surveillance:**

In year 1 of active surveillance:

- Every 3–4 months: measure prostate-specific antigen.
- Throughout active surveillance: monitor PSA kinetics (could include PSA density and velocity).
- At 12 months: perform a DRE.
- At 12–18 months: multiparametric MRI.

In year 2 and every year thereafter until active surveillance ends:

- Every 6 months: measure PSA.
- Throughout active surveillance: monitor PSA kinetics (could include PSA density and velocity).
- Every 12 months: perform a DRE.

If there is concern about clinical or PSA changes at any time during active surveillance, the person should be reassessed with multiparametric MRI and/or re-biopsy. For people who are being managed with radical treatment:

PSA levels should be measured no earlier than 6 weeks after treatment, then at least every 6 months for 2 years, and once a year thereafter. After at least 6 months' initial follow up, consider a remote follow-up strategy for people with a stable PSA who have had no significant treatment complications, unless they are taking part in a clinical trial that needs formal clinic-based follow up. Do not routinely offer DRE to people with localized prostate cancer who are not on active surveillance while their PSA remains at baseline levels.

**For all people with prostate cancer:**

Review and manage:

- Complications of the disease, including pain, lower urinary tract symptoms, and symptoms of spinal cord compression.
- Adverse effects from treatment, including sexual dysfunction and urinary incontinence. Adverse effects of androgen withdrawal include change in body shape and weight gain, tiredness, hot flushes, loss of libido, erectile dysfunction, gynaecomastia, and loss of bone density.
- Impact on quality of life.

If biochemical relapse (rising PSA level) occurs, calculate an estimate of PSA doubling time based on a minimum of three measurements over at least 6 months.

Refer men with evidence of significant disease progression (rapidly rising PSA level or bone pain) to a urological cancer specialist.

**How should I manage adverse effects of hormonal treatments in people with prostate cancer?**

**• Hot flushes**

- Offer medroxyprogesterone acetate 20 mg per day, initially for a period of 10 weeks (off-label use). Evaluate the effect at the end of this treatment period.
- Consider cyproterone acetate 50 mg twice a day for 4 weeks if medroxyprogesterone is not effective, or not tolerated.
- Explain that there is no strong evidence to support the use of complementary therapies to treat hot flushes.

**• Fatigue**

- Offer men who are starting or taking androgen deprivation therapy supervised resistance and aerobic exercise at least twice a week for 12 weeks to reduce fatigue and improve quality of life.

**Osteoporosis**

- Do not routinely offer bisphosphonates to prevent osteoporosis in men taking androgen deprivation therapy.
- Consider assessing the fracture risk in men taking androgen deprivation therapy.
- Offer bisphosphonates to men who have osteoporosis.
- Consider denosumab for men in whom bisphosphonates are contraindicated or not tolerated.

**• Gynaecomastia**

- For men starting long-term bicalutamide monotherapy (longer than 6 months), offer referral to a specialist for prophylactic radiotherapy to both breast buds within the first month of treatment.
- If radiotherapy is unsuccessful, consider weekly tamoxifen (off-label use).

**How should I manage radiation-induced enteropathy in a person with prostate cancer?**

Ensure that people with signs or symptoms of radiation-induced enteropathy are offered care from a healthcare team with expertise in radiation-induced enteropathy. Symptoms may include diarrhoea, faecal urgency, steatorrhoea, or rectal pain. Specialist assessment is likely to involve sigmoidoscopy to ascertain the nature of the radiation injury and exclude such conditions as inflammatory bowel disease and cancer of the large bowel.

**How should I manage sexual dysfunction in a person with prostate cancer?**

• Offer a phosphodiesterase-5 (PDE-5) inhibitor, such as sildenafil, tadalafil, or vardenafil, to men who experience loss of erectile function.

If PDE-5 inhibitors are ineffective or contraindicated, offer a choice of:

- Intraurethral inserts.
- Penile injections.
- Penile prostheses.
- Vacuum devices.

Ensure men have access to a sexual health specialist. Consider referring men (and their partner) for psychosexual counselling.

**How should I manage suspected spinal cord compression in a person with prostate cancer?**

• Suspected spinal cord compression is a medical emergency. It must be diagnosed and treated before there is significant neurological compromise.

**How should I manage urinary incontinence or retention in a person with prostate cancer?**

• Warn people undergoing radical treatment for prostate cancer of the likely effects of the treatment on their urinary function. Offer, men experiencing urinary symptoms before treatment a urological assessment. Ensure people with troublesome urinary symptoms after treatment have access to specialist continence services for assessment, diagnosis, and conservative treatment. Management options include coping strategies, pelvic floor muscle re-education, bladder retraining, and pharmacological treatment. Refer people with intractable urinary incontinence to a specialist surgeon for consideration of an artificial urinary sphincter.

**Investigation:**

The traditional investigation for suspected prostate cancer was a transrectal ultrasound-guided (TRUS) biopsy. However, recent guidelines from NICE have now advocated the increasing use of multiparametric MRI as a first-line investigation.

**Multiparametric MRI is now the first-line investigation for people with suspected clinically localised prostate cancer.**

- the results are reported using a 5-point Likert scale
  If the Likert scale is >=3 a multiparametric MRI-influenced prostate biopsy is offered
  If the Likert scale is 1-2 then NICE recommend discussing with the patient the pros and cons of having a biopsy.

TOPIC: acute prostatitis

**Definition:**

• Acute bacterial prostatitis is a severe, potentially life-threatening bacterial infection of the prostate.
• Acute bacterial prostatitis is accompanied by infection of the urinary tract.

**Causes**

• Acute bacterial prostatitis is caused by urinary pathogens, most commonly Escherichia coli (up to 50% of infections), followed by Pseudomonas aeruginosa, Klebsiella, Enterococcus, Enterobacter, Proteus and Serratia species.
• Rarely, it can occur secondary to a sexually transmitted infection (STI) such as Chlamydia trachomatis, or Neisseria gonorrhoea.
• Acute prostatitis can follow urethral instrumentation, trauma, bladder outflow obstruction, or dissemination of infection from elsewhere in the body. Men who have acute prostatitis following manipulation of the lower urinary tract are more likely to be infected with pathogens other than E. coli, to have multiple infections, and to develop a prostatic abscess.

**Prognosis**

• Acute bacterial prostatitis usually responds well to antibiotic treatment — time to resolution of symptoms depends on the initial severity of the disease. However:

**Complications**

- Acute urinary retention — this is a common complication of prostatitis and may be a presenting feature.
- Bacteraemia.
- Chronic prostatitis
- Epididymitis.
- Prostatic abscess — this is a rare complication which may require surgical intervention. Risk factors include long-term urinary catheterization, recent urethral manipulation, and an immunocompromised state.
- Pyelonephritis
- Sepsis.

**Diagnosis**

• **Suspect acute prostatitis in a man who presents with signs and symptoms of**:

A urinary tract infection (UTI):

- Dysuria, frequency, urgency.

Prostatitis:

- Perineal, penile, or rectal pain.
- Acute urinary retention, obstructive voiding symptoms (difficulty voiding, hesitancy, straining to urinate, weak stream).
- Low back pain, pain on ejaculation.
- Tender, swollen, warm prostate (on gentle rectal examination).

Bacteraemia:

- Rigors, arthralgia, or myalgia.
- Fever, tachycardia.

**How should I diagnose acute prostatitis?**

In men suspected of having acute prostatitis:

- Arrange collection of a mid-steam urine (MSU) sample to confirm urinary tract infection (UTI) by dipstick, culture and sensitivity.
  - Do not collect prostatic secretions as prostatic massage may lead to sepsis or prostatic abscess, is likely to be very painful, and is not needed for the diagnosis.
- Arrange blood cultures and full blood count.

- Conduct a physical examination — this should include the abdomen to detect a distended bladder or costovertebral angle tenderness, a genital examination, and a digital rectal examination (DRE).
    - ❖ In men with acute bacterial prostatitis, the prostate will be tender, enlarged, or boggy.
    - ❖ A DRE should be performed gently because vigorous prostatic massage can lead to sepsis.
- Consider screening for sexually transmitted infections (STIs), particularly in men considered to be at risk.
- Consider and exclude other causes of symptoms.

**Differential diagnosis**

- Benign prostatic hyperplasia (BPH) — typically presents with a gradual reduction in urinary flow, hesitancy, frequency, and nocturia. May also present with acute retention of urine.
- Chronic prostatitis — consider this if the symptoms have been present for several weeks or months.
- Urinary tract infection — there are usually no symptoms of bladder outflow obstruction unless there is coexisting BPH or prostatic malignancy.
- Acute unilateral or bilateral epididymo-orchitis — consider these if the scrotum, testis, or epididymis are painful or swollen, there will usually also be symptoms of dysuria and frequency.
- Prostate cancer — may present with similar symptoms to BPH.
- Bladder cancer — usually presents with haematuria, and there may be dysuria and urinary frequency.
- Colorectal cancer — typically presents with a change in bowel habit and there may be rectal bleeding or weight loss.

**Management**

**From age 18 years onwards (Male).**

• Admit to hospital if the man:

- Is unable to take oral antibiotics.
- Has severe symptoms.
- Has signs or symptoms of a more serious condition (for example sepsis, acute urinary retention or prostatic abscess).

• Consider urgent referral for any man who:

- Is immunocompromised or has diabetes mellitus.
- Has a pre-existing urological condition (such as benign prostatic hypertrophy or an indwelling catheter) — specialist urological management may be required.

Start oral antibiotic treatment, taking into account local antimicrobial resistance data. Prescribe an oral antibiotic for 14 days:

- Ciprofloxacin 500 mg twice daily or ofloxacin 200 mg twice daily first line, or if they are unsuitable trimethoprim 200 mg twice daily.
- Levofloxacin 500 mg once daily, or co-trimoxazole 960 mg twice daily (when there is bacteriological evidence of sensitivity and good reasons to prefer this combination to a single antibiotic) second-line.

Advise men:

- To take paracetamol (with or without a low-dose weak opioid, such as codeine) for pain, or ibuprofen if this is preferred and suitable.
- To drink enough fluids to avoid dehydration.
- About the usual course of acute prostatitis (several weeks).
- About possible adverse effects of the antibiotic — fluoroquinolones can very rarely cause long-lasting (up to months or years), disabling, and potentially irreversible adverse effects, sometimes affecting multiple systems, organ classes, and senses.

Advise them to stop treatment with a fluoroquinolone antibiotic at the first signs of a serious adverse reaction, such as tendonitis or tendon rupture, muscle pain, muscle weakness, joint pain, joint

swelling, peripheral neuropathy, and central nervous system effects, and to contact their doctor immediately. Give them the patient information sheet Fluoroquinolone antibiotics (-oxacins): what you need to know about side effects of tendons, muscles, joints, and nerves.

Arrange follow up after 48 hours to check the man's response to treatment and the urine culture results.

- Review the choice of antibiotic and change it according to susceptibility results if the bacteria are resistant — use a narrow-spectrum antibiotic wherever possible.
- Admit to hospital if symptoms have not improved 48 hours after starting antibiotic treatment.
- If a sexually transmitted infection (STI) is identified, refer urgently to a genito-urinary medicine (GUM) clinic. If there is likely to be a delay before the person is seen, seek advice from a GUM specialist regarding interim treatment.

- Review antibiotic treatment after 14 days and either stop treatment, or continue for an additional 14 days based on an assessment of history, symptoms, clinical examination, urine and blood tests.
- Following recovery, refer for investigation to exclude structural abnormality of the urinary tract.

TOPIC: chronic prostatitis

**Definition**

• Chronic prostatitis is defined as at least 3 months of urogenital pain, which may be perineal, suprapubic, inguinal, rectal, testicular, or penile and is often associated with lower urinary tract symptoms (such as dysuria, frequency, hesitancy, and urgency), and sexual dysfunction (erectile dysfunction, painful ejaculation, or postcoital pelvic discomfort).
• Chronic prostatitis can be further classified as:
  - Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS) (sometimes also referred to as abacterial prostatitis or prostate pain syndrome) — accounts for over 90% of men with chronic prostatitis (there is no proven bacterial infection).
  - Chronic bacterial prostatitis (CBP) — less than 10% of men with chronic prostatitis.
• Prostate pain syndrome — the European Association of Urology (EAU) uses the term 'prostate pain syndrome' (PPS) instead of CP/CPPS. This term is not used in the UK, therefore CKS uses the terms CP/CPPS in this topic.

**Management**

**From age 16 years onwards (Male).**

• Explain to the man that:
  - The cause is not always understood, but is thought to be multifactorial.
  - The condition is chronic and treatment can be difficult, but most men notice improvement within six months.
  - Treatment is often more about controlling symptoms rather than effecting an immediate cure.
• Reassure the man about the nature of the disease and that chronic prostatitis is not cancer and is very rarely caused by a sexually transmitted infection.
• For men with suspected chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS):
  - Consider using the Urinary, Psychosocial, Organ-specific, Infection, Neurological/systemic, and Tenderness (UPOINT) classification system to stratify patients into specific symptom-led phenotypes. The, UPOINT approach to symptom evaluation can direct which treatments to use. Most men with CP/CPPS require multimodal treatment aimed at the main symptoms (taking comorbidity into account). Options include offering: Paracetamol and/or a nonsteroidal anti-inflammatory drug (NSAID) for pain relief. **Do not prescribe opioids.**

An alpha-blocker for 4–6 weeks if significant lower urinary tract symptoms (LUTS) are present.
Targeted cognitive behavioural therapy, counselling and antidepressants for men with psychosocial symptoms (for example, depression, stress and poor coping mechanisms).
A single course of antibiotics, if symptoms have been present for less than 6 months. Antibiotic options include:
  - Trimethoprim 200 mg twice a day for 4–6 weeks, or
  - Doxycycline 100 mg twice daily for 4–6 weeks.

A stool softener such as lactulose or docusate if defecation is painful.
Acupuncture.
• Refer to a urologist if:
  - There is diagnostic uncertainty.
  - Symptoms are severe (use clinical judgement to determine the urgency of referral).
  - Symptoms persist after initial management.
• For men with suspected chronic bacterial prostatitis (a history of urinary tract infection, or an episode of acute prostatitis within the last 12 months):
  - Refer to a urologist for specialist assessment (use clinical judgement to determine the urgency of referral).
  - If defecation is painful — offer a stool softener such as lactulose or docusate.
  - If the man is in pain, prescribe paracetamol and/or a nonsteroidal anti-inflammatory drug (NSAID).

TOPIC: acute pyelonephritis

**Definition**

Pyelonephritis is an infection of one or both kidneys usually caused by bacteria from the bladder.

**Causes**

The most common causative pathogens of acute pyelonephritis are gram-negative bacteria:

- Escherichia coli (60-80% of uncomplicated infections).
- Klebsiella species (20%).
- Proteus mirablis (15%).
- Pseudomonas species.
- Enterobacter species.

**Complications**

- Sepsis.
- Parenchyma renal scarring.
- Recurrent urinary tract infections.
- Renal abscess formation.
- Preterm labour in pregnancy.
- Emphysematous pyelonephritis.

The risk of developing a complication is increased in people with:

- Severe illness, including hypotension, tachycardia, reduced levels of consciousness, or dehydration.
- Age over 65 years.
- Abnormalities of renal tract anatomy and function (such as vesico-ureteric reflux and polycystic kidney disease).
- Foreign body within the renal tract, including renal stones and urinary, ureteric, or nephrostomy catheters.
- Immunocompromise, for example, due to immunosuppressant drug use, cancer, cancer therapies, or AIDS.
- Diabetes mellitus.
- Pregnancy.
- Persistent pyelonephritis despite treatment.
- Renal impairment.

**Prognosis**

Acute pyelonephritis usually responds well to antibiotic therapy — time to resolution of symptoms depends largely on the initial severity of disease. In the majority of cases, prompt diagnosis and appropriate treatment result in a complete and uncomplicated recovery within days to weeks.

The prognosis is less favourable for older people and those with complicating factors or underlying renal disease.

**Diagnosis**

Acute pyelonephritis is diagnosed by taking a detailed medical history and a physical examination.

Suspect acute pyelonephritis in people with signs or symptoms of a urinary tract infection (for example, dysuria, frequency, urgency) accompanied by any signs or symptoms of pyelonephritis (including fever, nausea, vomiting, or flank pain). In all people suspected of having acute pyelonephritis, arrange collection of a mid-steam urine (MSU) or catheter specimen of urine (CSU), to determine the infecting micro-organism. Obtain a urine sample for culture before starting empirical drug treatment. Dipstick testing of urine for nitrites and leukocytes is not necessary, however it may be a useful adjunct to guide diagnosis in some clinical situations. It should not be used in people:

- With an indwelling catheter — make a working diagnosis based on clinical judgement.
- Aged over 65 years — dipsticks become more unreliable with increasing age over 65 years.

• A definitive diagnosis of acute pyelonephritis is made in people with loin pain and/or fever if a UTI is confirmed by culturing a urinary pathogen from the urine, and other causes of loin pain and/or fever have been excluded.

**What are the signs and symptoms of acute pyelonephritis?**

There are no clinical features or routine investigations that conclusively distinguish acute pyelonephritis from lower urinary tract infection. However, the triad of flank pain (typically unilateral), fever, and nausea and vomiting occurs much more often in people with pyelonephritis than in those with lower urinary tract infection. Onset is typically sudden with signs and symptoms of both systemic inflammation and bladder inflammation.

Common signs and symptoms of pyelonephritis include:

- Flank/renal angle pain and/or tenderness.
- Myalgia.
- Flu-like symptoms.
- Rigors or raised temperature of 37.9°C or higher (or below 36°C in people aged over 65 years). Most people have fever, although it may be absent early in people with early or mild cases, frail, older people, or in the immunocompromised.
- Nausea/vomiting.

**Differential diagnosis**

- Acute abdominal conditions — especially if nausea and vomiting are prominent.
- Acute prostatitis
- Gynaecological conditions — however, typically there is no tenderness in the costovertebral region.
- Musculoskeletal disorders — suspect when costovertebral pain is a conspicuous feature.
- Lower lobe pneumonia — symptoms include cough and pleuritic chest pain. Physical examination may show decreased breath sounds, rales, or rhonchi.
- Lower urinary tract infection
- Pelvic inflammatory disease
- Pelvic pain syndrome — recurrent symptoms, including dysuria, pain on intercourse, and pelvic pain, occur with negative cultures.
- Shingles.

**Management**

**From age 16 years onwards.**

Admit people to hospital if they have any symptoms or signs suggesting a more serious illness or condition (for example, sepsis). Consider referring or seeking specialist advice for people with acute pyelonephritis if they:

- Are significantly dehydrated or unable to take oral fluids and medicines.
- Are pregnant.
- Have a higher risk of developing complications — people with known or suspected structural or functional abnormality of the genitourinary tract or underlying disease (such as diabetes mellitus, or immunosuppression).
- Have recurrent episodes of UTI (for example, two or more episodes in a 6-month period).

Also consider referral for:

- Men, following a single episode without an obvious cause.
- Women with recurrent pyelonephritis.

**How should I manage a person with suspected acute pyelonephritis in primary care?**

Admit people with acute pyelonephritis to hospital if it is severe, or they have any signs or symptoms which suggest a more serious illness or condition, for example, sepsis.

Signs of sepsis, include:

- Significant tachycardia, hypotension, or breathlessness.

- Marked signs of illness (such as impaired level of consciousness, perfuse sweating, rigors, pallor, significantly reduced mobility).
- A temperature greater than 38°C or less than 36°C.

**For all other people, offer an antibiotic, taking into account:**

- The severity of symptoms.
- The risk of developing complications, which is higher in people with known or suspected structural or functional abnormality of the genitourinary tract or immunosuppression.
- Previous urine culture and susceptibility results.
- Previous antibiotic use, which may have led to resistant bacteria.

**For women who are not pregnant, men, and people with indwelling catheters, take account of local antimicrobial resistance data, and prescribe either of the following first line options:**

- Cefalexin 500mg twice or three times a day (up to 1– 1.5g three or four times a day for severe infections) for 7-10 days.
- Co-amoxiclav (only if appropriate in line with culture and sensitivity results) 500/125 mg three times a day for 7-10 days.
- Trimethoprim (only if appropriate in line with culture and sensitivity results) 200mg twice a day for 14 days.
- Ciprofloxacin 500 mg twice a day for 7 days.

• **For pregnant women who do not require admission, prescribe:**

- Cefalexin 500mg twice or three times a day (up to 1– 1.5g three or four times a day for severe infections) for 7-10 days.

**For people with indwelling urinary catheters:**

- Check that the catheter is correctly positioned, drains correctly and is not blocked.
- Consider removing or, if this cannot be done, changing the catheter as soon as possible if it has been in place for more than 7 days.

Do not allow catheter removal or change to delay antibiotic treatment.

Do not give antibiotic prophylaxis for catheter changes unless the person has a history of symptomatic UTIs due to catheter change.

Review culture and sensitivity results when they become available, and change the antibiotic if indicated. If the bacteria are resistant and symptoms are not already improving, use a narrow-spectrum antibiotic wherever possible.

**Reassess the person if symptoms worsen at any time, or do not start to improve within 48 hours of taking the antibiotic, and consider:**

- Other possible diagnoses.
- Any symptoms or signs suggesting a more serious illness or condition, such as sepsis.
- Whether previous antibiotic use may have led to resistant bacteria.
- Whether referral for specialist urological assessment is needed

**What information and advice should I provide to people with pyelonephritis?**

• Explain how to take a mid-stream urine sample.

• Give people advice about:

- Possible adverse effects of the antibiotic, particularly diarrhoea and nausea.

Nausea with vomiting may also be a possible indication of worsening pyelonephritis.

- When to seek medical help — for example if symptoms worsen at any time, do not start to improve within 48 hours of taking the antibiotic, or if they become systemically very unwell.

**Advise people to:**

- Use paracetamol for pain, or if preferred and suitable, ibuprofen if required.
- Drink sufficient fluids to avoid dehydration.

TOPIC: Urinary tract infection (lower) - men

**Scenario: Recurrent UTI in men**

**From age 16 years onwards (Male).**

**How should I manage recurrent urinary tract infection in men?**

- If a man presents with recurrent episodes of lower urinary tract infection (UTI), for example, two or more episodes in the last 6 months:
    - Treat each episode as acute lower UTI, and ensure urine culture is arranged on each occasion.

TOPIC: Urological cancers

**Presentation**

**Prostate cancer:**

- Prostate cancer usually presents with lower urinary tract symptoms, including nocturia, urinary frequency, and hesitancy. Haematuria can occur, as can erectile dysfunction. Some prostate cancers present with disseminated disease, typically metastases to bone.
- The lower urinary symptoms overlap with those of benign prostatic hyperplasia – and the two conditions can co-exist. Digital rectal examination can help to differentiate the two, with hardness of the prostate or individual nodules being features suggestive of cancer.
- Prostate-specific antigen (PSA) testing is generally available in primary care, with age-specific raised values suggestive of cancer.
- Definitive diagnosis requires biopsy, often guided by imaging. This is performed in secondary care.

**Bladder cancer:**

- Several symptoms have been reported, with haematuria being the most common. Dysuria and urinary frequency are also features, especially when persistent.
- As haematuria is a symptom of several cancers, investigation strategies may need to consider more than one possible cancer site, such as kidney, prostate, or endometrium. Similarly, dysuria and urinary frequency may be misattributed to urinary tract infection, especially in the elderly.
- A diagnosis of bladder cancer is generally made by cystoscopy with biopsy, performed in secondary care. As bladder cancer shares some symptoms with other urological cancers, most haematuria clinics investigate with ultrasound before proceeding to cystoscopy.

**Renal cancer:**

- Renal cancer symptoms include haematuria, loin pain, urinary tract infections, or a mass in the flank.
- The symptoms overlap with other urological cancers, particularly bladder cancer.
- Most renal cancers are visible on ultrasound of the kidneys – a test that is available on referral to primary care.
- Definitive diagnosis of renal cancer requires histology, performed in secondary care

**Testicular cancer:**

- Testicular cancer usually presents as a change in the shape or texture of the testis. This may be painful.
- It can present as disseminated disease, particularly with lymph node spread.
- Testicular cancer can be seen on ultrasound of the testis, a test that is available in primary care.

**Penile cancer:**

- Penile cancer is usually seen as a raised lesion.
- It can be difficult to differentiate penile cancer from the commoner lesions seen with some sexually transmitted diseases.
- It is often possible to diagnose a typical penile cancer visually, but confirmation of the diagnosis is generally made by excision biopsy in secondary care.

## TOPIC: Scrotal pain and swelling

**Causes**

- Testicular torsion (torsion of spermatic cord) or torsion of a testicular or epididymal appendage.
- **Testicular cancer**
- **Squamous cell carcinoma of the scrotum**
- **Inguinal hernia**
- **Epididymo-orchitis**
- **Haematocele**
- **Epididymal cyst or spermatocele**
- **Hydrocele**
- **Varicocele**

**Other possible causes of scrotal pain and/or swelling not included in this CKS topic include:**

- Allergic contact dermatitis.
- Hidradenitis suppurativa.
- Idiopathic scrotal oedema (mainly in children), causing self-limiting swelling of the skin of the scrotum, with normal underlying testis and epididymis.
- Syphilitic gumma of the testis (round, hard, insensitive mass indistinguishable from a tumour).
- Pancreatitis (owing to fluid tracking down the retroperitoneal compartment into the inguinal canal and scrotal sac).
- Crohn's disease skin manifestations of erythema nodosum and pyoderma gangrenosum.
- Malignant metastases (mainly from penile lesions).
- Complications of peritoneal dialysis and ventriculo-peritoneal shunts.
- Referred pain from ureteric stone, aortic or common iliac artery aneurysm, lumbar spine, interstitial cystitis, or appendicitis.

**Testicular torsion**

This is a surgical emergency that describes twisting of the spermatic cord and its contents:

- Testicular torsion can occur in children of any age, and incidence peaks in the neonatal period, with a second peak around puberty between 13–16 years of age. It may also occur prenatally.
- Extravaginal torsion predominates in utero and in neonates, where there is twisting of the entire cord, including the processus vaginalis. This occurs before the testis is fixed in the scrotum by the gubernaculum. The exact cause is unknown, and an anatomical defect is not usually identified. The prognosis for testicular viability in neonates is poor.
- In older children and adults, testicular torsion is usually intravaginal, where there is twisting of the cord within the tunica vaginalis. This is commonly due to a 'bell clapper deformity', where there is abnormal fixation of the posterior lateral aspect of the testes to the tunica vaginalis, allowing the testicle to swing and rotate within the tunica vaginalis. Presence of a testicular tumour increases the risk of testicular torsion.

Complications depend on the time between symptom onset and detorsion, and the degree of cord twisting, and include:

- Testicular atrophy, ischaemia, and necrosis.
- Segmental ischaemia of the testicle (if there is chronic intermittent torsion).
- Subfertility or infertility.
- Torsion of the contralateral testis (40% risk of torsion if fixation of the contralateral side is not performed).

Torsion of the testicular or epididymal appendage (also known as torsion of the appendix testis, appendix epididymis, or testicular hydatids)

The appendix testis and appendix epididymis are embryological remnants at the upper pole of the testis. Torsion of testicular or epididymal appendages can occur spontaneously at any age, but is more common in pre-pubertal children.

It is usually self-limiting and can be managed conservatively.

**Testicular cancer**

Testicular tumours account for 1–2% of all paediatric solid tumours, the majority of which are benign. Testicular cancer accounts for 1% of adult cancers and 5% of adult urological tumours.

It usually presents in people aged 18–40 years at diagnosis. It is the most common solid tumour detected in males aged 15–34 years. The majority of pre-pubertal testicular tumours are yolk sac tumours (malignant germ cell tumours) and teratomas (usually benign in pre-pubertal children). Around 90–95% of post-pubertal tumours are germ cell tumours, mainly teratomas and seminomas.

**Predisposing or associated factors include:**

- White ethnicity.
- Family history of testicular cancer in a first-degree relative.
- Undescended testis (cryptorchidism), even after orchidopexy surgery.
- Hypospadias.
- Sub- or infertility.
- Klinefelter's syndrome.
- A contralateral tumour.
- Testicular intraepithelial neoplasia.

**Squamous cell carcinoma of the scrotum**

Squamous cell carcinoma of the scrotum is extremely rare.

- It may be associated with occupational exposure to carcinogenic polycyclic aromatic hydrocarbons, for example in car mechanics, car and aeroplane manufacture, gas workers, engineers, steel manufacture, and aluminium workers.
- It can also occur as a complication of human papillomavirus (HPV) infection or PUVA (psoralen plus ultraviolet A) phototherapy for psoriasis.
- the median age at diagnosis is 52–57 years.

**Inguinal hernia**

An inguinal hernia is a protrusion of the abdominal or pelvic contents through a dilated internal inguinal ring alongside the spermatic cord, which also extends down the inguinal canal into the scrotum.

Inguinal hernias are categorized as either direct or indirect, based on the relationship of the hernia sac to the inferior epigastric artery.

Indirect inguinal hernias usually occur because of a persistent processus vaginalis. If an inguinal hernia extends into the scrotum, it is almost always indirect.

**Possible risk factors for inguinal hernia include:**

- Positive family history.
- Previous contralateral hernia.
- Increased age.
- Connective tissues disorders (such as Marfan's syndrome or Ehlers-Danlos syndrome with abnormal collagen metabolism).

**Complications are rare, but include;**

- Incarceration — this describes an inability to reduce the hernia mass into the abdomen, and typically presents with severe pain over a swelling in the groin or scrotum.
- Strangulation — this describes when the blood supply to the herniated tissues is compromised, causing ischaemia. Clinical features include an acutely painful, firm, tender, irreducible mass.
- Intestinal obstruction — if a hernia contains small bowel that becomes obstructed, clinical features include vomiting, abdominal pain, and swelling.

- Recurrence after surgical repair — recurrence rates of up to 15% have been reported in the literature.
- Chronic pain — pain lasting more than 3 months may occur in 10–12% of cases in the literature.

**Epididymo-orchitis**

Epididymo-orchitis is a clinical syndrome consisting of pain, swelling, and inflammation of the epididymis and/or testis.

Epididymitis or orchitis may occur alone or in combination. In studies, over 50% of people with epididymitis also have orchitis.

It may be due to infectious or non-infectious causes, depending on the person's age and risk factors:

- In pre-pubertal children, epididymo-orchitis is usually idiopathic and self-limiting.
- It may be secondary to recent adenovirus, enterovirus, or other viral infection.
- It may be due to reflux of urine into the ejaculatory ducts, with associated structural urogenital abnormalities.
- Rarely, it is caused by enteric organisms that cause urinary tract infections (UTIs), which may be associated with structural or functional abnormalities of the urinary tract in children.

In sexually active people aged 14–35 years, it is usually caused by local spread of sexually transmitted infections (STIs) such as Chlamydia trachomatis or Neisseria gonorrhoeae from the urethra.

In people aged 35 years or older with a low-risk sexual history, it is usually caused by enteric organisms such as Escherichia coli that cause UTIs which spread from the bladder.

People at increased risk include those with structural abnormalities of the urinary tract, recent urological surgery or instrumentation including urinary catheterization, positive urine dipstick test, and men who have insertive anal sex. Rare causes of epididymo-orchitis include mumps infection (causes unilateral or bilateral parotid swelling which precedes usually unilateral orchitis, often in young adults), Behçet's disease, vasculitis (such as Henoch-Schonlein purpura in children), brucellosis, candida infection, tuberculosis infection, or an adverse effect of amiodarone drug treatment.

**Complications are usually associated with enteric causative organisms, and include:**

- Sepsis.
- Abscess formation and ischaemia/infarction of the testicle.
- Reactive hydrocele.
- Testicular atrophy and subfertility, for example due to mumps orchitis.
- Chronic scrotal pain.

**Haematocele**

- A haematocele typically results from direct trauma to the scrotum (including iatrogenic trauma from aspiration of a hydrocele), although idiopathic cases can occur.

**Epididymal cyst or spermatocele**

- Epididymal cysts and spermatoceles are benign, usually small, non-painful cystic swellings of the epididymis, which may be multiple and are frequently bilateral. The cause is unknown. If the cyst contains spermatozoa (usually seen on histology only), it may be referred to as a spermatocele.

**Hydrocele**

A simple hydrocele is an abnormal collection of serous fluid between the parietal and visceral layers of the tunica vaginalis membrane that surrounds the testis, or along the spermatic cord.A communicating hydrocele results from patency of the processus vaginalis which allows peritoneal fluid to freely communicate with the scrotal portion of the processus. The processus vaginalis persists in 80–94% of newborns and 20% of adults. An abdomino-scrotal hydrocele is rare, when a simple hydrocele enlarges through the inguinal canal resulting in an abdominal component. A hydrocele of the spermatic cord occurs when the processus vaginalis closes segmentally, trapping fluid anywhere along the spermatic cord. Most congenital simple hydroceles spontaneously resolve by 12 months of age. Hydrocele can be complicated by non-incarcerated inguinal hernia due to a large open processus vaginalis which allows passage of abdominal viscera (rare).

**Possible causes of a late-onset, non-communicating hydrocele in adolescents and adults include:**

- Minor trauma.
- Infection and epididymitis.
- Testicular torsion.
- Varicocele surgery (due to ligation of the lymphatics).
- Testicular cancer.

**Varicocele**

A varicocele is an abnormal dilatation of the testicular veins in the pampiniform plexus of the spermatic cord, caused by venous reflux:

- It is often asymptomatic and is unusual in children under 10 years of age. It is present in about 15% of adolescents and adults.
- About 90% occur on the left side, and about 10% of cases are bilateral
- There may be associated fertility problems in 20% of affected adolescents

**Diagnosis:**

A diagnosis of testicular torsion should be suspected in any person presenting with acute scrotal pain and/or swelling, before other causes are considered.

**Ask about:**

- Any scrotal pain — the location (including unilateral or bilateral), nature, radiation to surrounding structures, speed of onset, duration, severity, exacerbating factors (such as activity or positional changes).
- Any associated scrotal swelling or mass.
- Any previous episodes (previous severe, self-limiting scrotal pain and swelling may occur with intermittent testicular torsion and detorsion).
- Associated symptoms, such as nausea or vomiting (common with torsion, possible with epididymo-orchitis); abdominal pain (may suggest torsion); gynaecomastia (may be associated with some testicular tumours); urinary symptoms (may suggest epididymo-orchitis); urethral discharge (may suggest epididymo-orchitis or urethritis); parotid swelling (suggests mumps orchitis); back or flank pain, breathlessness, or weight loss (may be associated with metastatic testicular cancer).
- Current and previous sexual history (if appropriate).
- History of trauma (commonly associated with haematocele; rarely associated with testicular torsion) or strenuous physical activity (rarely associated with testicular torsion).
- Previous abdominal or pelvic surgery including inguinal hernia repair, varicocele surgery, or surgery for undescended testis; previous urological instrumentation or urinary tract infection (UTI, may suggest epididymo-orchitis).

**Examine the scrotum and inguinal region with the person standing and lying down, starting with the normal, less painful side, to assess:**

- Position of the swelling in relation to the testis (testicular, extra-testicular).
- Testicular lie (suspect testicular torsion if high-riding or transverse).
- Size of the testis (may be enlarged with a testicular tumour).
- Symmetry of the testes (with torsion the epididymis may be located anteriorly).
- Testicular tenderness (present in torsion and epididymo-orchitis).
- Consistency of the swelling (may be solid with testicular cancer, soft with a hydrocele).
- The presence of the cremasteric reflex (gentle pinching or stroking of the medial thigh usually causes elevation of the ipsilateral testicle; may be absent in torsion).
- 'Prehn sign' (relief of pain with elevation of the testes) — may suggest epididymitis, but does not rule out testicular torsion.

- Transillumination of the testis (indicative of hydrocele; a haematocele may transilluminate to a lesser extent).
- Features of an inguinal hernia, such as a positive cough impulse and/or palpable bulge in the inguinal canal. If a hernia is confirmed, assess whether it is reducible or irreducible.
- Supraclavicular lymphadenopathy or abdominal mass (retroperitoneal lymphadenopathy) — may be associated with testicular cancer.

**Other clinical features, such as:**

- Fever or other systemic symptoms which may indicate a complication associated with epididymo-orchitis.
- An erythematous nodule, plaque, or ulcer on the scrotal skin which may indicate skin cancer (rare).
- Scrotal skin erythema and/or urethral discharge which may indicate epididymo-orchitis.
- A 'blue dot sign', where an inflamed and ischaemic torted appendage can be seen through the scrotal skin.

Arrange an urgent ultrasound scan of the scrotum, provided a scrotal swelling is not of acute onset, if:

- There is diagnostic uncertainty.
- There are persistent, unexplained testicular symptoms.
- It is not possible to distinguish if the scrotal swelling is testicular or extra-testicular.
- A hydrocele is detected in a person aged 18–40 years.
- There is a history of trauma and scrotal pain.
- A haematocele does not follow trauma or is chronic.

Do not arrange an ultrasound scan of the scrotum if:

- A scrotal swelling is of acute onset and there is suspected testicular torsion, acute epididymo-orchitis, or a strangulated inguinal hernia — urgent hospital admission or referral is indicated.
- A testicular tumour is suspected clinically — arrange an urgent urology referral (using a 2-week wait pathway).

Assess the need to test for sexually transmitted infections (STIs) if acute epididymo-orchitis is suspected in an adolescent or adult.

Assess the need to test for a UTI with a urine dipstick test and mid-stream urine (MSU) for microscopy and culture, if acute epididymo-orchitis is suspected in a pre-pubertal child, or an adult aged over 35 years with a low-risk sexual history.

A urine dipstick test positive for leucocytes and nitrites may suggest an enteric causative organism.

A urine dipstick test positive for leucocytes only may suggest an STI causative organism.

**Differentiating clinical features**

Table 1. The main clinical features of possible testicular causes of scrotal pain and swelling.

| Diagnosis | Clinical features |
|---|---|
| Testicular cancer | • Usually unilateral testis enlargement, or change in shape or texture (often painless, may be dull ache or dragging sensation in the scrotum).<br>• Typically, solid firm mass involving all or part of testis, may be scrotal swelling.<br>• May be associated supraclavicular lymphadenopathy or abdominal mass (retroperitoneal lymphadenopathy).<br>• May present with hydrocele or epididymo-orchitis. |
| Testicular torsion (torsion of spermatic cord) | • Usually acute, severe unilateral scrotal pain and swelling; pain may radiate to the groin or lower abdominal pain.<br>• May be associated with nausea and vomiting.<br>• May be history of previous episodes of severe, self-limiting pain (torsion and detorsion).<br>• The cremasteric reflex may be abnomal or absent, and the testis may be elevated in the scrotum and have a transverse lie.<br>• In neonates, extravaginal torsion may present with painless scrotal swelling or mass, with or without acute inflammation. |
| Torsion of appendix testis or appendix epididymis | • May be sudden onset or gradual onset over a few days.<br>• Typically, self-limiting pain and tenderness over the superior pole of the testis or the epididymis.<br>• Systemic symptoms and nausea and vomiting are unlikely.<br>• Cremasteric reflex may be present.<br>• May be a 'blue dot sign', where an inflamed and ischaemic torsed appendage can be seen through the scrotal skin over the superior pole. |
| Epididymo-orchitis | • Acute unilateral scrotal pain and/or swelling.<br>• May be fever, dysuria, urethral discharge, or parotid swelling (latter if mumps orchitis).<br>• Typically, palpable tender swelling of epididymis and/or testis, may be erythema of scrotal skin.<br>• Scrotal pain and tenderness may be relieved by elevation of testis ('Prehn sign').<br>• May present with hydrocele. |

| Diagnosis | Clinical features |
|---|---|
| Squamous cell carcinoma of the scrotum | • Gradual onset, often painless raised papule, plaque, ulcer, or non-healing skin lesion on the scrotal wall.<br>• May be associated inguinal lymphadenopathy. |
| Inguinal hernia | • May be sudden or gradual onset, may present with dull dragging discomfort in the scrotum.<br>• May be painful, particularly if incarcerated or strangulated.<br>• Unilateral inguinal bulge, palpable separate to the testicle; cannot palpate the spermatic cord or inguinal ring above the swollen area.<br>• May enlarge with increased intra-abdominal pressure (positive cough impulse), and disappear on lying down (if reducible). |
| Haematocele | • May be sudden or gradual onset, usually pain and tenderness following trauma.<br>• May transilluminate with a pen torch (to a lesser extent than a hydrocele). |
| Epididymal cyst or spermatocele | • Gradual onset, painless, soft, fluctuant, smooth round nodule palpated in the spermatic cord.<br>• Usually small, but may enlarge.<br>• May transilluminate with a pen torch. |
| Hydrocele | • May be sudden or gradual onset.<br>• Typically painless fluctuant, ovoid swelling enveloping the testis, or located above the testis along the spermatic cord.<br>• Communicating hydroceles typically fluctuate in size, related to activity or raised intra-abdominal pressure.<br>• Transilluminates with a pen torch.<br>• May be associated with a varicocele, testicular torsion, testicular cancer, or inguinal hernia. |
| Varicocele | • Gradual onset, usually painless but may be a dull, dragging discomfort in the scrotum when standing.<br>• The dilated tortuous veins have a 'bag of worms' consistency.<br>• Disappears on lying down; worse on standing, straining, or activity. |

**Management**

**Scenario: Testicular torsion**

**From birth onwards (Male).**

- If a person has sudden-onset scrotal pain and testicular torsion is suspected clinically, arrange emergency hospital admission to urology or paediatric surgery, depending on clinical judgement.
- If a person has no current scrotal swelling or pain, but has a history of previous episodes of severe, self-limiting scrotal pain or swelling, arrange a urology referral, the urgency depending on the frequency and duration of episodes.

**Scenario: Testicular cancer**

**From birth onwards (Male).**

If a person has suspected testicular cancer on clinical examination (or incidentally following ultrasound of the scrotum):

- Arrange an urgent urology referral (using a 2-week suspected cancer pathway).
- Consider arranging blood tests for tumour markers (alpha-fetoprotein [AFP], human chorionic gonadotrophin [hCG], and lactate dehydrogenase [LDH] levels) at the time of referral, depending on local referral guidelines and availability.

**Scenario: Squamous cell carcinoma of the scrotum**

**From birth onwards (Male).**

If a person has suspected squamous cell carcinoma of the scrotum:

- Arrange an urgent referral to a urologist or dermatologist (using a 2-week suspected cancer pathway), depending on clinical judgement.

**Scenario: Inguinal hernia**

**From birth onwards (Male).**

If a person presents with a suspected inguinal hernia:

- If there is a suspected serious complication such as strangulation or intestinal obstruction, arrange emergency hospital admission.

**If hospital admission is not needed:**

for children and young people aged less than 18 years:

- Arrange urgent referral to a paediatric surgeon, preferably to be seen within 2 weeks.

for adults aged 18 years and above:

- Arrange urgent referral to paediatric or general surgery to consider surgical management if the hernia is irreducible, or only partially reducible.
- Offer routine referral to paediatric or general surgery to consider surgical management if the hernia is symptomatic but reducible, depending on clinical judgement. Note: referral may not be needed if a hernia is minimally symptomatic, and the person has significant comorbidity and declines surgery.

**Scenario: Epididymo-orchitis**

**From birth onwards (Male).**

If an adolescent or adult has suspected epididymo-orchitis:

- If symptoms are severe, the person is systemically unwell, or there is a suspected serious complication, arrange emergency hospital admission.
- If hospital admission is not needed, identify the most likely causative organism based on the person's age, urine dipstick test results, and risk factors to guide management.

Assess the risk of a sexually transmitted infection (STI), such as chlamydia or gonorrhoea.
Assess the risk of an enteric organism associated with lower urinary tract infection (UTI).
Consider the possibility of other rare causes such as mumps orchitis, and manage appropriately.
If an STI is the most likely cause, advise urgent referral to a local specialist sexual health clinic for STI testing, treatment, and possible contact tracing.
If urgent referral to a sexual health specialist is not possible.
If an enteric organism is the most likely cause:

- Ensure a urine dipstick test and mid-stream urine (MSU) sample for microscopy and culture have been arranged.

- Consider treating empirically with oral ofloxacin 200 mg twice daily for 14 days, or oral levofloxacin 500 mg once daily for 10 days. If there is any uncertainty about management in primary care, seek urgent specialist advice.

If a quinolone antibiotic is contraindicated, treat with oral co-amoxiclav 500/125 mg three times a day for 10 days.

If treating with fluoroquinolone antibiotics in primary care, advise about potential adverse effects:

- Advise the person that this class of antibiotic can very rarely cause long-lasting, potentially irreversible adverse effects affecting the tendons, muscles, joints, and/or central nervous system.
- Advise the person to stop fluoroquinolone antibiotic treatment immediately and seek immediate medical advice if they develop serious tendon, muscle, or joint pain or inflammation.

Give advice on self-management measures and when to arrange follow up.

Advise on rest, scrotal support (such as with supportive underwear), and the use of analgesia such as paracetamol and/or nonsteroidal anti-inflammatory drugs (NSAIDs) until symptoms improve.

If symptoms worsen, or are not improving after 3 days of antibiotic treatment, advise the person to seek urgent medical review.

**Antibiotic treatment regimens**

If urgent referral to a local specialist sexual health clinic is not possible, start empirical antibiotic treatment in primary care. Advise the person to abstain from sexual contact until they and any partner(s) have completed treatment and follow up if there is a confirmed or suspected sexually transmitted infection (STI). Ideally follow up and contact tracing should be arranged by a local specialist sexual health clinic.

If epididymo-orchitis is most likely due to any STI:

- Treat empirically with ceftriaxone 1 g intramuscular (IM) injection as a single dose, depending on local prescribing protocols, plus oral doxycycline 100 mg twice daily for 10–14 days.
- If a cephalosporin and/or tetracycline antibiotic is contraindicated, treat with oral ofloxacin 200 mg twice daily for 14 days.

If epididymo-orchitis is most likely due to chlamydia or other non-gonococcal organisms (if no risk factors for gonorrhoea):

- Treat empirically with oral doxycycline 100 mg twice daily for 10–14 days, or oral ofloxacin 200 mg twice daily for 14 days.
- If a quinolone antibiotic is contraindicated, treat with oral co-amoxiclav 500/125 mg three times a day for 10 days.

If epididymitis is most likely due to an STI and/or enteric organism (for example men who have insertive anal sex):

- Consider treating empirically with ceftriaxone 1 g IM, depending on local prescribing protocols, plus oral ofloxacin 200 mg twice daily for 14 days.
- If a quinolone antibiotic is contraindicated, treat with oral co-amoxiclav 500/125 mg three times a day for 10 days.

**How should I follow up a person with epididymo-orchitis?**

**If a person has a diagnosis of epididymo-orchitis and has started first-line management:**

- If symptoms have worsened or are not improving after 3 days of antibiotic treatment:
  Check compliance with treatment.
  Check results of any initial investigations, and amend treatment as necessary when results are available.
- If an enteric organism is confirmed, arrange a urology referral to investigate for an underlying structural or functional abnormality or urinary tract obstruction.
- If a sexually transmitted infection (STI) is suspected or confirmed, check sexual abstinence and ensure that partner notification is complete if not already undertaken.

Consider an alternative diagnosis including an infective or non-infective cause, and manage appropriately.

Ensure the person is reviewed after 2 weeks:

- Check compliance with treatment.
- Check symptoms have improved or resolved.

Reassure that scrotal swelling completely resolves in more than 80% of people by 3 months after antibiotic treatment.

**How should I manage a pre-pubertal child with suspected epididymo-orchitis?**

If a pre-pubertal child has suspected epididymo-orchitis:

- If symptoms are severe, the child is systemically unwell, or there is a suspected serious complication, arrange emergency hospital admission.
- If a urinary tract infection (UTI) is confirmed, seek specialist advice on the choice of antibiotic, dose, and duration of treatment, and arrange onward referral to a paediatrician as appropriate.
- If a UTI is not confirmed, do not start antibiotic treatment routinely in primary care.
- Reassure parents/carers that epididymitis is usually a self-limiting condition which resolves with rest and analgesia such as paracetamol and/or nonsteroidal anti-inflammatory drugs (NSAIDs) as needed.
- If there is any uncertainty about management in primary care, seek specialist advice or consider arranging referral to paediatric urology, the urgency depending on clinical judgement.

**Scenario: Haematocele**

**From birth onwards (Male).**

If the person has a suspected haematocele:

- If it follows acute trauma, arrange emergency hospital admission.
- If it is atraumatic or chronic, arrange referral for an urgent ultrasound of the scrotum and onward urology referral as appropriate.
- small haematoceles (smaller than three times the size of the contralateral testis) can be managed conservatively.
- large haematoceles may need surgical management.
- If there is any uncertainty regarding management, seek specialist advice.

**Scenario: Epididymal cyst/spermatocele**

**From birth onwards (Male).**

If there is a confirmed diagnosis of epididymal cyst (for example clinically or following ultrasound of the scrotum):

- Reassure the person it is a common harmless finding, is usually asymptomatic, and rarely needs treatment.
- If the person has bothersome symptoms, offer a routine referral to urology.

**Scenario: Hydrocele**

**From birth onwards (Male).**

If an infant has a suspected congenital hydrocele since birth:

- Reassure the parents/carers that the hydrocele is likely to resolve without treatment by 12 months of age.

Arrange referral to a paediatric surgeon, the urgency depending on clinical judgement, if:

- A simple, non-communicating hydrocele either is not decreasing in size, or is still present after 12 months of age.
- A concomitant inguinal hernia or underlying testicular pathology is suspected.
- The hydrocele is localized to the spermatic cord.

- There is also a palpable abdominal mass (suggesting an abdomino-scrotal hydrocele).

**How should I manage a non-congenital hydrocele?**

If there is a suspected non-congenital hydrocele in an adolescent or adult:

- Assess whether the hydrocele may be due to an underlying cause, such as testicular torsion, testicular cancer, epididymo-orchitis, trauma, or varicocele surgery.
- Arrange emergency hospital admission or specialist referral, depending on the possible underlying cause.
- If the person is 18–40 years of age, or the testis cannot be palpated, arrange an urgent ultrasound scan of the scrotum.

## TOPIC: Varicocele

**Definition**

A varicocele is a scrotal swelling consisting of a collection of dilated veins of the pampiniform plexus in the spermatic cord. It may be caused by incompetent or absent valves in the testicular (spermatic) vein or, rarely, may be secondary to a tumour or other pathological process obstructing the spermatic vein.

Varicoceles are graded according to their size:

- Sub-clinical — detected only by Doppler ultrasound.
- Grade I (small) — palpable only with Valsalva manoeuvre.
- Grade II (moderate) — palpable without Valsalva manoeuvre.
- Grade III (large) — visible through the scrotal skin.

**Risk factors**

• The risk factors for varicocele include:

- Being tall and lean, with lower body mass index (BMI) than age-matched controls.
- Being underweight.
- Family history of varicocele — especially in first degree relatives.

**Complications**

- Pain.
- Failure of ipsilateral testicular growth and development, hypogonadism.
  Left testicular volume loss has been noted in 70% of adolescent males with grade II and III varicocele. In late adolescence, the contralateral right testis may also become smaller.
- Fertility problems.
  Clinically detectable varicoceles can be associated with abnormal gonadotrophin levels, impaired spermatogenesis, and histological changes to sperm. Approximately 20% of males diagnosed with varicocele will have fertility problems. The incidence of varicocele among men with primary infertility is up to 44%. Increased scrotal temperature, hypoxia, and reflux of toxic metabolites leading to testicular damage are possible underlying mechanisms.

**Prognosis**

• Most varicoceles are asymptomatic, do not affect fertility, and do not require treatment.

• Varicocelectomy is overall associated with:

- Reduced pain, improved testicular volume.
- Improved semen parameters — although data regarding whether this translates to improved pregnancy rates are highly conflicting.

• A greater degree of clinical improvement following varicocelectomy is associated with a higher grade of varicocele.

**Diagnosis**

Suspect varicocele in adolescent boys, and men, if there is:

- Painless scrotal swelling on the left side — 90% of varicoceles appear on the left and the remaining 10% are bilateral.
- Scrotal or groin pain (uncommon) — fewer than 3% of men with varicocele have pain, dragging, or heavy sensations in the scrotum.

Typical presentation of a 'bag of worms' within the spermatic cord above the testis on the left side of the scrotum:

- The scrotum on the side of the varicocele may be seen to hang lower than on the normal side.
- Dilation and tortuosity of the veins are increased on standing and are usually decreased when the man lies down. The varicocele cannot usually be palpated lying down.
- Performing the Valsalva manoeuvre whilst standing increases dilation.
- There may be a cough impulse.

Small testis — larger varicoceles are associated with a higher incidence of testicular growth arrest in adolescents, Surgical correction may reverse this.
Infertility — approximately one in four men with abnormal semen parameters will have a varicocele, and approximately 40% of men with infertility have a varicocele.

**How should I assess a man or boy with a suspected varicocele?**

The diagnosis of varicocele is usually clinical.

- Ultrasound with colour flow Doppler imaging may be useful if there is diagnostic uncertainty, or in larger men where thick scrotal skin or increased amounts of scrotal tissue may make examination difficult.

Examine all adolescents in the supine and standing positions to identify varicocele and assess whether testicular growth arrest has occurred.

- If the testes are symmetrical, examine the boy annually through puberty and refer to a urologist if there is testicular asymmetry (left smaller than right) or impaired testicular growth.

For men with varicocele, consider carrying out semen analyses and evaluation of serum follicle-stimulating hormone (FSH) and testosterone levels to assess testicular function.

- Abnormal sperm production with an elevated FSH is consistent with impaired spermatogenesis.

**Management**

**From age 10 years onwards (Male).**

For adolescents with: Subclinical or grade I varicocele — no treatment is necessary.
Provide advice and reassurance: Grade II or III varicocele and symmetrical testes — observe with annual examinations.
The primary indication for surgery is testicular growth arrest: Grade II or III and asymmetrical testes — refer to a urologist for possible surgery.
For men with: Sub-clinical or grade I varicocele — no treatment is necessary.
Offer semen analysis if fertility is a concern.

- Grade II or III asymptomatic varicocele and normal semen parameters — consider observing with semen analysis every 1–2 years if clinically appropriate.
- Grade II or III symptomatic varicocele, or with abnormal semen parameters — refer to a urologist for possible surgery.

Refer urgently to a urologist if:

- A varicocele appears suddenly and is painful, The varicocele does not drain when lying down.

Refer routinely to a urologist if there is pain or discomfort.
Refer adolescents with a varicocele to a urologist:

- If there are concerns about reduced ipsilateral testicular volume.
- If the boy or parents/guardians are concerned by the appearance, or symptoms, and cannot be fully reassured in primary care.

Do not routinely refer the male partner of an infertile couple for varicocele surgery solely as a form of fertility treatment.

- However, consider seeking specialist advice for men with clinical varicocele, abnormal semen parameters, and otherwise unexplained infertility.

Refer to a urologist if there is uncertainty about the nature of a scrotal swelling.
Where referral and treatment are not required:

- Reassure the man that the varicocele is not likely to cause any symptoms or long-term complications.

Explain that:

- Any associated discomfort may be initially managed with supportive underwear and simple analgesia.
- Although varicoceles may be associated with fertility problems, nearly two-thirds of men who have a varicocele have no difficulty in fathering children.
- If surgery is being considered, explain that the urologist will discuss the risks and benefits of the available procedures.